

API SPEC V1 – Enterprise API Specification

Section 1: API Specification Overview & Governance

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Owner: Platform Engineering

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Classification: Internal Confidential

1. Purpose

The purpose of this document is to establish the enterprise API governance framework, API standards, lifecycle management controls, interoperability requirements, security requirements, audit requirements, and compliance obligations governing all application programming interfaces (APIs) within the Transparent Opportunity Decision Platform (TODP).

This API specification serves as the authoritative reference for all internal APIs, external APIs, administrative APIs, governance APIs, audit APIs, replay APIs, and future platform integration interfaces.

The document ensures that all APIs are designed, implemented, operated, monitored, secured, governed, and maintained according to enterprise-grade standards.

2. Objectives

OBJ-API-001

Establish consistent API standards across TODP.

OBJ-API-002

Provide controlled interoperability between platform modules.

OBJ-API-003

Support enterprise security requirements.

OBJ-API-004

Enable traceable API interactions.

OBJ-API-005

Support auditability and governance.

OBJ-API-006

Enable scalable platform integration.

OBJ-API-007

Support replayable API activity reconstruction.

OBJ-API-008

Maintain enterprise compliance readiness.

3. Scope

This specification governs APIs supporting:

- Opportunity Module
- Authority Module
- Evidence Module
- Retrieval Module
- Ranking Module
- Recommendation Module
- Governance Module
- Audit Module
- Replay Module

The scope includes:

- Internal service APIs
- Administrative APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Integration APIs
- Future partner APIs
- Future customer APIs

4. API Governance Principles

All TODP APIs shall comply with the following principles.

PRN-API-001

Consistency

APIs shall follow standardized design patterns.

PRN-API-002

Security

APIs shall implement enterprise security controls.

PRN-API-003

Traceability

API activity shall be fully traceable.

PRN-API-004

Auditability

API activity shall generate audit records.

PRN-API-005

Governance

API changes shall follow governance workflows.

PRN-API-006

Versioning

APIs shall support controlled evolution.

PRN-API-007

Reliability

APIs shall support enterprise operational requirements.

PRN-API-008

Replayability

API interactions shall support replay reconstruction requirements.

5. API Governance Authority Model

The API governance structure shall include:

Platform Engineering

API ownership authority.

Architecture Review Board

Architecture governance authority.

Security Governance Team

Security review authority.

Compliance Governance Team

Compliance review authority.

Product Governance Council

Final approval authority.

Engineering Leadership

Implementation authority.

All API ownership assignments shall be documented.

6. API Ownership Requirements

Each API shall maintain assigned ownership.

Required Ownership Roles

API Owner

Business and technical accountability.

Technical Owner

Implementation accountability.

Security Reviewer

Security oversight.

Governance Reviewer

Governance oversight.

Operational Owner

Production support accountability.

Ownership assignments shall be maintained throughout the API lifecycle.

7. API Lifecycle Model

All APIs shall follow the approved lifecycle.

Lifecycle States

Proposed

Under Review

Approved

Implemented

Active

Deprecated

Retired

Archived

Lifecycle state transitions shall be auditable.

8. API Classification Framework

APIs shall be classified according to business purpose.

API-CLASS-001

Core Business API

API-CLASS-002

Governance API

API-CLASS-003

Audit API

API-CLASS-004

Administrative API

API-CLASS-005

Integration API

API-CLASS-006

Replay API

API-CLASS-007

Operational API

API-CLASS-008

Security API

Each API shall have one primary classification.

9. API Design Standards

All APIs shall adhere to approved enterprise design standards.

Design Requirements

- Consistent naming
- Versioned interfaces
- Standard response formats
- Standard error handling
- Documented contracts
- Traceable operations

API design deviations require formal approval.

10. API Versioning Standards

All APIs shall implement version management.

Version Format

v1

v2

v3

Versioning Requirements

- Backward compatibility evaluation
- Change documentation
- Approval records
- Migration guidance

API versions shall remain traceable.

11. API Contract Governance

Every API shall maintain a formally approved contract.

Required Contract Elements

- Endpoint definition
- Request schema
- Response schema
- Error schema
- Security requirements
- Ownership information

Contracts shall be version controlled.

12. Security Requirements

All APIs shall comply with enterprise security requirements.

Minimum Requirements

- Authentication
- Authorization
- Encryption
- Input validation
- Output validation
- Audit logging
- Monitoring

Security controls shall be reviewed regularly.

13. Traceability Requirements

All API interactions shall support traceability.

Required Attributes

- Request identifier
- Correlation identifier
- User identifier
- Service identifier
- Timestamp
- Outcome

Traceability records shall support investigations and replay.

14. Audit Requirements

All API activity shall generate audit records.

Auditable Activities

- Requests
- Responses
- Failures
- Configuration changes

- Administrative actions
- Security events

Audit records shall be retained according to retention policies.

15. Compliance Requirements

API governance shall support:

- SOC 2
- ISO 27001
- ISO 9001
- Enterprise Governance Standards
- Customer Security Reviews
- Procurement Assessments

Compliance mappings shall be maintained.

16. API Availability Requirements

APIs shall support enterprise operational requirements.

Availability Requirements

- Service health monitoring
- Fault detection
- Recovery procedures
- Operational visibility

Availability objectives shall be defined for production APIs.

17. API Performance Requirements

API performance governance shall include:

- Response latency
- Throughput
- Error rates
- Resource utilization

Performance targets shall be documented for each API.

18. API Scalability Requirements

APIs shall support projected enterprise growth.

Growth Factors

- User growth
- Opportunity growth

- Evidence growth
- Retrieval growth
- Governance growth

Scalability assumptions shall be documented.

19. API Change Management

All API changes shall follow formal governance procedures.

Governed Changes

- New APIs
- Endpoint modifications
- Schema changes
- Security changes
- Version changes

Change approvals shall be retained.

20. API Documentation Requirements

Every API shall maintain documentation including:

- Purpose
- Scope
- Endpoint definitions
- Security requirements
- Error handling
- Version information
- Operational guidance

Documentation shall be maintained as a controlled artifact.

21. API Monitoring Requirements

Monitoring shall include:

- Availability monitoring
- Performance monitoring
- Error monitoring
- Security monitoring
- Usage monitoring

Monitoring controls shall support operational management.

22. API Risk Management

API risks shall be identified and managed.

Risk Categories

- Security risk
- Operational risk
- Availability risk
- Compliance risk
- Scalability risk

Risk ownership shall be assigned.

23. API Exception Management

API design exceptions shall require:

- Business justification
- Risk assessment
- Architecture review
- Governance approval

Exceptions shall be tracked and reviewed.

24. Architectural Constraints

All APIs shall comply with:

API-CON-001

Maintain security controls.

API-CON-002

Support traceability.

API-CON-003

Support auditability.

API-CON-004

Support governance controls.

API-CON-005

Maintain version control.

API-CON-006

Support replayability.

API-CON-007

Support scalability.

API-CON-008

Support enterprise readiness.

25. Metrics

The following metrics shall be monitored.

MET-API-001

API availability.

MET-API-002

API latency.

MET-API-003

API error rate.

MET-API-004

API security incidents.

MET-API-005

API governance compliance.

MET-API-006

API audit coverage.

MET-API-007

API traceability completeness.

MET-API-008

API documentation completeness.

26. Audit Controls

API audits shall verify:

- Contract compliance
- Security compliance
- Traceability completeness
- Monitoring coverage

- Governance approvals
- Documentation quality

Audit findings shall be tracked to closure.

27. Repository Requirements

API specifications shall be maintained in a controlled repository.

Repository Controls

- Version control
- Access controls
- Change history
- Approval records

Repository integrity shall be preserved.

28. Related Documents

This document maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

SEC V1

GOV V1

AUD V1

REL V1

29. Future Evolution Requirements

The API governance framework shall support:

- New modules
- New integrations
- External consumers
- Enterprise customers
- Regulatory requirements

Architecture evolution shall remain governed.

30. Section Summary

This section establishes the API governance foundation for TODP. It defines API ownership, lifecycle management, classification standards, design principles, security requirements, traceability controls, audit obligations, compliance expectations, change management procedures, monitoring requirements, and governance responsibilities. These controls ensure that all TODP APIs remain secure, consistent, auditable, traceable, scalable, replayable, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 2: API Architecture & Platform Standards

Document ID: API-V1-S02

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Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the architectural standards, platform conventions, communication models, interface principles, interoperability requirements, and technical constraints governing all APIs within the Transparent Opportunity Decision Platform (TODP).

This section establishes a consistent enterprise-grade API architecture that supports modularity, scalability, security, observability, governance, auditability, and replayability across all TODP domains.

2. Objectives

OBJ-API-ARCH-001

Establish a unified API architecture model.

OBJ-API-ARCH-002

Standardize communication across platform modules.

OBJ-API-ARCH-003

Support modular architecture principles.

OBJ-API-ARCH-004

Enable enterprise scalability.

OBJ-API-ARCH-005

Support operational reliability.

OBJ-API-ARCH-006

Maintain interoperability standards.

OBJ-API-ARCH-007

Support governance and audit requirements.

OBJ-API-ARCH-008

Enable future platform expansion.

3. Scope

This section applies to all APIs supporting:

- Opportunity Module
- Authority Module
- Evidence Module
- Retrieval Module
- Ranking Module
- Recommendation Module
- Governance Module
- Audit Module
- Replay Module

The standards apply to:

- Internal APIs
- Administrative APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Integration APIs
- Future customer APIs
- Future partner APIs

4. Architectural Principles

All APIs shall comply with the following architecture principles.

API-PRN-001

Contract First Design

API contracts shall be defined before implementation.

API-PRN-002

Loose Coupling

Consumers shall remain independent from implementation details.

API-PRN-003

Versioned Evolution

APIs shall evolve through controlled versioning.

API-PRN-004

Security by Design

Security controls shall be embedded into API architecture.

API-PRN-005

Observability by Design

APIs shall support monitoring and diagnostics.

API-PRN-006

Governance by Design

Governance requirements shall be incorporated into API design.

API-PRN-007

Auditability by Design

API interactions shall generate auditable records.

API-PRN-008

Replayability by Design

API activity shall support replay reconstruction.

5. API Communication Architecture

TODP APIs shall utilize a service-oriented communication architecture.

Communication Characteristics

- Service-to-service communication
- Request-response interactions
- Controlled asynchronous processing
- Contract-based integration
- Version-controlled interfaces

Direct database integrations between modules are prohibited unless explicitly approved through an ADR.

6. API Architectural Boundaries

Each TODP module shall expose controlled API interfaces.

Approved Domains

Opportunity APIs

Authority APIs

Evidence APIs

Retrieval APIs

Ranking APIs

Recommendation APIs

Governance APIs

Audit APIs

Replay APIs

Modules shall communicate through approved interfaces only.

7. Canonical API Design Model

The platform shall maintain a canonical API model.

Required Elements

- Resource identification
- Standard operations
- Version identifiers

- Correlation identifiers
- Security context
- Audit context

The canonical model shall be used consistently across all APIs.

8. API Endpoint Structure Standards

Endpoint naming shall follow standardized conventions.

Format

/api/{version}/{domain}/{resource}

Examples

/api/v1/opportunities

/api/v1/evidence

/api/v1/recommendations

/api/v1/governance

Endpoint structures shall remain predictable and discoverable.

9. API Resource Architecture

Resources shall represent business entities rather than implementation details.

Approved Resource Categories

- Opportunities
- Authorities
- Evidence
- Retrieval Requests
- Rankings
- Recommendations
- Governance Reviews
- Audit Records
- Replay Executions

Internal implementation objects shall not be exposed externally.

10. Standard API Operations

The following operation patterns shall be supported where applicable.

CREATE

Resource creation

READ

Resource retrieval

UPDATE

Resource modification

DELETE

Controlled removal

SEARCH

Resource discovery

VALIDATE

Business validation

REPLAY

Historical reconstruction

Operations shall be documented within individual API specifications.

11. API Request Standards

All requests shall include:

Required Fields

- Request identifier
- Correlation identifier
- Timestamp
- Authentication context

Optional Fields

- Governance context
- Replay context
- Audit context

Request validation shall be mandatory.

12. API Response Standards

All responses shall follow a standardized format.

Response Components

- Status
- Result
- Metadata
- Traceability identifiers
- Audit references

Responses shall support consistent consumer behavior.

13. Error Handling Architecture

All APIs shall implement standardized error handling.

Required Error Elements

- Error code
- Error category
- Error message
- Traceability identifier
- Timestamp

Errors shall be machine-readable.

14. API Correlation Framework

Every API interaction shall support request correlation.

Correlation Attributes

- Request ID
- Correlation ID
- User ID
- Session ID
- Opportunity ID where applicable

Correlation records shall support investigations and replay.

15. Integration Architecture Standards

External integrations shall use controlled integration APIs.

Requirements

- Contract governance
- Security validation
- Monitoring controls
- Audit controls
- Version controls

Unmanaged integrations are prohibited.

16. Event Architecture Standards

APIs may generate platform events.

Event Requirements

- Unique identifier
- Event type
- Timestamp
- Source service
- Outcome

Events shall support audit and replay functions.

17. API Security Architecture Standards

All APIs shall support:

- Authentication
- Authorization
- Encryption
- Input validation
- Output validation
- Audit logging

Security standards are further defined within SEC V1.

18. Observability Architecture Standards

All APIs shall generate observability data.

Required Telemetry

- Metrics
- Logs
- Events
- Traces

Telemetry standards shall be applied consistently.

19. Scalability Architecture Standards

APIs shall support enterprise scalability objectives.

Scalability Areas

- User growth
- Transaction growth

- Evidence growth
- Governance growth
- Audit growth

Capacity assumptions shall be documented.

20. Availability Architecture Standards

API architecture shall support:

- Fault tolerance
- Recovery procedures
- Monitoring
- Service continuity

Availability requirements shall be defined per API.

21. Data Exchange Standards

APIs shall use standardized data exchange formats.

Requirements

- Structured payloads
- Versioned schemas
- Validation rules
- Traceable identifiers

Schema governance shall be maintained.

22. Governance Architecture Standards

API interactions shall support governance controls.

Governance Requirements

- Approval workflows
- Policy validation
- Exception tracking
- Audit support

Governance controls shall be enforceable.

23. Audit Architecture Standards

API activity shall support:

- Audit event generation
- Audit traceability

- Historical retention
- Investigation support

Audit controls shall be mandatory.

24. Replay Architecture Standards

API interactions shall preserve information necessary for replay.

Replay Elements

- Request context
- Response context
- Version information
- Traceability identifiers

Replay reconstruction shall be supported across the platform.

25. API Dependency Management

API dependencies shall be governed.

Requirements

- Dependency documentation
- Version tracking
- Risk assessment
- Change governance

Hidden dependencies are prohibited.

26. Architectural Constraints

All APIs shall comply with:

API-ARCH-CON-001

Maintain modular boundaries.

API-ARCH-CON-002

Support contract-first development.

API-ARCH-CON-003

Support traceability.

API-ARCH-CON-004

Support auditability.

API-ARCH-CON-005

Support replayability.

API-ARCH-CON-006

Maintain security controls.

API-ARCH-CON-007

Support observability.

API-ARCH-CON-008

Support enterprise scalability.

27. Metrics

The following architecture metrics shall be monitored.

MET-API-ARCH-001

API contract compliance rate.

MET-API-ARCH-002

Cross-module interoperability success rate.

MET-API-ARCH-003

Dependency governance compliance.

MET-API-ARCH-004

API observability coverage.

MET-API-ARCH-005

Replay coverage.

MET-API-ARCH-006

Audit coverage.

MET-API-ARCH-007

Architecture exception rate.

MET-API-ARCH-008

Scalability readiness score.

28. Audit Requirements

Architecture audits shall verify:

- Contract-first compliance
- Modular boundary enforcement
- Dependency governance
- Traceability coverage
- Audit controls
- Replay readiness
- Security compliance
- Observability coverage

Audit findings shall be documented and remediated.

29. Compliance Requirements

The API architecture standards shall support:

- SOC 2
- ISO 27001
- ISO 9001
- Enterprise Governance Standards
- Customer Security Reviews
- Procurement Assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, COMP, and REL artifacts.

30. Section Summary

This section defines the architectural standards governing all TODP APIs, including communication architecture, endpoint standards, resource modeling, request and response conventions, error handling, observability, security controls, governance requirements, audit requirements, replay support, scalability expectations, and dependency management controls. These standards establish a consistent and enterprise-ready API architecture that supports interoperability, transparency, traceability, auditability, governance, and long-term platform evolution.

API SPEC V1 – Enterprise API Specification

Section 3: Common API Conventions, Standards & Shared Contracts

Document ID: API-V1-S03

Document Title: Common API Conventions, Standards & Shared Contracts

Version: 1.0

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Owner: Platform Engineering
Review Authority: Architecture Review Board (ARB)
Approval Authority: Product Governance Council
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1. Purpose

The purpose of this section is to define the common conventions, shared contracts, standardized payload structures, identifier requirements, validation rules, response models, error models, metadata standards, and interoperability controls applicable to all APIs within the Transparent Opportunity Decision Platform (TODP).

This section establishes a common language and integration model across all platform APIs to ensure consistency, maintainability, interoperability, governance compliance, auditability, traceability, and replayability.

2. Objectives

OBJ-API-COM-001

Standardize API interaction patterns.

OBJ-API-COM-002

Establish common request structures.

OBJ-API-COM-003

Establish common response structures.

OBJ-API-COM-004

Provide consistent error handling.

OBJ-API-COM-005

Support platform-wide traceability.

OBJ-API-COM-006

Support audit and governance requirements.

OBJ-API-COM-007

Enable interoperability between modules.

OBJ-API-COM-008

Support future API expansion.

3. Scope

This section applies to all APIs supporting:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

The standards apply to:

- Internal APIs
- Administrative APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Integration APIs
- External APIs

4. Standard Content Types

Approved content types shall include:

application/json

Primary API payload format.

application/problem+json

Error response format.

text/plain

Operational diagnostic responses where approved.

Additional content types require Architecture Review Board approval.

5. Standard Character Encoding

All API payloads shall use:

UTF-8

Character encoding standard.

All services shall consistently support UTF-8 processing.

6. Standard Request Headers

Required Request Headers

X-Request-ID

Unique request identifier.

X-Correlation-ID

Cross-service correlation identifier.

Authorization

Authentication token.

Content-Type

Payload format.

Optional Request Headers

X-Replay-ID

Replay context identifier.

X-Governance-ID

Governance review identifier.

X-Audit-ID

Audit correlation identifier.

7. Standard Response Headers

Required Response Headers

X-Request-ID

Request identifier.

X-Correlation-ID

Correlation identifier.

X-Service-Version

API implementation version.

X-Response-Time

Response processing duration.

Optional Response Headers

X-Replay-Reference

Replay reference.

X-Governance-Reference

Governance reference.

8. Standard Identifier Model

All business entities shall use immutable identifiers.

Supported Identifiers

Opportunity ID

Authority ID

Evidence ID

Retrieval ID

Ranking ID

Recommendation ID

Governance ID

Audit ID

Replay ID

Identifiers shall remain globally unique within their domains.

9. Identifier Format Standards

Identifier Requirements

- Immutable
- Unique
- Non-reusable

- Traceable
- Machine-readable

Identifiers shall support audit and replay reconstruction.

10. Standard Request Structure

All requests shall follow a common structure.

Request Components

Metadata

Context information.

Payload

Business request data.

Traceability

Correlation information.

Governance

Governance context where applicable.

Audit

Audit context where applicable.

11. Standard Response Structure

All successful responses shall contain:

status

Request outcome.

data

Business result.

metadata

Operational metadata.

traceability

Correlation information.

timestamp

Response generation time.

Responses shall remain consistent across APIs.

12. Standard Metadata Structure

Metadata shall include:

- Request ID
- Correlation ID
- Service Name
- Service Version
- Processing Timestamp

Additional metadata may be included when required.

13. Standard Traceability Model

Traceability information shall include:

- Request ID
- Correlation ID
- User ID
- Service ID
- Opportunity ID where applicable

Traceability data shall support investigations and replay operations.

14. Standard Error Model

Error responses shall contain:

errorCode

Machine-readable code.

errorCategory

Error classification.

errorMessage

Human-readable message.

requestId

Request identifier.

timestamp

Occurrence timestamp.

15. Error Classification Framework

Error Categories

Validation Error

Authentication Error

Authorization Error

Business Rule Error

Governance Error

System Error

Integration Error

Replay Error

Each error shall map to one category.

16. Validation Standards

All APIs shall implement:

- Schema validation
- Input validation
- Business rule validation
- Security validation
- Governance validation

Validation failures shall generate standardized responses.

17. Pagination Standards

APIs returning collections shall support pagination.

Pagination Elements

page

Current page.

pageSize

Records per page.

totalRecords

Total available records.

totalPages

Total available pages.

Pagination behavior shall remain consistent.

18. Sorting Standards

Collection APIs shall support sorting.

Requirements

- Explicit sort fields
- Ascending order
- Descending order

Unsupported sorting requests shall generate validation errors.

19. Filtering Standards

APIs shall support filtering where applicable.

Filtering Categories

- Status filtering
- Date filtering
- Ownership filtering
- Governance filtering
- Classification filtering

Filtering rules shall be documented per API.

20. Search Standards

Search-capable APIs shall support:

- Exact match search
- Partial search
- Identifier search
- Metadata search

Search behavior shall be documented.

21. Timestamp Standards

All timestamps shall use:

ISO 8601

UTC timezone

Example

2026-01-01T12:00:00Z

Timestamp consistency is mandatory.

22. Version Negotiation Standards

Version management shall support:

- Explicit versioning
- Backward compatibility review
- Controlled deprecation

Version negotiation behavior shall be documented.

23. Governance Metadata Standards

Governance-enabled APIs shall support:

- Governance ID
- Policy version
- Approval reference
- Governance outcome

Governance metadata shall remain traceable.

24. Audit Metadata Standards

Auditable APIs shall support:

- Audit ID
- Audit timestamp
- Audit category
- Audit reference

Audit metadata shall support investigations.

25. Replay Metadata Standards

Replay-enabled APIs shall support:

- Replay ID
- Historical version references

- Replay context
- Reconstruction identifiers

Replay metadata shall support historical reconstruction.

26. Security Metadata Standards

Security-related metadata shall include:

- Authentication context
- Authorization context
- Security policy reference
- Security event reference

Security metadata shall remain auditable.

27. Common Contract Governance

Shared contracts shall be governed through:

- Version control
- Change review
- Architecture review
- Security review
- Governance approval

Uncontrolled contract changes are prohibited.

28. Architectural Constraints

All APIs shall comply with:

API-COM-CON-001

Use standardized identifiers.

API-COM-CON-002

Use standardized request structures.

API-COM-CON-003

Use standardized response structures.

API-COM-CON-004

Support traceability.

API-COM-CON-005

Support auditability.

API-COM-CON-006

Support governance controls.

API-COM-CON-007

Support replayability.

API-COM-CON-008

Maintain interoperability.

29. Metrics

The following metrics shall be monitored.

MET-API-COM-001

Contract compliance rate.

MET-API-COM-002

Standard response compliance.

MET-API-COM-003

Validation success rate.

MET-API-COM-004

Traceability completeness.

MET-API-COM-005

Audit metadata coverage.

MET-API-COM-006

Replay metadata coverage.

MET-API-COM-007

Governance metadata coverage.

MET-API-COM-008

Interoperability compliance rate.

30. Section Summary

This section establishes the common conventions, shared contracts, identifier standards, request and response models, validation requirements, metadata standards, pagination rules, filtering standards, error handling conventions, governance metadata requirements, audit metadata requirements, replay metadata requirements, and interoperability controls applicable to all TODP APIs. These standards ensure consistency, maintainability, traceability, auditability, governance compliance, and enterprise interoperability across the entire API ecosystem.

API SPEC V1 – Enterprise API Specification

Section 4: Opportunity Module API Specification

Document ID: API-V1-S04

Document Title: Opportunity Module API Specification

Version: 1.0

Status: Approved

Owner: Opportunity Domain Team

Review Authority: Architecture Review Board (ARB)

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Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Opportunity Module within the Transparent Opportunity Decision Platform (TODP).

The Opportunity APIs provide controlled interfaces for opportunity creation, retrieval, modification, lifecycle management, ownership management, classification management, validation, audit traceability, governance integration, and replay support.

These APIs establish Opportunity as the authoritative entry point into the TODP decision lifecycle.

2. Objectives

OBJ-API-OPP-001

Support opportunity lifecycle management.

OBJ-API-OPP-002

Provide controlled opportunity access.

OBJ-API-OPP-003

Support opportunity validation.

OBJ-API-OPP-004

Maintain opportunity traceability.

OBJ-API-OPP-005

Enable governance integration.

OBJ-API-OPP-006

Support audit requirements.

OBJ-API-OPP-007

Enable replay reconstruction.

OBJ-API-OPP-008

Support enterprise scalability.

3. Scope

The Opportunity API domain governs:

- Opportunity creation
- Opportunity retrieval
- Opportunity updates
- Opportunity ownership management
- Opportunity classification management
- Opportunity lifecycle management
- Opportunity validation
- Opportunity audit access
- Opportunity replay support

4. API Domain Information

Domain Name

Opportunity

Domain Identifier

OPP

Primary Resource

Opportunity

Primary Identifier

Opportunity ID

Ownership

Opportunity Domain Team

Related Modules

Authority

Evidence

Governance

Audit

Replay

5. Opportunity Resource Model

Primary Attributes

- Opportunity ID
- Opportunity Name
- Opportunity Type
- Opportunity Description
- Opportunity Status
- Opportunity Classification
- Opportunity Owner
- Creation Timestamp
- Last Modified Timestamp

Governance Attributes

- Governance Status
- Governance Reference

Audit Attributes

- Audit Reference

6. API-OPP-001

Create Opportunity

Endpoint

POST /api/v1/opportunities

Purpose

Create a new opportunity record.

Required Inputs

- Opportunity Name
- Opportunity Type
- Opportunity Owner

Optional Inputs

- Description
- Classification
- Metadata

Success Response

201 Created

Generated Outputs

- Opportunity ID
- Initial Status
- Audit Reference

Authorization Requirement

Opportunity Create Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-OPP-002

Get Opportunity

Endpoint

GET /api/v1/opportunities/{opportunityId}

Purpose

Retrieve opportunity information.

Required Inputs

- Opportunity ID

Success Response

200 OK

Generated Outputs

- Opportunity Details
- Lifecycle Information
- Governance Information

Authorization Requirement

Opportunity Read Permission

Audit Requirement

Mandatory

8. API-OPP-003

Update Opportunity

Endpoint

PUT /api/v1/opportunities/{opportunityId}

Purpose

Modify approved opportunity attributes.

Required Inputs

- Opportunity ID
- Updated Attributes

Governance Validation

Required

Success Response

200 OK

Audit Requirement

Mandatory

Replay Requirement

Mandatory

9. API-OPP-004

Search Opportunities

Endpoint

GET /api/v1/opportunities

Purpose

Search opportunity inventory.

Supported Filters

- Status
- Classification
- Owner
- Date Range

Supported Sorting

- Creation Date
- Last Modified Date
- Status

Success Response

200 OK

Authorization Requirement

Opportunity Search Permission

10. API-OPP-005

Validate Opportunity

Endpoint

POST /api/v1/opportunities/{opportunityId}/validate

Purpose

Execute validation rules.

Validation Categories

- Schema Validation
- Business Validation
- Governance Validation

Success Response

200 OK

Validation Results Returned

11. API-OPP-006

Transition Opportunity Status

Endpoint

POST /api/v1/opportunities/{opportunityId}/status

Purpose

Move opportunity between lifecycle states.

Supported States

- Draft
- Submitted
- Validated
- Active
- Under Review
- Recommended
- Rejected
- Closed
- Archived

Governance Validation

Required

Audit Requirement

Mandatory

12. API-OPP-007

Assign Opportunity Owner

Endpoint

POST /api/v1/opportunities/{opportunityId}/owner

Purpose

Assign or transfer ownership.

Required Inputs

- Opportunity ID
- New Owner

Audit Requirement

Mandatory

Governance Validation

Required

13. API-OPP-008

Manage Opportunity Classification

Endpoint

POST /api/v1/opportunities/{opportunityId}/classification

Purpose

Update classification metadata.

Classification Categories

- Business Classification
- Governance Classification
- Operational Classification

Audit Requirement

Mandatory

14. API-OPP-009

Get Opportunity History

Endpoint

GET /api/v1/opportunities/{opportunityId}/history

Purpose

Retrieve lifecycle history.

Outputs

- State Changes
- Ownership Changes
- Governance Actions

Replay Requirement

Supported

15. API-OPP-010

Get Opportunity Audit Trail

Endpoint

GET /api/v1/opportunities/{opportunityId}/audit

Purpose

Retrieve audit history.

Outputs

- Audit References
- Event History
- Governance Actions

Authorization Requirement

Audit Access Permission

16. API-OPP-011

Get Opportunity Governance Status

Endpoint

GET /api/v1/opportunities/{opportunityId}/governance

Purpose

Retrieve governance status.

Outputs

- Governance State
- Governance Decisions
- Policy References

17. API-OPP-012

Replay Opportunity State

Endpoint

POST /api/v1/opportunities/{opportunityId}/replay

Purpose

Support historical reconstruction.

Required Inputs

- Opportunity ID
- Historical Timestamp

Outputs

- Historical State
- Version References

Replay Authorization

Required

18. Request Validation Requirements

Opportunity APIs shall validate:

- Identifier format
- Required attributes
- Ownership assignments
- Lifecycle state transitions
- Governance constraints

Validation failures shall return standardized error responses.

19. Response Standards

Opportunity APIs shall return:

- Status
- Data
- Metadata
- Traceability Information
- Audit References

Responses shall comply with API-V1-S03 standards.

20. Error Handling Standards

Supported Error Categories

- Validation Error
- Authorization Error
- Governance Error
- Business Rule Error
- System Error

Errors shall be standardized.

21. Security Requirements

Opportunity APIs shall implement:

- Authentication
- Authorization
- Input validation
- Audit logging
- Governance validation

Security requirements align with SEC V1.

22. Traceability Requirements

All Opportunity API interactions shall generate:

- Request ID
- Correlation ID
- Opportunity ID
- User ID
- Timestamp

Traceability data shall support replay and investigations.

23. Audit Requirements

The following activities shall generate audit records:

- Creation
- Modification
- Ownership transfer
- Status transitions
- Validation
- Governance actions
- Replay requests

Audit records shall be immutable.

24. Governance Requirements

Governance validation shall be enforced for:

- Ownership changes
- Classification changes
- Status transitions
- Opportunity closure
- Opportunity archival

Governance exceptions require approval.

25. Replay Requirements

Replay support shall preserve:

- Historical states
- Historical ownership
- Historical classifications
- Governance decisions

Replay reconstruction shall remain supported throughout retention periods.

26. Operational Requirements

Operational controls shall include:

- Monitoring
- Logging
- Health checks
- Error reporting

Operational telemetry shall be collected.

27. Metrics

The following metrics shall be monitored.

MET-OPP-API-001

Opportunity creation volume.

MET-OPP-API-002

Opportunity retrieval volume.

MET-OPP-API-003

Validation success rate.

MET-OPP-API-004

Status transition volume.

MET-OPP-API-005

Governance validation success rate.

MET-OPP-API-006

Audit coverage.

MET-OPP-API-007

Replay request volume.

MET-OPP-API-008

API response latency.

28. Audit Controls

API audits shall verify:

- Access control compliance
- Lifecycle integrity
- Governance enforcement
- Audit completeness
- Replay support

Findings shall be tracked through remediation processes.

29. Compliance Requirements

Opportunity APIs shall support:

- SOC 2 controls
- ISO 27001 controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Opportunity Module APIs, including opportunity creation, retrieval, update, lifecycle management, ownership management, classification management, governance integration, audit support, and replay capabilities. These APIs establish Opportunity as the controlled entry point into the TODP decision lifecycle and ensure that all opportunity interactions remain secure, governed, traceable, auditable, and replayable.

API SPEC V1 – Enterprise API Specification

Section 5: Authority Module API Specification

Document ID: API-V1-S05

Document Title: Authority Module API Specification

Version: 1.0

Status: Approved

Owner: Authority Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Authority Module within the Transparent Opportunity Decision Platform (TODP).

The Authority APIs provide controlled interfaces for authority evaluation, role management, permission management, authorization decisions, delegation management, governance validation, audit traceability, and replay support.

These APIs establish Authority as the centralized authorization and decision-entitlement layer for all TODP modules.

2. Objectives

OBJ-API-AUTH-001

Provide centralized authorization services.

OBJ-API-AUTH-002

Support role and permission management.

OBJ-API-AUTH-003

Enable governance-controlled access decisions.

OBJ-API-AUTH-004

Maintain authorization traceability.

OBJ-API-AUTH-005

Support audit and compliance requirements.

OBJ-API-AUTH-006

Enable delegation management.

OBJ-API-AUTH-007

Support replay reconstruction.

OBJ-API-AUTH-008

Provide enterprise-scale authorization services.

3. Scope

The Authority API domain governs:

- Authorization decisions
- Role management
- Permission management
- Delegation management
- Authority validation
- Access reviews
- Governance validation
- Audit support
- Replay support

4. API Domain Information

Domain Name

Authority

Domain Identifier

AUTH

Primary Resource

Authority Decision

Primary Identifier

Authority ID

Ownership

Authority Domain Team

Related Modules

Opportunity

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

5. Authority Resource Model

Primary Attributes

- Authority ID
- Subject ID
- Role ID
- Permission Set
- Authority Scope
- Decision Outcome
- Effective Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Review History

6. API-AUTH-001

Validate Authority

Endpoint

POST /api/v1/authority/validate

Purpose

Evaluate whether a subject possesses required authority.

Required Inputs

- Subject ID
- Requested Action
- Resource Context

Outputs

- Authorization Decision
- Decision Rationale
- Policy References

Success Response

200 OK

Authorization Requirement

System-Level Authorization

Audit Requirement

Mandatory

7. API-AUTH-002

Get Authority Decision

Endpoint

GET /api/v1/authority/{authorityId}

Purpose

Retrieve authorization decision details.

Required Inputs

- Authority ID

Outputs

- Decision Details
- Policy References
- Audit References

Success Response

200 OK

8. API-AUTH-003

Create Role

Endpoint

POST /api/v1/authority/roles

Purpose

Create a new role definition.

Required Inputs

- Role Name
- Description
- Permission Set

Governance Validation

Required

Success Response

201 Created

9. API-AUTH-004

Update Role

Endpoint

PUT /api/v1/authority/roles/{roleId}

Purpose

Modify role definitions.

Governed Elements

- Permissions
- Scope
- Restrictions

Governance Validation

Mandatory

Audit Requirement

Mandatory

10. API-AUTH-005

Get Role

Endpoint

GET /api/v1/authority/roles/{roleId}

Purpose

Retrieve role details.

Outputs

- Role Definition
- Assigned Permissions
- Governance Status

11. API-AUTH-006

Assign Role

Endpoint

POST /api/v1/authority/roles/{roleId}/assign

Purpose

Assign role to subject.

Required Inputs

- Subject ID
- Role ID

Governance Validation

Required

Audit Requirement

Mandatory

12. API-AUTH-007

Remove Role

Endpoint

POST /api/v1/authority/roles/{roleId}/remove

Purpose

Remove role assignment.

Required Inputs

- Subject ID
- Role ID

Audit Requirement

Mandatory

Governance Validation

Required

13. API-AUTH-008

Create Delegation

Endpoint

POST /api/v1/authority/delegations

Purpose

Create authority delegation.

Required Inputs

- Delegator
- Delegate
- Delegation Scope
- Expiration Date

Governance Validation

Mandatory

Outputs

- Delegation ID

14. API-AUTH-009

Revoke Delegation

Endpoint

POST /api/v1/authority/delegations/{delegationId}/revoke

Purpose

Revoke delegated authority.

Governance Validation

Required

Audit Requirement

Mandatory

15. API-AUTH-010

Get Delegation

Endpoint

GET /api/v1/authority/delegations/{delegationId}

Purpose

Retrieve delegation information.

Outputs

- Delegation Details
- Delegation History
- Governance Status

16. API-AUTH-011

Get Permission Inventory

Endpoint

GET /api/v1/authority/permissions

Purpose

Retrieve permission catalog.

Outputs

- Permission Definitions
- Scope Definitions
- Governance References

17. API-AUTH-012

Review Authority Access

Endpoint

POST /api/v1/authority/review

Purpose

Perform governance access review.

Inputs

- Subject ID
- Review Scope

Outputs

- Review Results
- Governance Findings

18. API-AUTH-013

Get Authority History

Endpoint

GET /api/v1/authority/{authorityId}/history

Purpose

Retrieve historical authorization information.

Outputs

- Historical Decisions
- Role Changes
- Delegation Changes

Replay Support

Enabled

19. API-AUTH-014

Get Authority Audit Trail

Endpoint

GET /api/v1/authority/{authorityId}/audit

Purpose

Retrieve audit history.

Outputs

- Audit Records
- Access Reviews
- Governance Actions

Authorization Requirement

Audit Access Permission

20. API-AUTH-015

Replay Authority Evaluation

Endpoint

POST /api/v1/authority/replay

Purpose

Reconstruct historical authorization decisions.

Required Inputs

- Historical Timestamp
- Subject ID
- Requested Action

Outputs

- Historical Decision
- Policy Versions
- Governance References

Replay Authorization

Required

21. Request Validation Requirements

Authority APIs shall validate:

- Subject identifiers
- Role assignments

- Permission references
- Delegation validity
- Governance constraints

Validation failures shall generate standardized error responses.

22. Security Requirements

Authority APIs shall enforce:

- Strong authentication
- Authorization validation
- Separation-of-duties controls
- Governance approval controls
- Audit logging

Security controls align with SEC V1.

23. Traceability Requirements

All Authority API interactions shall generate:

- Request ID
- Correlation ID
- Authority ID
- Subject ID
- Policy Reference
- Timestamp

Traceability records shall support investigations and replay.

24. Governance Requirements

Governance validation shall be enforced for:

- Role creation
- Role modification
- Permission changes
- Delegation creation
- Delegation revocation
- Access reviews

Governance exceptions require approval.

25. Audit Requirements

Audit records shall be generated for:

- Authorization decisions
- Role assignments

- Role removals
- Delegations
- Access reviews
- Governance actions
- Replay requests

Audit records shall be immutable.

26. Replay Requirements

Replay support shall preserve:

- Historical roles
- Historical permissions
- Historical delegations
- Historical policy versions
- Historical authority decisions

Replay reconstruction shall remain supported throughout retention periods.

27. Operational Requirements

Operational controls shall include:

- Monitoring
- Logging
- Health checks
- Security monitoring
- Performance monitoring

Operational telemetry shall be collected continuously.

28. Metrics

The following metrics shall be monitored.

MET-AUTH-API-001

Authorization request volume.

MET-AUTH-API-002

Authorization success rate.

MET-AUTH-API-003

Authorization latency.

MET-AUTH-API-004

Role assignment volume.

MET-AUTH-API-005

Delegation volume.

MET-AUTH-API-006

Governance review completion rate.

MET-AUTH-API-007

Audit coverage.

MET-AUTH-API-008

Replay request volume.

29. Compliance Requirements

Authority APIs shall support:

- SOC 2 access control requirements
- ISO 27001 access management controls
- ISO 27001 segregation-of-duty controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Authority Module APIs, including authorization evaluation, role management, permission management, delegation management, governance access reviews, audit support, and replay capabilities. These APIs establish Authority as the centralized authorization layer of TODP and ensure that all access decisions remain governed, traceable, auditable, secure, explainable, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 6: Evidence Module API Specification

Document ID: API-V1-S06

Document Title: Evidence Module API Specification

Version: 1.0

Status: Approved

Owner: Evidence Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Evidence Module within the Transparent Opportunity Decision Platform (TODP).

The Evidence APIs provide controlled interfaces for evidence ingestion, evidence registration, evidence validation, evidence classification, evidence lifecycle management, evidence lineage tracking, evidence access governance, audit traceability, and replay support.

These APIs establish Evidence as the authoritative source of factual information used throughout the TODP decision lifecycle.

2. Objectives

OBJ-API-EVD-001

Provide controlled evidence management.

OBJ-API-EVD-002

Support evidence lifecycle governance.

OBJ-API-EVD-003

Enable evidence validation and classification.

OBJ-API-EVD-004

Maintain evidence lineage.

OBJ-API-EVD-005

Support governance-controlled evidence access.

OBJ-API-EVD-006

Enable audit and compliance requirements.

OBJ-API-EVD-007

Support replay reconstruction.

OBJ-API-EVD-008

Provide enterprise-scale evidence services.

3. Scope

The Evidence API domain governs:

- Evidence registration
- Evidence ingestion
- Evidence retrieval
- Evidence validation
- Evidence classification
- Evidence lifecycle management
- Evidence lineage management
- Evidence audit support
- Evidence replay support

4. API Domain Information

Domain Name

Evidence

Domain Identifier

EVD

Primary Resource

Evidence

Primary Identifier

Evidence ID

Ownership

Evidence Domain Team

Related Modules

Opportunity

Authority

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

5. Evidence Resource Model

Primary Attributes

- Evidence ID
- Evidence Name
- Evidence Type
- Evidence Source
- Evidence Status
- Evidence Classification
- Evidence Owner
- Registration Timestamp

Governance Attributes

- Governance Status
- Retention Classification
- Policy Reference

Audit Attributes

- Audit Reference
- Lineage Reference

6. API-EVD-001

Register Evidence

Endpoint

POST /api/v1/evidence

Purpose

Register new evidence.

Required Inputs

- Evidence Name
- Evidence Type
- Evidence Source

Optional Inputs

- Metadata
- Classification
- Ownership Information

Success Response

201 Created

Generated Outputs

- Evidence ID
- Registration Status
- Audit Reference

Authorization Requirement

Evidence Create Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-EVD-002

Get Evidence

Endpoint

GET /api/v1/evidence/{evidenceId}

Purpose

Retrieve evidence information.

Required Inputs

- Evidence ID

Outputs

- Evidence Details
- Classification Information
- Governance Information

Success Response

200 OK

8. API-EVD-003

Update Evidence

Endpoint

PUT /api/v1/evidence/{evidenceId}

Purpose

Modify approved evidence attributes.

Governance Validation

Required

Audit Requirement

Mandatory

Replay Requirement

Mandatory

Success Response

200 OK

9. API-EVD-004

Search Evidence

Endpoint

GET /api/v1/evidence

Purpose

Search evidence inventory.

Supported Filters

- Evidence Type
- Classification
- Status
- Source
- Owner
- Date Range

Supported Sorting

- Registration Date
- Last Modified Date
- Classification

Success Response

200 OK

10. API-EVD-005

Validate Evidence

Endpoint

POST /api/v1/evidence/{evidenceId}/validate

Purpose

Execute evidence validation rules.

Validation Categories

- Schema Validation
- Source Validation
- Metadata Validation
- Governance Validation

Outputs

- Validation Results
- Validation Status

Success Response

200 OK

11. API-EVD-006

Classify Evidence

Endpoint

POST /api/v1/evidence/{evidenceId}/classification

Purpose

Manage evidence classification.

Classification Categories

- Business Classification
- Security Classification
- Governance Classification
- Retention Classification

Governance Validation

Required

Audit Requirement

Mandatory

12. API-EVD-007

Transition Evidence Lifecycle

Endpoint

POST /api/v1/evidence/{evidenceId}/status

Purpose

Transition evidence lifecycle state.

Supported States

- Registered
- Validated
- Active
- Restricted
- Retired
- Archived
- Disposed

Governance Validation

Required

Audit Requirement

Mandatory

13. API-EVD-008

Assign Evidence Owner

Endpoint

POST /api/v1/evidence/{evidenceId}/owner

Purpose

Assign or transfer evidence ownership.

Required Inputs

- Evidence ID
- New Owner

Governance Validation

Required

Audit Requirement

Mandatory

14. API-EVD-009

Get Evidence Lineage

Endpoint

GET /api/v1/evidence/{evidenceId}/lineage

Purpose

Retrieve evidence lineage information.

Outputs

- Source Information
- Transformation History
- Usage History
- Retention History

Replay Support

Enabled

15. API-EVD-010

Get Evidence Versions

Endpoint

GET /api/v1/evidence/{evidenceId}/versions

Purpose

Retrieve historical evidence versions.

Outputs

- Version History
- Change History
- Historical Metadata

Replay Support

Enabled

16. API-EVD-011

Get Evidence Audit Trail

Endpoint

GET /api/v1/evidence/{evidenceId}/audit

Purpose

Retrieve audit history.

Outputs

- Audit Records
- Governance Actions
- Access History

Authorization Requirement

Audit Access Permission

17. API-EVD-012

Get Evidence Governance Status

Endpoint

GET /api/v1/evidence/{evidenceId}/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Retention Status

18. API-EVD-013

Record Evidence Usage

Endpoint

POST /api/v1/evidence/{evidenceId}/usage

Purpose

Record evidence consumption activity.

Usage Categories

- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

Audit Requirement

Mandatory

19. API-EVD-014

Replay Evidence State

Endpoint

POST /api/v1/evidence/{evidenceId}/replay

Purpose

Reconstruct historical evidence state.

Required Inputs

- Evidence ID
- Historical Timestamp

Outputs

- Historical State
- Historical Metadata
- Historical Classification

Replay Authorization

Required

20. API-EVD-015

Export Evidence Metadata

Endpoint

GET /api/v1/evidence/{evidenceId}/metadata

Purpose

Retrieve evidence metadata profile.

Outputs

- Metadata Attributes
- Governance Metadata
- Lineage References

Authorization Requirement

Evidence Read Permission

21. Request Validation Requirements

Evidence APIs shall validate:

- Evidence identifiers
- Source references
- Ownership assignments
- Classification values
- Governance constraints
- Lifecycle transitions

Validation failures shall generate standardized responses.

22. Security Requirements

Evidence APIs shall enforce:

- Authentication
- Authorization
- Input validation

- Governance validation
- Audit logging
- Data access controls

Security requirements align with SEC V1.

23. Traceability Requirements

All Evidence API interactions shall generate:

- Request ID
- Correlation ID
- Evidence ID
- User ID
- Source Reference
- Timestamp

Traceability records shall support investigations and replay.

24. Governance Requirements

Governance validation shall be enforced for:

- Classification changes
- Lifecycle transitions
- Ownership transfers
- Retention changes
- Evidence archival
- Evidence disposal

Governance exceptions require approval.

25. Audit Requirements

Audit records shall be generated for:

- Registration
- Modification
- Classification changes
- Lifecycle transitions
- Ownership transfers
- Evidence usage
- Governance actions
- Replay requests

Audit records shall be immutable.

26. Replay Requirements

Replay support shall preserve:

- Historical evidence versions
- Historical classifications
- Historical ownership
- Historical governance status
- Historical lineage information

Replay reconstruction shall remain supported throughout retention periods.

27. Operational Requirements

Operational controls shall include:

- Monitoring
- Logging
- Health checks
- Usage tracking
- Performance monitoring

Operational telemetry shall be collected continuously.

28. Metrics

The following metrics shall be monitored.

MET-EVD-API-001

Evidence registration volume.

MET-EVD-API-002

Evidence retrieval volume.

MET-EVD-API-003

Validation success rate.

MET-EVD-API-004

Classification completeness.

MET-EVD-API-005

Lineage coverage.

MET-EVD-API-006

Governance compliance rate.

MET-EVD-API-007

Replay request volume.

MET-EVD-API-008

API response latency.

29. Compliance Requirements

Evidence APIs shall support:

- SOC 2 information management controls
- ISO 27001 asset management controls
- ISO 27001 information classification controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Evidence Module APIs, including evidence registration, retrieval, validation, classification, lifecycle management, lineage tracking, governance integration, audit support, usage tracking, and replay capabilities. These APIs establish Evidence as the authoritative factual foundation of TODP and ensure that all evidence interactions remain governed, traceable, auditable, secure, explainable, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 7: Retrieval Module API Specification

Document ID: API-V1-S07

Document Title: Retrieval Module API Specification

Version: 1.0

Status: Approved

Owner: Retrieval Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Retrieval Module within the Transparent Opportunity Decision Platform (TODP).

The Retrieval APIs provide controlled interfaces for evidence discovery, retrieval request execution, retrieval strategy selection, relevance filtering, governance-aware evidence selection, retrieval traceability, audit support, and replay reconstruction.

These APIs establish Retrieval as the governed evidence access layer connecting Evidence with Ranking and Recommendation workflows.

2. Objectives

OBJ-API-RET-001

Provide controlled evidence retrieval services.

OBJ-API-RET-002

Support multiple retrieval strategies.

OBJ-API-RET-003

Enable governance-aware evidence access.

OBJ-API-RET-004

Maintain retrieval traceability.

OBJ-API-RET-005

Support audit and compliance requirements.

OBJ-API-RET-006

Provide explainable retrieval outcomes.

OBJ-API-RET-007

Support replay reconstruction.

OBJ-API-RET-008

Provide enterprise-scale retrieval services.

3. Scope

The Retrieval API domain governs:

- Retrieval request creation
- Evidence discovery
- Retrieval execution
- Retrieval filtering
- Retrieval result management
- Retrieval explainability
- Retrieval audit support
- Retrieval replay support

4. API Domain Information

Domain Name

Retrieval

Domain Identifier

RET

Primary Resource

Retrieval Request

Primary Identifier

Retrieval ID

Ownership

Retrieval Domain Team

Related Modules

Authority

Evidence

Ranking

Recommendation

Governance

Audit

Replay

5. Retrieval Resource Model

Primary Attributes

- Retrieval ID
- Opportunity ID
- Retrieval Strategy
- Retrieval Criteria
- Retrieval Status
- Retrieved Evidence Count
- Execution Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Traceability Reference

6. API-RET-001

Create Retrieval Request

Endpoint

POST /api/v1/retrieval

Purpose

Create a retrieval request.

Required Inputs

- Opportunity ID
- Retrieval Criteria
- Retrieval Strategy

Optional Inputs

- Filters
- Governance Context
- Metadata

Success Response

201 Created

Generated Outputs

- Retrieval ID
- Request Status
- Audit Reference

Authorization Requirement

Retrieval Execute Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-RET-002

Execute Retrieval

Endpoint

POST /api/v1/retrieval/{retrievalId}/execute

Purpose

Execute evidence retrieval.

Required Inputs

- Retrieval ID

Outputs

- Retrieved Evidence Set
- Retrieval Summary
- Strategy Information

Success Response

200 OK

Governance Validation

Required

8. API-RET-003

Get Retrieval Request

Endpoint

GET /api/v1/retrieval/{retrievalId}

Purpose

Retrieve retrieval request information.

Outputs

- Retrieval Configuration
- Request Status
- Execution Metadata

Success Response

200 OK

9. API-RET-004

Search Retrieval Requests

Endpoint

GET /api/v1/retrieval

Purpose

Search retrieval history.

Supported Filters

- Opportunity ID
- Retrieval Strategy
- Status
- Date Range

Supported Sorting

- Execution Date
- Request Date
- Status

Success Response

200 OK

10. API-RET-005

Get Retrieval Results

Endpoint

GET /api/v1/retrieval/{retrievalId}/results

Purpose

Retrieve evidence returned from retrieval execution.

Outputs

- Evidence References
- Retrieval Scores
- Matching Information

Success Response

200 OK

11. API-RET-006

Apply Retrieval Filters

Endpoint

POST /api/v1/retrieval/{retrievalId}/filters

Purpose

Apply additional retrieval filters.

Supported Filters

- Classification Filters
- Governance Filters
- Source Filters
- Trust-Level Filters

Governance Validation

Required

12. API-RET-007

Get Retrieval Explainability

Endpoint

GET /api/v1/retrieval/{retrievalId}/explainability

Purpose

Retrieve retrieval rationale.

Outputs

- Strategy Used
- Matching Criteria
- Applied Filters
- Governance Constraints

Success Response

200 OK

13. API-RET-008

Validate Retrieval Request

Endpoint

POST /api/v1/retrieval/{retrievalId}/validate

Purpose

Validate retrieval request configuration.

Validation Categories

- Strategy Validation
- Governance Validation
- Authority Validation

Outputs

- Validation Results

14. API-RET-009

Get Retrieval History

Endpoint

GET /api/v1/retrieval/{retrievalId}/history

Purpose

Retrieve historical retrieval activity.

Outputs

- Execution History
- Filter Changes
- Governance Actions

Replay Support

Enabled

15. API-RET-010

Get Retrieval Audit Trail

Endpoint

GET /api/v1/retrieval/{retrievalId}/audit

Purpose

Retrieve retrieval audit history.

Outputs

- Audit Records
- Governance Actions
- Access History

Authorization Requirement

Audit Access Permission

16. API-RET-011

Get Retrieval Governance Status

Endpoint

GET /api/v1/retrieval/{retrievalId}/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Governance Decisions

17. API-RET-012

Export Retrieval Results

Endpoint

GET /api/v1/retrieval/{retrievalId}/export

Purpose

Export retrieval results.

Supported Formats

- JSON
- CSV
- Governance Export Package

Governance Validation

Required

18. API-RET-013

Record Retrieval Usage

Endpoint

POST /api/v1/retrieval/{retrievalId}/usage

Purpose

Record downstream consumption activity.

Usage Categories

- Ranking
- Recommendation
- Governance
- Audit
- Replay

Audit Requirement

Mandatory

19. API-RET-014

Replay Retrieval Execution

Endpoint

POST /api/v1/retrieval/{retrievalId}/replay

Purpose

Reconstruct historical retrieval execution.

Required Inputs

- Retrieval ID
- Historical Timestamp

Outputs

- Historical Results
- Historical Filters
- Historical Governance Context

Replay Authorization

Required

20. API-RET-015

Compare Retrieval Executions

Endpoint

POST /api/v1/retrieval/{retrievalId}/compare

Purpose

Compare retrieval executions.

Comparison Areas

- Evidence Differences
- Filter Differences
- Governance Differences
- Strategy Differences

Outputs

- Comparison Report

21. Request Validation Requirements

Retrieval APIs shall validate:

- Opportunity references
- Retrieval strategies
- Filter configurations
- Authority permissions
- Governance constraints

Validation failures shall generate standardized responses.

22. Security Requirements

Retrieval APIs shall enforce:

- Authentication
- Authorization
- Governance validation
- Input validation
- Audit logging
- Evidence access controls

Security requirements align with SEC V1.

23. Traceability Requirements

All Retrieval API interactions shall generate:

- Request ID
- Correlation ID
- Retrieval ID
- Opportunity ID
- User ID
- Timestamp

Traceability records shall support investigations and replay.

24. Governance Requirements

Governance validation shall be enforced for:

- Retrieval execution
- Filter application
- Result exports
- Restricted evidence access
- Replay requests

Governance exceptions require approval.

25. Audit Requirements

Audit records shall be generated for:

- Retrieval requests
- Retrieval executions
- Filter changes
- Governance actions
- Result exports
- Usage tracking
- Replay requests

Audit records shall be immutable.

26. Replay Requirements

Replay support shall preserve:

- Retrieval criteria
- Retrieval strategies
- Applied filters
- Governance constraints
- Historical evidence references

Replay reconstruction shall remain supported throughout retention periods.

27. Operational Requirements

Operational controls shall include:

- Monitoring
- Logging
- Health checks
- Usage tracking
- Performance monitoring

Operational telemetry shall be collected continuously.

28. Metrics

The following metrics shall be monitored.

MET-RET-API-001

Retrieval request volume.

MET-RET-API-002

Retrieval execution volume.

MET-RET-API-003

Retrieval success rate.

MET-RET-API-004

Retrieval latency.

MET-RET-API-005

Governance validation success rate.

MET-RET-API-006

Retrieval explainability coverage.

MET-RET-API-007

Replay request volume.

MET-RET-API-008

Evidence retrieval coverage.

29. Compliance Requirements

Retrieval APIs shall support:

- SOC 2 access governance controls
- ISO 27001 access control requirements
- ISO 27001 information retrieval controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Retrieval Module APIs, including retrieval request management, retrieval execution, filtering, explainability, governance integration, audit support, result export, usage tracking, comparison capabilities, and replay functionality. These APIs establish Retrieval as the governed evidence access layer of TODP and ensure that evidence discovery and selection remain secure, traceable, auditable, explainable, governed, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 8: Ranking Module API Specification

Document ID: API-V1-S08

Document Title: Ranking Module API Specification

Version: 1.0

Status: Approved

Owner: Ranking Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Ranking Module within the Transparent Opportunity Decision Platform (TODP).

The Ranking APIs provide controlled interfaces for ranking execution, ranking configuration management, score generation, prioritization processing, explainability retrieval, governance validation, audit traceability, and replay support.

These APIs establish Ranking as the transparent evaluation layer that transforms retrieved evidence into prioritized and explainable evaluation outcomes.

2. Objectives

OBJ-API-RNK-001

Provide controlled ranking services.

OBJ-API-RNK-002

Support explainable ranking outcomes.

OBJ-API-RNK-003

Enable evidence-based prioritization.

OBJ-API-RNK-004

Support governance-controlled ranking operations.

OBJ-API-RNK-005

Maintain ranking traceability.

OBJ-API-RNK-006

Support audit and compliance requirements.

OBJ-API-RNK-007

Enable replay reconstruction.

OBJ-API-RNK-008

Provide enterprise-scale ranking services.

3. Scope

The Ranking API domain governs:

- Ranking request creation
- Ranking execution
- Ranking configuration management
- Ranking result retrieval
- Ranking explainability

- Ranking governance validation
- Ranking audit support
- Ranking replay support

4. API Domain Information

Domain Name

Ranking

Domain Identifier

RNK

Primary Resource

Ranking Evaluation

Primary Identifier

Ranking ID

Ownership

Ranking Domain Team

Related Modules

Retrieval

Recommendation

Governance

Audit

Replay

5. Ranking Resource Model

Primary Attributes

- Ranking ID
- Opportunity ID
- Ranking Strategy
- Ranking Version
- Ranking Status
- Score Output
- Execution Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Traceability Reference

6. API-RNK-001

Create Ranking Request

Endpoint

POST /api/v1/rankings

Purpose

Create a ranking evaluation request.

Required Inputs

- Opportunity ID
- Retrieval ID
- Ranking Strategy

Optional Inputs

- Ranking Parameters
- Governance Context
- Metadata

Success Response

201 Created

Generated Outputs

- Ranking ID
- Ranking Status
- Audit Reference

Authorization Requirement

Ranking Execute Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-RNK-002

Execute Ranking

Endpoint

POST /api/v1/rankings/{rankingId}/execute

Purpose

Execute ranking evaluation.

Required Inputs

- Ranking ID

Outputs

- Ranking Results
- Score Distribution
- Evaluation Summary

Success Response

200 OK

Governance Validation

Required

8. API-RNK-003

Get Ranking Evaluation

Endpoint

GET /api/v1/rankings/{rankingId}

Purpose

Retrieve ranking evaluation details.

Outputs

- Evaluation Configuration
- Status
- Version Information

Success Response

200 OK

9. API-RNK-004

Search Rankings

Endpoint

GET /api/v1/rankings

Purpose

Search ranking evaluations.

Supported Filters

- Opportunity ID
- Strategy
- Status
- Date Range

Supported Sorting

- Execution Date
- Creation Date
- Status

Success Response

200 OK

10. API-RNK-005

Get Ranking Results

Endpoint

GET /api/v1/rankings/{rankingId}/results

Purpose

Retrieve ranking output.

Outputs

- Ranked Results
- Scores
- Evidence References

Success Response

200 OK

11. API-RNK-006

Get Ranking Explainability

Endpoint

GET /api/v1/rankings/{rankingId}/explainability

Purpose

Retrieve ranking rationale.

Outputs

- Scoring Factors
- Weighting Information
- Evidence References
- Ranking Rationale

Success Response

200 OK

12. API-RNK-007

Validate Ranking Configuration

Endpoint

POST /api/v1/rankings/{rankingId}/validate

Purpose

Validate ranking configuration.

Validation Categories

- Strategy Validation
- Governance Validation
- Configuration Validation

Outputs

- Validation Results

13. API-RNK-008

Manage Ranking Configuration

Endpoint

POST /api/v1/rankings/configurations

Purpose

Create or update ranking configurations.

Governed Elements

- Scoring Rules
- Weights
- Thresholds
- Prioritization Logic

Governance Validation

Mandatory

Audit Requirement

Mandatory

14. API-RNK-009

Get Ranking Configuration

Endpoint

GET /api/v1/rankings/configurations/{configurationId}

Purpose

Retrieve ranking configuration.

Outputs

- Configuration Details
- Version Information
- Governance Status

15. API-RNK-010

Get Ranking History

Endpoint

GET /api/v1/rankings/{rankingId}/history

Purpose

Retrieve ranking history.

Outputs

- Execution History
- Configuration Changes
- Governance Actions

Replay Support

Enabled

16. API-RNK-011

Get Ranking Audit Trail

Endpoint

GET /api/v1/rankings/{rankingId}/audit

Purpose

Retrieve ranking audit history.

Outputs

- Audit Records
- Governance Actions
- Execution History

Authorization Requirement

Audit Access Permission

17. API-RNK-012

Get Ranking Governance Status

Endpoint

GET /api/v1/rankings/{rankingId}/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Governance Decisions

18. API-RNK-013

Export Ranking Results

Endpoint

GET /api/v1/rankings/{rankingId}/export

Purpose

Export ranking output.

Supported Formats

- JSON
- CSV
- Governance Export Package

Governance Validation

Required

19. API-RNK-014

Replay Ranking Execution

Endpoint

POST /api/v1/rankings/{rankingId}/replay

Purpose

Reconstruct historical ranking execution.

Required Inputs

- Ranking ID
- Historical Timestamp

Outputs

- Historical Results
- Historical Configuration
- Historical Governance Context

Replay Authorization

Required

20. API-RNK-015

Compare Ranking Executions

Endpoint

POST /api/v1/rankings/{rankingId}/compare

Purpose

Compare ranking executions.

Comparison Areas

- Score Differences
- Evidence Differences
- Configuration Differences
- Governance Differences

Outputs

- Comparison Report

21. Request Validation Requirements

Ranking APIs shall validate:

- Opportunity references
- Retrieval references
- Ranking strategies
- Configuration references
- Governance constraints

Validation failures shall generate standardized responses.

22. Security Requirements

Ranking APIs shall enforce:

- Authentication
- Authorization
- Governance validation

- Input validation
- Audit logging
- Configuration controls

Security requirements align with SEC V1.

23. Traceability Requirements

All Ranking API interactions shall generate:

- Request ID
- Correlation ID
- Ranking ID
- Opportunity ID
- User ID
- Timestamp

Traceability records shall support investigations and replay.

24. Governance Requirements

Governance validation shall be enforced for:

- Ranking execution
- Configuration changes
- Result exports
- Threshold modifications
- Replay requests

Governance exceptions require approval.

25. Audit Requirements

Audit records shall be generated for:

- Ranking requests
- Ranking executions
- Configuration changes
- Governance actions
- Result exports
- Comparison requests
- Replay requests

Audit records shall be immutable.

26. Replay Requirements

Replay support shall preserve:

- Ranking configurations
- Ranking strategies
- Scoring rules
- Governance constraints
- Historical evidence references

Replay reconstruction shall remain supported throughout retention periods.

27. Operational Requirements

Operational controls shall include:

- Monitoring
- Logging
- Health checks
- Usage tracking
- Performance monitoring

Operational telemetry shall be collected continuously.

28. Metrics

The following metrics shall be monitored.

MET-RNK-API-001

Ranking execution volume.

MET-RNK-API-002

Ranking success rate.

MET-RNK-API-003

Ranking latency.

MET-RNK-API-004

Explainability coverage.

MET-RNK-API-005

Governance validation success rate.

MET-RNK-API-006

Configuration change volume.

MET-RNK-API-007

Replay request volume.

MET-RNK-API-008

Ranking audit coverage.

29. Compliance Requirements

Ranking APIs shall support:

- SOC 2 governance controls
- ISO 27001 change management controls
- ISO 27001 integrity controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Ranking Module APIs, including ranking execution, configuration management, score generation, explainability services, governance integration, audit support, comparison capabilities, export functionality, and replay operations. These APIs establish Ranking as the transparent evaluation layer of TODP and ensure that prioritization outcomes remain governed, traceable, auditable, explainable, secure, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 9: Recommendation Module API Specification

Document ID: API-V1-S09

Document Title: Recommendation Module API Specification

Version: 1.0

Status: Approved

Owner: Recommendation Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Recommendation Module within the Transparent Opportunity Decision Platform (TODP).

The Recommendation APIs provide controlled interfaces for recommendation generation, recommendation retrieval, recommendation explainability, confidence evaluation,

recommendation governance validation, recommendation comparison, audit traceability, and replay support.

These APIs establish Recommendation as the explainable decision-support layer that transforms ranking outcomes into transparent and actionable recommendations for governance review and human decision-making.

2. Objectives

OBJ-API-REC-001

Provide controlled recommendation services.

OBJ-API-REC-002

Support explainable recommendations.

OBJ-API-REC-003

Enable evidence-based decision support.

OBJ-API-REC-004

Provide recommendation confidence indicators.

OBJ-API-REC-005

Support governance-controlled recommendation workflows.

OBJ-API-REC-006

Maintain recommendation traceability.

OBJ-API-REC-007

Support replay reconstruction.

OBJ-API-REC-008

Provide enterprise-scale recommendation services.

3. Scope

The Recommendation API domain governs:

- Recommendation request creation
- Recommendation generation
- Recommendation retrieval
- Recommendation explainability
- Recommendation confidence assessment

- Recommendation governance validation
- Recommendation audit support
- Recommendation replay support

4. API Domain Information

Domain Name

Recommendation

Domain Identifier

REC

Primary Resource

Recommendation

Primary Identifier

Recommendation ID

Ownership

Recommendation Domain Team

Related Modules

Ranking

Governance

Audit

Replay

5. Recommendation Resource Model

Primary Attributes

- Recommendation ID
- Opportunity ID
- Ranking ID
- Recommendation Strategy
- Recommendation Version
- Recommendation Outcome
- Recommendation Confidence
- Recommendation Status
- Execution Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Traceability Reference

6. API-REC-001

Create Recommendation Request

Endpoint

POST /api/v1/recommendations

Purpose

Create a recommendation generation request.

Required Inputs

- Opportunity ID
- Ranking ID
- Recommendation Strategy

Optional Inputs

- Recommendation Parameters
- Governance Context
- Metadata

Success Response

201 Created

Generated Outputs

- Recommendation ID
- Recommendation Status
- Audit Reference

Authorization Requirement

Recommendation Execute Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-REC-002

Generate Recommendation

Endpoint

POST /api/v1/recommendations/{recommendationId}/execute

Purpose

Generate recommendation output.

Required Inputs

- Recommendation ID

Outputs

- Recommendation Outcome
- Recommendation Summary
- Confidence Assessment

Success Response

200 OK

Governance Validation

Required

8. API-REC-003

Get Recommendation

Endpoint

GET /api/v1/recommendations/{recommendationId}

Purpose

Retrieve recommendation details.

Outputs

- Recommendation Configuration
- Recommendation Status
- Version Information

Success Response

200 OK

9. API-REC-004

Search Recommendations

Endpoint

GET /api/v1/recommendations

Purpose

Search recommendation inventory.

Supported Filters

- Opportunity ID
- Status
- Strategy
- Confidence Level
- Date Range

Supported Sorting

- Creation Date
- Execution Date
- Confidence Score

Success Response

200 OK

10. API-REC-005

Get Recommendation Results

Endpoint

GET /api/v1/recommendations/{recommendationId}/results

Purpose

Retrieve recommendation outcome.

Outputs

- Recommendation Outcome
- Supporting Evidence References
- Ranking References
- Confidence Indicators

Success Response

200 OK

11. API-REC-006

Get Recommendation Explainability

Endpoint

GET /api/v1/recommendations/{recommendationId}/explainability

Purpose

Retrieve recommendation rationale.

Outputs

- Recommendation Rationale
- Supporting Evidence
- Ranking Factors
- Governance Considerations

Success Response

200 OK

12. API-REC-007

Get Recommendation Confidence

Endpoint

GET /api/v1/recommendations/{recommendationId}/confidence

Purpose

Retrieve confidence evaluation details.

Outputs

- Confidence Score
- Confidence Factors

- Evidence Quality Indicators
- Data Freshness Indicators

Success Response

200 OK

13. API-REC-008

Validate Recommendation

Endpoint

POST /api/v1/recommendations/{recommendationId}/validate

Purpose

Validate recommendation configuration and outcome.

Validation Categories

- Strategy Validation
- Governance Validation
- Explainability Validation
- Confidence Validation

Outputs

- Validation Results

14. API-REC-009

Manage Recommendation Configuration

Endpoint

POST /api/v1/recommendations/configurations

Purpose

Create or update recommendation configurations.

Governed Elements

- Recommendation Logic
- Thresholds
- Confidence Rules
- Outcome Categories

Governance Validation

Mandatory

Audit Requirement

Mandatory

15. API-REC-010

Get Recommendation Configuration

Endpoint

GET /api/v1/recommendations/configurations/{configurationId}

Purpose

Retrieve recommendation configuration.

Outputs

- Configuration Details
- Version Information
- Governance Status

16. API-REC-011

Get Recommendation History

Endpoint

GET /api/v1/recommendations/{recommendationId}/history

Purpose

Retrieve recommendation history.

Outputs

- Execution History
- Configuration Changes
- Governance Actions

Replay Support

Enabled

17. API-REC-012

Get Recommendation Audit Trail

Endpoint

GET /api/v1/recommendations/{recommendationId}/audit

Purpose

Retrieve recommendation audit history.

Outputs

- Audit Records
- Governance Actions
- Review History

Authorization Requirement

Audit Access Permission

18. API-REC-013

Get Recommendation Governance Status

Endpoint

GET /api/v1/recommendations/{recommendationId}/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Governance Decisions

19. API-REC-014

Export Recommendation Results

Endpoint

GET /api/v1/recommendations/{recommendationId}/export

Purpose

Export recommendation output.

Supported Formats

- JSON
- CSV
- Governance Export Package

Governance Validation

Required

20. API-REC-015

Replay Recommendation Execution

Endpoint

POST /api/v1/recommendations/{recommendationId}/replay

Purpose

Reconstruct historical recommendation execution.

Required Inputs

- Recommendation ID
- Historical Timestamp

Outputs

- Historical Recommendation
- Historical Configuration
- Historical Governance Context

Replay Authorization

Required

21. API-REC-016

Compare Recommendations

Endpoint

POST /api/v1/recommendations/{recommendationId}/compare

Purpose

Compare recommendation executions.

Comparison Areas

- Outcome Differences
- Confidence Differences
- Evidence Differences
- Governance Differences

Outputs

- Comparison Report

22. Request Validation Requirements

Recommendation APIs shall validate:

- Opportunity references
- Ranking references
- Recommendation strategies
- Configuration references
- Governance constraints

Validation failures shall generate standardized responses.

23. Security Requirements

Recommendation APIs shall enforce:

- Authentication
- Authorization
- Governance validation
- Input validation
- Audit logging
- Configuration controls

Security requirements align with SEC V1.

24. Traceability Requirements

All Recommendation API interactions shall generate:

- Request ID
- Correlation ID
- Recommendation ID
- Opportunity ID
- Ranking ID
- User ID
- Timestamp

Traceability records shall support investigations and replay.

25. Governance Requirements

Governance validation shall be enforced for:

- Recommendation generation
- Configuration changes
- Confidence rule modifications
- Result exports
- Replay requests

Governance exceptions require approval.

26. Audit Requirements

Audit records shall be generated for:

- Recommendation requests
- Recommendation executions
- Configuration changes
- Governance actions
- Result exports
- Comparison requests
- Replay requests

Audit records shall be immutable.

27. Replay Requirements

Replay support shall preserve:

- Recommendation configurations
- Recommendation strategies
- Confidence rules
- Governance constraints
- Historical ranking references
- Historical evidence references

Replay reconstruction shall remain supported throughout retention periods.

28. Metrics

The following metrics shall be monitored.

MET-REC-API-001

Recommendation execution volume.

MET-REC-API-002

Recommendation success rate.

MET-REC-API-003

Recommendation latency.

MET-REC-API-004

Explainability coverage.

MET-REC-API-005

Confidence evaluation coverage.

MET-REC-API-006

Governance validation success rate.

MET-REC-API-007

Replay request volume.

MET-REC-API-008

Recommendation audit coverage.

29. Compliance Requirements

Recommendation APIs shall support:

- SOC 2 governance controls
- ISO 27001 change management controls
- ISO 27001 integrity controls
- Enterprise governance requirements
- Customer security reviews
- Procurement assessments

Compliance mappings shall be maintained within RTM, SEC, GOV, AUD, and COMP artifacts.

30. Section Summary

This section defines the Recommendation Module APIs, including recommendation generation, confidence evaluation, explainability services, configuration management, governance integration, audit support, comparison capabilities, export functionality, and replay operations. These APIs establish Recommendation as the explainable decision-support layer of TODP and ensure that recommendation outcomes remain transparent, governed, traceable, auditable, secure, explainable, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 10: Governance Module API Specification

Document ID: API-V1-S10

Document Title: Governance Module API Specification

Version: 1.0

Status: Approved

Owner: Governance Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Governance Module within the Transparent Opportunity Decision Platform (TODP).

The Governance APIs provide controlled interfaces for policy validation, governance review workflows, approval management, exception processing, risk validation, compliance validation, governance decision recording, audit traceability, and replay support.

These APIs establish Governance as the independent oversight and policy enforcement layer responsible for ensuring that TODP decisions remain compliant, accountable, transparent, and aligned with enterprise governance requirements.

2. Objectives

OBJ-API-GOV-001

Provide centralized governance services.

OBJ-API-GOV-002

Support policy-driven decision oversight.

OBJ-API-GOV-003

Enable governance review workflows.

OBJ-API-GOV-004

Support exception management.

OBJ-API-GOV-005

Maintain governance traceability.

OBJ-API-GOV-006

Support audit and compliance requirements.

OBJ-API-GOV-007

Enable replay reconstruction.

OBJ-API-GOV-008

Provide enterprise-scale governance services.

3. Scope

The Governance API domain governs:

- Governance review requests
- Policy validation
- Governance approvals
- Governance rejections
- Exception management
- Risk validation
- Compliance validation
- Governance audit support
- Governance replay support

4. API Domain Information

Domain Name

Governance

Domain Identifier

GOV

Primary Resource

Governance Review

Primary Identifier

Governance ID

Ownership

Governance Domain Team

Related Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Audit

Replay

5. Governance Resource Model

Primary Attributes

- Governance ID
- Review Type
- Review Status
- Review Outcome
- Policy Version
- Governance Reviewer
- Review Timestamp

Risk Attributes

- Risk Assessment Reference
- Exception Reference

Audit Attributes

- Audit Reference
- Traceability Reference

6. API-GOV-001

Create Governance Review

Endpoint

POST /api/v1/governance/reviews

Purpose

Create governance review request.

Required Inputs

- Review Type
- Resource Reference
- Governance Context

Optional Inputs

- Risk Information
- Supporting Evidence
- Metadata

Success Response

201 Created

Generated Outputs

- Governance ID
- Review Status
- Audit Reference

Authorization Requirement

Governance Submission Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-GOV-002

Execute Governance Review

Endpoint

POST /api/v1/governance/reviews/{governanceId}/execute

Purpose

Perform governance evaluation.

Required Inputs

- Governance ID

Outputs

- Review Outcome
- Policy Findings
- Governance Summary

Success Response

200 OK

8. API-GOV-003

Get Governance Review

Endpoint

GET /api/v1/governance/reviews/{governanceId}

Purpose

Retrieve governance review details.

Outputs

- Review Information
- Review Outcome
- Policy References

Success Response

200 OK

9. API-GOV-004

Search Governance Reviews

Endpoint

GET /api/v1/governance/reviews

Purpose

Search governance reviews.

Supported Filters

- Review Type
- Status
- Reviewer
- Date Range

Supported Sorting

- Review Date
- Status
- Outcome

Success Response

200 OK

10. API-GOV-005

Validate Governance Policy

Endpoint

POST /api/v1/governance/policies/validate

Purpose

Validate resource against governance policies.

Required Inputs

- Resource Reference
- Policy Scope

Outputs

- Validation Results
- Policy Findings

Success Response

200 OK

11. API-GOV-006

Approve Governance Review

Endpoint

POST /api/v1/governance/reviews/{governanceId}/approve

Purpose

Approve governance review.

Outputs

- Approval Outcome
- Approval Reference
- Governance Status

Governance Authorization

Required

Audit Requirement

Mandatory

12. API-GOV-007

Reject Governance Review

Endpoint

POST /api/v1/governance/reviews/{governanceId}/reject

Purpose

Reject governance review.

Required Inputs

- Rejection Reason

Outputs

- Rejection Outcome
- Remediation Requirements

Audit Requirement

Mandatory

13. API-GOV-008

Create Governance Exception

Endpoint

POST /api/v1/governance/exceptions

Purpose

Create governance exception request.

Required Inputs

- Exception Type
- Business Justification
- Risk Assessment

Outputs

- Exception ID
- Review Status

Governance Approval

Required

14. API-GOV-009

Approve Governance Exception

Endpoint

POST /api/v1/governance/exceptions/{exceptionId}/approve

Purpose

Approve governance exception.

Outputs

- Approval Outcome
- Expiration Information

Audit Requirement

Mandatory

15. API-GOV-010

Get Governance Exception

Endpoint

GET /api/v1/governance/exceptions/{exceptionId}

Purpose

Retrieve exception details.

Outputs

- Exception Information
- Approval Status
- Risk Information

16. API-GOV-011

Perform Risk Validation

Endpoint

POST /api/v1/governance/risk/validate

Purpose

Validate governance risk exposure.

Risk Categories

- Operational Risk
- Security Risk
- Compliance Risk
- Business Risk

Outputs

- Risk Findings
- Risk Severity
- Recommended Actions

17. API-GOV-012

Perform Compliance Validation

Endpoint

POST /api/v1/governance/compliance/validate

Purpose

Validate compliance obligations.

Supported Frameworks

- SOC 2
- ISO 27001
- Internal Governance Policies

Outputs

- Compliance Findings
- Compliance Status

18. API-GOV-013

Get Governance History

Endpoint

GET /api/v1/governance/reviews/{governanceId}/history

Purpose

Retrieve governance history.

Outputs

- Review History
- Approval History
- Exception History

Replay Support

Enabled

19. API-GOV-014

Get Governance Audit Trail

Endpoint

GET /api/v1/governance/reviews/{governanceId}/audit

Purpose

Retrieve governance audit history.

Outputs

- Audit Records
- Governance Actions
- Reviewer Actions

Authorization Requirement

Audit Access Permission

20. API-GOV-015

Get Governance Policies

Endpoint

GET /api/v1/governance/policies

Purpose

Retrieve governance policy inventory.

Outputs

- Policy Definitions
- Policy Versions
- Effective Dates

21. API-GOV-016

Get Governance Status

Endpoint

GET /api/v1/governance/reviews/{governanceId}/status

Purpose

Retrieve governance status.

Outputs

- Current Status
- Outstanding Actions
- Governance Outcome

22. API-GOV-017

Replay Governance Review

Endpoint

POST /api/v1/governance/reviews/{governanceId}/replay

Purpose

Reconstruct historical governance review.

Required Inputs

- Governance ID
- Historical Timestamp

Outputs

- Historical Review Outcome
- Historical Policy Versions
- Historical Risk Findings

Replay Authorization

Required

23. API-GOV-018

Compare Governance Reviews

Endpoint

POST /api/v1/governance/reviews/{governanceId}/compare

Purpose

Compare governance reviews.

Comparison Areas

- Policy Differences
- Risk Differences
- Outcome Differences
- Exception Differences

Outputs

- Comparison Report

24. Request Validation Requirements

Governance APIs shall validate:

- Review references
- Policy references
- Risk assessments
- Exception requests
- Governance authority assignments

Validation failures shall generate standardized responses.

25. Security Requirements

Governance APIs shall enforce:

- Authentication
- Authorization
- Governance role validation

- Input validation
- Audit logging
- Policy access controls

Security requirements align with SEC V1.

26. Traceability Requirements

All Governance API interactions shall generate:

- Request ID
- Correlation ID
- Governance ID
- Policy Reference
- Reviewer ID
- Timestamp

Traceability records shall support investigations and replay.

27. Audit Requirements

Audit records shall be generated for:

- Governance reviews
- Policy validations
- Approvals
- Rejections
- Exception requests
- Risk validations
- Compliance validations
- Replay requests

Audit records shall be immutable.

28. Replay Requirements

Replay support shall preserve:

- Historical governance reviews
- Historical policy versions
- Historical risk assessments
- Historical exceptions
- Historical governance outcomes

Replay reconstruction shall remain supported throughout retention periods.

29. Metrics

The following metrics shall be monitored.

MET-GOV-API-001

Governance review volume.

MET-GOV-API-002

Approval rate.

MET-GOV-API-003

Exception volume.

MET-GOV-API-004

Policy validation success rate.

MET-GOV-API-005

Compliance validation success rate.

MET-GOV-API-006

Risk assessment volume.

MET-GOV-API-007

Replay request volume.

MET-GOV-API-008

Governance audit coverage.

30. Section Summary

This section defines the Governance Module APIs, including governance reviews, policy validation, approvals, rejections, exception management, risk validation, compliance validation, audit support, comparison capabilities, and replay operations. These APIs establish Governance as the independent oversight and policy enforcement layer of TODP and ensure that governance decisions remain transparent, traceable, auditable, compliant, secure, explainable, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 11: Audit Module API Specification

Document ID: API-V1-S11

Document Title: Audit Module API Specification

Version: 1.0

Status: Approved

Owner: Audit Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Audit Module within the Transparent Opportunity Decision Platform (TODP).

The Audit APIs provide controlled interfaces for audit event collection, audit record retrieval, audit search, audit correlation, compliance evidence management, investigation support, audit reporting, traceability validation, and replay support.

These APIs establish Audit as the authoritative evidentiary and oversight layer responsible for preserving platform accountability, transparency, governance verification, compliance readiness, and forensic investigation capabilities.

2. Objectives

OBJ-API-AUD-001

Provide centralized audit services.

OBJ-API-AUD-002

Support immutable audit evidence management.

OBJ-API-AUD-003

Enable investigation and forensic analysis.

OBJ-API-AUD-004

Support compliance evidence collection.

OBJ-API-AUD-005

Maintain audit traceability.

OBJ-API-AUD-006

Support governance oversight.

OBJ-API-AUD-007

Enable replay reconstruction.

OBJ-API-AUD-008

Provide enterprise-scale audit services.

3. Scope

The Audit API domain governs:

- Audit event ingestion
- Audit record retrieval
- Audit search
- Audit correlation
- Audit evidence management
- Audit reporting
- Investigation support
- Audit traceability validation
- Audit replay support

4. API Domain Information

Domain Name

Audit

Domain Identifier

AUD

Primary Resource

Audit Record

Primary Identifier

Audit ID

Ownership

Audit Domain Team

Related Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Replay

5. Audit Resource Model

Primary Attributes

- Audit ID
- Event Type
- Event Source
- Event Category
- Event Timestamp
- Actor Reference
- Resource Reference

Governance Attributes

- Governance Reference
- Policy Reference

Investigation Attributes

- Correlation Reference
- Investigation Reference

6. API-AUD-001

Create Audit Event

Endpoint

POST /api/v1/audit/events

Purpose

Register audit event.

Required Inputs

- Event Type
- Event Source
- Resource Reference
- Event Context

Outputs

- Audit ID
- Registration Status

Success Response

201 Created

Authorization Requirement

System-Level Audit Permission

Audit Requirement

Mandatory

Replay Requirement

Mandatory

7. API-AUD-002

Get Audit Record

Endpoint

GET /api/v1/audit/records/{auditId}

Purpose

Retrieve audit record.

Required Inputs

- Audit ID

Outputs

- Audit Details
- Event Metadata
- Correlation Information

Success Response

200 OK

8. API-AUD-003

Search Audit Records

Endpoint

GET /api/v1/audit/records

Purpose

Search audit repository.

Supported Filters

- Event Type
- Resource Reference
- Actor
- Date Range
- Source System

Supported Sorting

- Timestamp
- Event Type
- Source

Success Response

200 OK

9. API-AUD-004

Get Audit Timeline

Endpoint

GET /api/v1/audit/timeline

Purpose

Retrieve chronological audit activity.

Outputs

- Event Timeline
- Correlated Activities
- Investigation References

Success Response

200 OK

10. API-AUD-005

Get Audit Correlation

Endpoint

GET /api/v1/audit/correlation/{correlationId}

Purpose

Retrieve correlated audit records.

Outputs

- Related Events
- Resource Relationships
- Activity Chains

Success Response

200 OK

11. API-AUD-006

Validate Audit Integrity

Endpoint

POST /api/v1/audit/integrity/validate

Purpose

Validate audit repository integrity.

Validation Categories

- Record Integrity
- Chain Integrity
- Timestamp Integrity
- Retention Integrity

Outputs

- Validation Results
- Integrity Findings

12. API-AUD-007

Create Investigation

Endpoint

POST /api/v1/audit/investigations

Purpose

Create investigation case.

Required Inputs

- Investigation Scope
- Investigation Reason
- Investigation Owner

Outputs

- Investigation ID
- Investigation Status

Audit Requirement

Mandatory

13. API-AUD-008

Get Investigation

Endpoint

GET /api/v1/audit/investigations/{investigationId}

Purpose

Retrieve investigation information.

Outputs

- Investigation Details
- Related Audit Records
- Investigation Findings

14. API-AUD-009

Generate Audit Report

Endpoint

POST /api/v1/audit/reports

Purpose

Generate audit report.

Supported Report Types

- Activity Report
- Compliance Report
- Governance Report
- Investigation Report

Outputs

- Report Reference
- Report Status

15. API-AUD-010

Export Audit Records

Endpoint

GET /api/v1/audit/export

Purpose

Export audit records.

Supported Formats

- JSON
- CSV
- Compliance Package

Governance Validation

Required

16. API-AUD-011

Get Compliance Evidence

Endpoint

GET /api/v1/audit/compliance/evidence

Purpose

Retrieve compliance evidence.

Supported Frameworks

- SOC 2
- ISO 27001
- Internal Governance Standards

Outputs

- Evidence References
- Compliance Mappings

17. API-AUD-012

Validate Traceability

Endpoint

POST /api/v1/audit/traceability/validate

Purpose

Validate end-to-end traceability.

Validation Areas

- Opportunity Lineage
- Evidence Lineage
- Governance Lineage
- Replay Lineage

Outputs

- Validation Findings

18. API-AUD-013

Get Audit Metrics

Endpoint

GET /api/v1/audit/metrics

Purpose

Retrieve audit operational metrics.

Outputs

- Volume Metrics
- Integrity Metrics
- Compliance Metrics

19. API-AUD-014

Get Audit History

Endpoint

GET /api/v1/audit/records/{auditId}/history

Purpose

Retrieve audit record history.

Outputs

- Historical Events
- Historical References
- Validation History

Replay Support

Enabled

20. API-AUD-015

Get Audit Governance Status

Endpoint

GET /api/v1/audit/governance

Purpose

Retrieve governance information related to audit operations.

Outputs

- Governance Status
- Policy References
- Exception Information

21. API-AUD-016

Replay Audit Investigation

Endpoint

POST /api/v1/audit/replay

Purpose

Reconstruct historical audit activity.

Required Inputs

- Audit Scope
- Historical Timestamp

Outputs

- Historical Audit Events
- Historical Correlations
- Historical Governance Context

Replay Authorization

Required

22. API-AUD-017

Compare Audit Snapshots

Endpoint

POST /api/v1/audit/compare

Purpose

Compare audit states.

Comparison Areas

- Event Differences
- Integrity Differences
- Governance Differences
- Compliance Differences

Outputs

- Comparison Report

23. Request Validation Requirements

Audit APIs shall validate:

- Audit references
- Investigation references

- Correlation identifiers
- Governance constraints
- Compliance references

Validation failures shall generate standardized responses.

24. Security Requirements

Audit APIs shall enforce:

- Authentication
- Authorization
- Audit access controls
- Input validation
- Governance validation
- Export controls

Security requirements align with SEC V1.

25. Traceability Requirements

All Audit API interactions shall generate:

- Request ID
- Correlation ID
- Audit ID
- Resource Reference
- User ID
- Timestamp

Traceability records shall support investigations and replay.

26. Governance Requirements

Governance validation shall be enforced for:

- Audit exports
- Investigation creation
- Compliance evidence exports
- Audit replay requests
- Retention-related activities

Governance exceptions require approval.

27. Audit Requirements

Audit records shall be generated for:

- Audit event creation
- Investigation activities

- Integrity validations
- Compliance evidence retrieval
- Report generation
- Export activities
- Replay requests

Audit records shall be immutable.

28. Replay Requirements

Replay support shall preserve:

- Historical audit records
- Historical correlations
- Historical investigations
- Historical governance context
- Historical compliance evidence

Replay reconstruction shall remain supported throughout retention periods.

29. Metrics

The following metrics shall be monitored.

MET-AUD-API-001

Audit event volume.

MET-AUD-API-002

Audit search volume.

MET-AUD-API-003

Investigation volume.

MET-AUD-API-004

Integrity validation success rate.

MET-AUD-API-005

Compliance evidence coverage.

MET-AUD-API-006

Traceability validation success rate.

MET-AUD-API-007

Replay request volume.

MET-AUD-API-008

Audit export volume.

30. Section Summary

This section defines the Audit Module APIs, including audit event management, audit search, correlation analysis, investigation support, compliance evidence management, traceability validation, audit reporting, governance integration, comparison capabilities, and replay functionality. These APIs establish Audit as the authoritative evidentiary layer of TODP and ensure that all audit activities remain transparent, traceable, auditable, compliant, secure, governed, and replayable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 12: Replay Module API Specification

Document ID: API-V1-S12

Document Title: Replay Module API Specification

Version: 1.0

Status: Approved

Owner: Replay Domain Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the APIs governing the Replay Module within the Transparent Opportunity Decision Platform (TODP).

The Replay APIs provide controlled interfaces for historical decision reconstruction, state restoration, replay execution, replay validation, replay comparison, investigation support, audit traceability, governance validation, and historical analysis.

These APIs establish Replay as the transparency and reconstruction layer responsible for reproducing historical decision flows across the entire TODP lifecycle.

2. Objectives

OBJ-API-RPL-001

Provide controlled replay services.

OBJ-API-RPL-002

Support historical decision reconstruction.

OBJ-API-RPL-003

Enable investigation and forensic analysis.

OBJ-API-RPL-004

Support governance validation.

OBJ-API-RPL-005

Maintain replay traceability.

OBJ-API-RPL-006

Support audit and compliance requirements.

OBJ-API-RPL-007

Preserve historical accountability.

OBJ-API-RPL-008

Provide enterprise-scale replay services.

3. Scope

The Replay API domain governs:

- Replay request management
- Historical state reconstruction
- Replay execution
- Replay validation
- Replay comparison
- Replay reporting
- Investigation support
- Replay audit support
- Replay governance support

4. API Domain Information

Domain Name

Replay

Domain Identifier

RPL

Primary Resource

Replay Execution

Primary Identifier

Replay ID

Ownership

Replay Domain Team

Related Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

5. Replay Resource Model

Primary Attributes

- Replay ID
- Replay Scope
- Replay Type
- Historical Timestamp
- Replay Status
- Replay Outcome
- Execution Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Traceability Reference

6. API-RPL-001

Create Replay Request

Endpoint

POST /api/v1/replay

Purpose

Create replay execution request.

Required Inputs

- Replay Scope
- Historical Timestamp
- Resource Reference

Optional Inputs

- Replay Parameters
- Governance Context
- Metadata

Success Response

201 Created

Generated Outputs

- Replay ID
- Replay Status
- Audit Reference

Authorization Requirement

Replay Execute Permission

Audit Requirement

Mandatory

7. API-RPL-002

Execute Replay

Endpoint

POST /api/v1/replay/{replayId}/execute

Purpose

Execute historical reconstruction.

Required Inputs

- Replay ID

Outputs

- Replay Results
- Historical State
- Reconstruction Summary

Success Response

200 OK

Governance Validation

Required

8. API-RPL-003

Get Replay Execution

Endpoint

GET /api/v1/replay/{replayId}

Purpose

Retrieve replay execution details.

Outputs

- Replay Configuration
- Replay Status
- Historical Scope

Success Response

200 OK

9. API-RPL-004

Search Replay Executions

Endpoint

GET /api/v1/replay

Purpose

Search replay history.

Supported Filters

- Replay Type
- Replay Scope
- Status
- Date Range

Supported Sorting

- Execution Date
- Historical Timestamp
- Status

Success Response

200 OK

10. API-RPL-005

Get Replay Results

Endpoint

GET /api/v1/replay/{replayId}/results

Purpose

Retrieve replay output.

Outputs

- Historical Reconstruction
- Historical Decisions
- Historical References

Success Response

200 OK

11. API-RPL-006

Validate Replay Execution

Endpoint

POST /api/v1/replay/{replayId}/validate

Purpose

Validate replay accuracy.

Validation Categories

- Data Integrity Validation
- Version Validation
- Governance Validation
- Traceability Validation

Outputs

- Validation Results
- Validation Findings

12. API-RPL-007

Compare Replay Results

Endpoint

POST /api/v1/replay/{replayId}/compare

Purpose

Compare replay outcomes.

Comparison Areas

- Historical Differences
- Decision Differences
- Evidence Differences
- Governance Differences

Outputs

- Comparison Report

13. API-RPL-008

Get Replay Explainability

Endpoint

GET /api/v1/replay/{replayId}/explainability

Purpose

Retrieve replay rationale.

Outputs

- Reconstruction Logic
- Historical Dependencies
- Version References
- Governance References

Success Response

200 OK

14. API-RPL-009

Get Replay Timeline

Endpoint

GET /api/v1/replay/{replayId}/timeline

Purpose

Retrieve reconstructed timeline.

Outputs

- Event Timeline
- Decision Timeline
- Governance Timeline

Replay Support

Enabled

15. API-RPL-010

Generate Replay Report

Endpoint

POST /api/v1/replay/{replayId}/report

Purpose

Generate replay report.

Supported Reports

- Reconstruction Report
- Investigation Report
- Governance Report
- Compliance Report

Outputs

- Report Reference

16. API-RPL-011

Get Replay Audit Trail

Endpoint

GET /api/v1/replay/{replayId}/audit

Purpose

Retrieve replay audit history.

Outputs

- Audit Records
- Replay Activities
- Governance Actions

Authorization Requirement

Audit Access Permission

17. API-RPL-012

Get Replay Governance Status

Endpoint

GET /api/v1/replay/{replayId}/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Governance Decisions

18. API-RPL-013

Export Replay Results

Endpoint

GET /api/v1/replay/{replayId}/export

Purpose

Export replay output.

Supported Formats

- JSON
- CSV
- Investigation Package

Governance Validation

Required

19. API-RPL-014

Get Historical Versions

Endpoint

GET /api/v1/replay/{replayId}/versions

Purpose

Retrieve historical versions used during replay.

Outputs

- Policy Versions
- Ranking Versions
- Recommendation Versions
- Governance Versions

20. API-RPL-015

Reconstruct Decision Flow

Endpoint

POST /api/v1/replay/{replayId}/decision-flow

Purpose

Reconstruct end-to-end decision flow.

Outputs

- Opportunity Flow
- Evidence Flow
- Ranking Flow
- Recommendation Flow
- Governance Flow

21. API-RPL-016

Reconstruct Evidence Lineage

Endpoint

POST /api/v1/replay/{replayId}/lineage

Purpose

Reconstruct historical evidence lineage.

Outputs

- Source Lineage
- Transformation History
- Usage History

22. API-RPL-017

Reconstruct Governance Outcome

Endpoint

POST /api/v1/replay/{replayId}/governance-outcome

Purpose

Reconstruct governance decision history.

Outputs

- Governance Reviews
- Policy Evaluations
- Approval History

23. API-RPL-018

Replay Investigation Support

Endpoint

POST /api/v1/replay/{replayId}/investigation

Purpose

Support investigation workflows.

Outputs

- Investigation Findings
- Historical Correlations
- Supporting References

24. Request Validation Requirements

Replay APIs shall validate:

- Replay scope
- Historical timestamps
- Resource references
- Governance permissions
- Version references

Validation failures shall generate standardized responses.

25. Security Requirements

Replay APIs shall enforce:

- Authentication
- Authorization
- Governance validation
- Input validation
- Audit logging
- Historical data access controls

Security requirements align with SEC V1.

26. Traceability Requirements

All Replay API interactions shall generate:

- Request ID
- Correlation ID
- Replay ID
- Resource Reference
- User ID
- Timestamp

Traceability records shall support investigations and accountability.

27. Governance Requirements

Governance validation shall be enforced for:

- Replay execution
- Historical reconstruction
- Export operations
- Investigation activities
- Replay reporting

Governance exceptions require approval.

28. Audit Requirements

Audit records shall be generated for:

- Replay requests
- Replay executions
- Validation activities
- Comparison activities
- Export operations
- Investigation activities
- Governance actions

Audit records shall be immutable.

29. Replay Requirements

Replay support shall preserve:

- Historical states
- Historical versions
- Historical policies
- Historical governance outcomes
- Historical audit references

Replay reconstruction shall remain supported throughout retention periods.

30. Section Summary

This section defines the Replay Module APIs, including replay execution, historical reconstruction, validation services, comparison capabilities, explainability services, reporting functions, investigation support, governance integration, export functionality, and audit support. These APIs establish Replay as the transparency and reconstruction layer of TODP and ensure that historical decision outcomes remain traceable, auditable, explainable, governed, compliant, secure, and reproducible throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 13: Authentication & Authorization API Specification

Document ID: API-V1-S13

Document Title: Authentication & Authorization API Specification

Version: 1.0

Status: Approved

Owner: Platform Security Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the authentication and authorization APIs governing secure access to the Transparent Opportunity Decision Platform (TODP).

These APIs provide centralized identity validation, authentication management, token issuance, authorization verification, session management, access governance integration, audit traceability, and replay support.

These APIs establish the security control plane used by all TODP modules and services.

2. Objectives

OBJ-API-SEC-001

Provide centralized authentication services.

OBJ-API-SEC-002

Support secure authorization decisions.

OBJ-API-SEC-003

Enable identity verification.

OBJ-API-SEC-004

Support governance-controlled access.

OBJ-API-SEC-005

Maintain access traceability.

OBJ-API-SEC-006

Support audit and compliance requirements.

OBJ-API-SEC-007

Enable security investigation support.

OBJ-API-SEC-008

Provide enterprise-grade access controls.

3. Scope

The Authentication and Authorization API domain governs:

- User authentication
- Service authentication
- Token management
- Session management
- Authorization validation
- Access governance
- Security audit support
- Replay support

The APIs apply across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Security Domain Information

Domain Name

Security

Domain Identifier

SEC

Primary Resource

Security Session

Primary Identifier

Session ID

Ownership

Platform Security Team

Related Modules

Authority

Governance

Audit

Replay

5. Security Resource Model

Primary Attributes

- Session ID
- Subject ID
- Authentication Method
- Authentication Status
- Authorization Status
- Session Timestamp

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Security Event Reference

6. API-SEC-001

Authenticate User

Endpoint

POST /api/v1/security/authenticate

Purpose

Authenticate platform user.

Required Inputs

- Username
- Authentication Credential

Outputs

- Authentication Result
- Session ID
- Access Token

Success Response

200 OK

Audit Requirement

Mandatory

7. API-SEC-002

Authenticate Service

Endpoint

POST /api/v1/security/service-authenticate

Purpose

Authenticate service account.

Required Inputs

- Service Identifier
- Service Credential

Outputs

- Service Session
- Access Token

Success Response

200 OK

8. API-SEC-003

Refresh Access Token

Endpoint

POST /api/v1/security/token/refresh

Purpose

Refresh authentication token.

Required Inputs

- Refresh Token

Outputs

- New Access Token
- Expiration Information

Success Response

200 OK

9. API-SEC-004

Validate Access Token

Endpoint

POST /api/v1/security/token/validate

Purpose

Validate token status.

Outputs

- Validation Result
- Subject Information
- Authorization Context

Success Response

200 OK

10. API-SEC-005

Revoke Access Token

Endpoint

POST /api/v1/security/token/revoke

Purpose

Invalidate active token.

Required Inputs

- Token Reference

Outputs

- Revocation Status

Audit Requirement

Mandatory

11. API-SEC-006

Create Session

Endpoint

POST /api/v1/security/sessions

Purpose

Create authenticated session.

Outputs

- Session ID
- Session Status
- Expiration Information

Success Response

201 Created

12. API-SEC-007

Get Session

Endpoint

GET /api/v1/security/sessions/{sessionId}

Purpose

Retrieve session information.

Outputs

- Session Details
- Authentication Information
- Activity Information

13. API-SEC-008

Terminate Session

Endpoint

POST /api/v1/security/sessions/{sessionId}/terminate

Purpose

Terminate active session.

Outputs

- Termination Status

Audit Requirement

Mandatory

14. API-SEC-009

Authorize Request

Endpoint

POST /api/v1/security/authorize

Purpose

Validate access authorization.

Required Inputs

- Subject ID
- Requested Action
- Resource Reference

Outputs

- Authorization Decision
- Policy References

Success Response

200 OK

15. API-SEC-010

Get Authorization Context

Endpoint

GET /api/v1/security/authorization/context

Purpose

Retrieve authorization information.

Outputs

- Roles
- Permissions
- Policy References

16. API-SEC-011

Perform Access Review

Endpoint

POST /api/v1/security/access-review

Purpose

Perform access review.

Inputs

- Review Scope
- Subject Reference

Outputs

- Review Findings
- Governance References

17. API-SEC-012

Register Security Event

Endpoint

POST /api/v1/security/events

Purpose

Record security event.

Event Categories

- Authentication
- Authorization
- Session
- Access Review
- Security Violation

Outputs

- Security Event ID

18. API-SEC-013

Get Security Events

Endpoint

GET /api/v1/security/events

Purpose

Retrieve security events.

Supported Filters

- Event Type
- Subject ID
- Date Range

Outputs

- Security Event Records

19. API-SEC-014

Get Security Audit Trail

Endpoint

GET /api/v1/security/audit

Purpose

Retrieve security audit information.

Outputs

- Security Audit Records
- Access History
- Governance Actions

Authorization Requirement

Audit Access Permission

20. API-SEC-015

Get Security Governance Status

Endpoint

GET /api/v1/security/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Exception Information

21. API-SEC-016

Validate Security Policy

Endpoint

POST /api/v1/security/policies/validate

Purpose

Validate compliance with security policies.

Outputs

- Validation Results
- Policy Findings

22. API-SEC-017

Replay Security Activity

Endpoint

POST /api/v1/security/replay

Purpose

Reconstruct historical security activity.

Required Inputs

- Historical Timestamp
- Subject Reference

Outputs

- Historical Sessions
- Historical Authorizations
- Historical Security Events

Replay Authorization

Required

23. API-SEC-018

Compare Security States

Endpoint

POST /api/v1/security/compare

Purpose

Compare security states.

Comparison Areas

- Authorization Differences
- Policy Differences
- Session Differences

Outputs

- Comparison Report

24. Request Validation Requirements

Security APIs shall validate:

- Credentials
- Session identifiers
- Token references
- Authorization requests
- Governance permissions

Validation failures shall generate standardized responses.

25. Security Requirements

Security APIs shall enforce:

- Strong authentication
- Authorization validation
- Session protection
- Credential protection
- Audit logging
- Governance validation

Security requirements align with SEC V1.

26. Traceability Requirements

All Security API interactions shall generate:

- Request ID
- Correlation ID
- Session ID
- Subject ID
- Security Event ID
- Timestamp

Traceability records shall support investigations and replay.

27. Governance Requirements

Governance validation shall be enforced for:

- Access reviews
- Security policy changes
- Authorization exceptions
- Security replay requests

Governance exceptions require approval.

28. Audit Requirements

Audit records shall be generated for:

- Authentication events
- Authorization decisions
- Session activities
- Access reviews
- Security events
- Policy validations
- Replay requests

Audit records shall be immutable.

29. Metrics

The following metrics shall be monitored.

MET-SEC-API-001

Authentication volume.

MET-SEC-API-002

Authentication success rate.

MET-SEC-API-003

Authorization volume.

MET-SEC-API-004

Access review volume.

MET-SEC-API-005

Security event volume.

MET-SEC-API-006

Session termination volume.

MET-SEC-API-007

Replay request volume.

MET-SEC-API-008

Security audit coverage.

30. Section Summary

This section defines the Authentication and Authorization APIs, including identity validation, token management, session management, authorization evaluation, access reviews, security event management, governance integration, audit support, and replay functionality. These APIs establish the centralized security control plane of TODP and ensure that all platform access remains governed, traceable, auditable, compliant, secure, and reproducible throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 14: Administrative & Platform Management API Specification

Document ID: API-V1-S14

Document Title: Administrative & Platform Management API Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the administrative and platform management APIs governing operational administration, platform configuration, system maintenance, environment management, service administration, feature control, operational governance, audit traceability, and replay support within the Transparent Opportunity Decision Platform (TODP).

These APIs establish the controlled operational management layer for enterprise administration activities.

2. Objectives

OBJ-API-ADM-001

Provide centralized administrative services.

OBJ-API-ADM-002

Support controlled platform management.

OBJ-API-ADM-003

Enable configuration governance.

OBJ-API-ADM-004

Support operational oversight.

OBJ-API-ADM-005

Maintain administrative traceability.

OBJ-API-ADM-006

Support audit and compliance requirements.

OBJ-API-ADM-007

Enable governance-controlled administration.

OBJ-API-ADM-008

Provide enterprise operational readiness.

3. Scope

The Administrative API domain governs:

- Platform configuration management
- Environment administration
- Service administration
- Feature management
- Operational controls
- Administrative governance
- Audit support
- Replay support

The APIs support all TODP domains and platform services.

4. Administrative Domain Information

Domain Name

Administration

Domain Identifier

ADM

Primary Resource

Administrative Action

Primary Identifier

Administration ID

Ownership

Platform Engineering

Related Modules

Authority

Governance

Audit

Replay

5. Administrative Resource Model

Primary Attributes

- Administration ID
- Administrative Action
- Administrative Category
- Administrative Status
- Execution Timestamp
- Administrative Owner

Governance Attributes

- Governance Reference
- Policy Reference

Audit Attributes

- Audit Reference
- Change Reference

6. API-ADM-001

Get Platform Configuration

Endpoint

GET /api/v1/admin/configuration

Purpose

Retrieve platform configuration.

Outputs

- Configuration Settings
- Version Information
- Governance Status

Authorization Requirement

Administrative Read Permission

7. API-ADM-002

Update Platform Configuration

Endpoint

PUT /api/v1/admin/configuration

Purpose

Modify platform configuration.

Governed Elements

- Platform Settings
- Operational Parameters
- Service Controls

Governance Validation

Mandatory

Audit Requirement

Mandatory

8. API-ADM-003

Get Environment Information

Endpoint

GET /api/v1/admin/environments

Purpose

Retrieve environment inventory.

Outputs

- Environment Details
- Environment Status
- Version Information

9. API-ADM-004

Manage Environment State

Endpoint

POST /api/v1/admin/environments/{environmentId}/state

Purpose

Manage environment lifecycle.

Supported Actions

- Enable
- Disable
- Maintenance Mode
- Recovery Mode

Governance Validation

Required

10. API-ADM-005

Get Service Inventory

Endpoint

GET /api/v1/admin/services

Purpose

Retrieve service inventory.

Outputs

- Service Details
- Service Status
- Dependency Information

11. API-ADM-006

Manage Service State

Endpoint

POST /api/v1/admin/services/{serviceId}/state

Purpose

Manage service lifecycle.

Supported Actions

- Start
- Stop
- Restart
- Suspend

Governance Validation

Required

Audit Requirement

Mandatory

12. API-ADM-007

Get Feature Flags

Endpoint

GET /api/v1/admin/features

Purpose

Retrieve feature inventory.

Outputs

- Feature Definitions
- Feature Status
- Governance Information

13. API-ADM-008

Manage Feature Flag

Endpoint

POST /api/v1/admin/features/{featureId}

Purpose

Enable or disable feature controls.

Supported Actions

- Enable
- Disable
- Rollout
- Rollback

Governance Validation

Required

14. API-ADM-009

Get Platform Health

Endpoint

GET /api/v1/admin/health

Purpose

Retrieve platform health information.

Outputs

- Service Health
- Dependency Health
- Platform Status

15. API-ADM-010

Get Platform Metrics

Endpoint

GET /api/v1/admin/metrics

Purpose

Retrieve operational metrics.

Outputs

- Performance Metrics
- Capacity Metrics
- Availability Metrics

16. API-ADM-011

Execute Maintenance Action

Endpoint

POST /api/v1/admin/maintenance

Purpose

Execute administrative maintenance activity.

Supported Actions

- Cache Refresh
- Index Rebuild

- Metadata Refresh
- Validation Tasks

Governance Validation

Required

17. API-ADM-012

Get Administrative History

Endpoint

GET /api/v1/admin/history

Purpose

Retrieve administrative activity history.

Outputs

- Administrative Actions
- Configuration Changes
- Governance Actions

Replay Support

Enabled

18. API-ADM-013

Get Administrative Audit Trail

Endpoint

GET /api/v1/admin/audit

Purpose

Retrieve administrative audit history.

Outputs

- Audit Records
- Administrative Actions
- Change History

Authorization Requirement

Audit Access Permission

19. API-ADM-014

Get Administrative Governance Status

Endpoint

GET /api/v1/admin/governance

Purpose

Retrieve governance information.

Outputs

- Governance Status
- Policy References
- Exception Information

20. API-ADM-015

Validate Administrative Configuration

Endpoint

POST /api/v1/admin/configuration/validate

Purpose

Validate configuration compliance.

Validation Categories

- Governance Validation
- Security Validation
- Operational Validation

Outputs

- Validation Results

21. API-ADM-016

Export Administrative Data

Endpoint

GET /api/v1/admin/export

Purpose

Export administrative information.

Supported Formats

- JSON
- CSV
- Administrative Package

Governance Validation

Required

22. API-ADM-017

Replay Administrative Activity

Endpoint

POST /api/v1/admin/replay

Purpose

Reconstruct historical administrative activity.

Required Inputs

- Historical Timestamp
- Administrative Scope

Outputs

- Historical Configuration
- Historical Actions
- Historical Governance Context

Replay Authorization

Required

23. API-ADM-018

Compare Administrative States

Endpoint

POST /api/v1/admin/compare

Purpose

Compare platform administrative states.

Comparison Areas

- Configuration Differences
- Service Differences
- Governance Differences

Outputs

- Comparison Report

24. Request Validation Requirements

Administrative APIs shall validate:

- Configuration references
- Environment references
- Service references
- Administrative permissions
- Governance constraints

Validation failures shall generate standardized responses.

25. Security Requirements

Administrative APIs shall enforce:

- Strong authentication
- Administrative authorization
- Governance validation
- Input validation
- Audit logging
- Change controls

Security requirements align with SEC V1.

26. Traceability Requirements

All Administrative API interactions shall generate:

- Request ID
- Correlation ID
- Administration ID
- User ID
- Change Reference
- Timestamp

Traceability records shall support investigations and replay.

27. Governance Requirements

Governance validation shall be enforced for:

- Configuration changes
- Service lifecycle changes
- Environment changes
- Feature flag modifications
- Maintenance actions
- Replay requests

Governance exceptions require approval.

28. Audit Requirements

Audit records shall be generated for:

- Configuration changes
- Environment actions
- Service actions
- Feature modifications
- Maintenance activities
- Administrative exports
- Replay requests

Audit records shall be immutable.

29. Metrics

The following metrics shall be monitored.

MET-ADM-API-001

Administrative action volume.

MET-ADM-API-002

Configuration change volume.

MET-ADM-API-003

Service lifecycle activity volume.

MET-ADM-API-004

Feature flag change volume.

MET-ADM-API-005

Maintenance execution volume.

MET-ADM-API-006

Governance validation success rate.

MET-ADM-API-007

Replay request volume.

MET-ADM-API-008

Administrative audit coverage.

30. Section Summary

This section defines the Administrative and Platform Management APIs, including configuration management, environment administration, service administration, feature management, operational monitoring, governance integration, audit support, comparison capabilities, export functionality, and replay operations. These APIs establish the controlled operational management layer of TODP and ensure that administrative activities remain governed, traceable, auditable, compliant, secure, and reproducible throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 15: API Error Handling & Exception Management Specification

Document ID: API-V1-S15

Document Title: API Error Handling & Exception Management Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the enterprise-wide standards governing API error handling, exception processing, fault reporting, error classification, operational diagnostics, governance exceptions, recovery mechanisms, audit requirements, and replay support across the Transparent Opportunity Decision Platform (TODP).

This specification ensures consistent and predictable behavior when API operations encounter validation failures, governance violations, security failures, system faults, integration errors, or operational exceptions.

2. Objectives

OBJ-API-ERR-001

Establish standardized error handling.

OBJ-API-ERR-002

Provide consistent exception management.

OBJ-API-ERR-003

Support operational troubleshooting.

OBJ-API-ERR-004

Maintain auditability of failures.

OBJ-API-ERR-005

Enable governance oversight of exceptions.

OBJ-API-ERR-006

Support security incident investigations.

OBJ-API-ERR-007

Enable replay reconstruction of failures.

OBJ-API-ERR-008

Improve platform reliability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The standards apply to all API consumers and providers.

4. Error Handling Principles

All APIs shall comply with the following principles.

ERR-PRN-001

Consistency

Errors shall follow standardized structures.

ERR-PRN-002

Traceability

Errors shall be traceable.

ERR-PRN-003

Auditability

Errors shall generate audit records.

ERR-PRN-004

Security

Errors shall not expose sensitive information.

ERR-PRN-005

Governance

Governance violations shall be explicitly identified.

ERR-PRN-006

Replayability

Error events shall support replay reconstruction.

ERR-PRN-007

Recoverability

Errors shall support recovery workflows.

ERR-PRN-008

Observability

Errors shall support monitoring and diagnostics.

5. Standard Error Response Model

All API errors shall return:

errorCode

Unique machine-readable identifier.

errorCategory

Standard error classification.

errorMessage

Human-readable message.

requestId

Request identifier.

correlationId

Cross-service identifier.

timestamp

Error occurrence timestamp.

serviceName

Originating service.

6. Error Classification Framework

ERR-CAT-001

Validation Error

ERR-CAT-002

Authentication Error

ERR-CAT-003

Authorization Error

ERR-CAT-004

Governance Error

ERR-CAT-005

Business Rule Error

ERR-CAT-006

Integration Error

ERR-CAT-007

System Error

ERR-CAT-008

Replay Error

ERR-CAT-009

Audit Error

ERR-CAT-010

Compliance Error

7. Validation Error Standards

Validation errors occur when requests fail validation controls.

Examples

- Missing required attributes
- Invalid identifiers
- Invalid schema
- Invalid request structure

Standard Response

HTTP 400

Error Category

Validation Error

8. Authentication Error Standards

Authentication errors occur when identity validation fails.

Examples

- Invalid credentials
- Expired tokens
- Invalid sessions

Standard Response

HTTP 401

Error Category

Authentication Error

9. Authorization Error Standards

Authorization errors occur when access rights are insufficient.

Examples

- Missing permission
- Restricted resource access
- Unauthorized operation

Standard Response

HTTP 403

Error Category

Authorization Error

10. Governance Error Standards

Governance errors occur when policy requirements are violated.

Examples

- Policy violation
- Governance rejection
- Missing approval

Standard Response

HTTP 403

Error Category

Governance Error

11. Business Rule Error Standards

Business rule errors occur when domain constraints fail.

Examples

- Invalid lifecycle transition
- Invalid ownership assignment
- Invalid recommendation outcome

Standard Response

HTTP 422

Error Category

Business Rule Error

12. Integration Error Standards

Integration errors occur during external or internal service communication failures.

Examples

- Dependency unavailable
- Timeout
- Contract mismatch

Standard Response

HTTP 502

Error Category

Integration Error

13. System Error Standards

System errors occur due to platform failures.

Examples

- Service failure
- Resource exhaustion
- Unexpected exception

Standard Response

HTTP 500

Error Category

System Error

14. Replay Error Standards

Replay errors occur during historical reconstruction activities.

Examples

- Missing historical version
- Missing replay data
- Replay validation failure

Standard Response

HTTP 422

Error Category

Replay Error

15. Audit Error Standards

Audit errors occur during audit operations.

Examples

- Missing audit reference
- Audit integrity failure
- Audit export failure

Standard Response

HTTP 500

Error Category

Audit Error

16. Compliance Error Standards

Compliance errors occur when compliance controls fail.

Examples

- Compliance policy violation
- Regulatory restriction
- Retention violation

Standard Response

HTTP 403

Error Category

Compliance Error

17. Exception Management Framework

API exceptions shall be formally managed.

Exception Categories

- Security Exceptions
- Governance Exceptions
- Operational Exceptions
- Integration Exceptions
- Compliance Exceptions

Exceptions shall be documented and tracked.

18. Exception Lifecycle

Exception States

Raised

Under Review

Approved

Rejected

Implemented

Resolved

Archived

All transitions shall be auditable.

19. Fault Correlation Requirements

Errors shall support cross-service correlation.

Required Attributes

- Request ID
- Correlation ID
- Service Reference
- Resource Reference

Correlation data shall support investigations.

20. Error Logging Requirements

All errors shall generate logs.

Required Log Attributes

- Error Code
- Error Category
- Timestamp
- Request ID
- Correlation ID

Logs shall comply with audit requirements.

21. Security Error Handling Requirements

Security-related errors shall:

- Minimize information disclosure
- Generate audit events
- Trigger security monitoring
- Support investigations

Security events shall be reviewed.

22. Governance Exception Requirements

Governance exceptions shall require:

- Business justification
- Risk assessment
- Governance approval
- Expiration date

Exceptions shall remain traceable.

23. Recovery Management Requirements

APIs shall support recovery workflows.

Recovery Categories

- Retry Recovery
- Manual Recovery
- Governance Recovery
- Operational Recovery

Recovery guidance shall be documented.

24. Error Monitoring Requirements

Monitoring shall include:

- Error volume
- Error rate
- Critical errors
- Service failures
- Recovery success rate

Monitoring alerts shall be configured.

25. Replay Requirements

Replay support shall preserve:

- Error records
- Exception records
- Recovery actions
- Governance actions

Historical failures shall remain reconstructable.

26. Audit Requirements

Audit records shall be generated for:

- Critical errors
- Governance failures
- Security failures
- Compliance failures
- Exception approvals
- Recovery actions

Audit records shall be immutable.

27. Compliance Requirements

Error handling shall support:

- SOC 2 monitoring controls
- ISO 27001 incident controls
- ISO 27001 operational controls
- Enterprise governance standards

Compliance mappings shall be maintained.

28. Architectural Constraints

All APIs shall comply with:

ERR-CON-001

Use standardized error responses.

ERR-CON-002

Maintain traceability.

ERR-CON-003

Support auditability.

ERR-CON-004

Protect sensitive information.

ERR-CON-005

Support governance oversight.

ERR-CON-006

Support replay reconstruction.

ERR-CON-007

Support operational monitoring.

ERR-CON-008

Support compliance requirements.

29. Metrics

The following metrics shall be monitored.

MET-ERR-001

Error volume.

MET-ERR-002

Error rate.

MET-ERR-003

Critical error count.

MET-ERR-004

Recovery success rate.

MET-ERR-005

Governance exception volume.

MET-ERR-006

Security error volume.

MET-ERR-007

Replay failure volume.

MET-ERR-008

Compliance violation count.

30. Section Summary

This section defines the enterprise standards governing API error handling and exception management, including error classifications, response structures, fault correlation, exception governance, recovery processes, audit controls, monitoring requirements, compliance obligations, and replay support. These controls ensure that all TODP APIs handle failures consistently, securely, transparently, traceably, auditably, and in a manner aligned with enterprise governance and operational resilience requirements.

API SPEC V1 – Enterprise API Specification

Section 16: API Versioning & Lifecycle Management Specification

Document ID: API-V1-S16

Document Title: API Versioning & Lifecycle Management Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API versioning, lifecycle management, release governance, compatibility controls, deprecation management, retirement procedures, migration requirements, audit obligations, and compliance controls across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that APIs evolve in a controlled, predictable, traceable, and enterprise-governed manner while minimizing disruption to platform services, internal consumers, and future external integrations.

2. Objectives

OBJ-API-LCM-001

Establish standardized API versioning.

OBJ-API-LCM-002

Support controlled API evolution.

OBJ-API-LCM-003

Minimize consumer disruption.

OBJ-API-LCM-004

Maintain backward compatibility governance.

OBJ-API-LCM-005

Support auditability of API changes.

OBJ-API-LCM-006

Enable controlled API retirement.

OBJ-API-LCM-007

Support replay reconstruction requirements.

OBJ-API-LCM-008

Maintain enterprise governance compliance.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The standards apply to all API contracts, interfaces, schemas, and integrations.

4. API Lifecycle Principles

All APIs shall comply with the following principles.

LCM-PRN-001

Controlled Evolution

API changes shall be governed.

LCM-PRN-002

Traceability

Version changes shall be traceable.

LCM-PRN-003

Auditability

Version changes shall be auditable.

LCM-PRN-004

Compatibility

Consumer impact shall be evaluated.

LCM-PRN-005

Governance

Material changes require approval.

LCM-PRN-006

Transparency

Version changes shall be documented.

LCM-PRN-007

Replayability

Historical versions shall remain reconstructable.

LCM-PRN-008

Sustainability

API evolution shall support long-term maintainability.

5. API Lifecycle States

All APIs shall follow the approved lifecycle.

Lifecycle States

Proposed

Design

Review

Approved

Implemented

Testing

Released

Active

Deprecated

Retired

Archived

Lifecycle state transitions shall be formally governed.

6. Version Numbering Standard

TODP APIs shall use explicit major versioning.

Examples

v1

v2

v3

Version identifiers shall appear in:

- API endpoints
- API contracts
- Documentation
- Audit records
- Governance records

7. Major Version Definition

A major version shall be created when:

- Breaking contract changes occur
- Resource models change materially
- Authentication models change materially
- Governance requirements change materially

Major versions require Architecture Review Board approval.

8. Minor Version Definition

Minor version changes may include:

- New optional attributes
- New endpoints
- New response metadata
- Additional capabilities

Minor changes shall remain backward compatible.

9. Patch Version Definition

Patch changes may include:

- Defect corrections
- Documentation updates
- Non-breaking optimizations
- Security remediations

Patch changes shall not alter API behavior materially.

10. Version Governance Requirements

Version creation shall require:

- Impact assessment
- Architecture review
- Security review
- Governance approval
- Documentation updates

Approval records shall be retained.

11. Compatibility Management Framework

Compatibility evaluations shall include:

- Contract compatibility
- Schema compatibility
- Security compatibility
- Governance compatibility
- Replay compatibility

Compatibility findings shall be documented.

12. Backward Compatibility Standards

Backward compatibility shall be maintained whenever practical.

Protected Elements

- Existing endpoints
- Existing response structures
- Existing identifiers
- Existing authentication mechanisms

Compatibility exceptions require approval.

13. Breaking Change Governance

Breaking changes shall require:

- Formal impact assessment
- Consumer notification
- Migration plan
- Governance approval

Breaking changes shall be documented.

14. API Deprecation Framework

Deprecated APIs shall remain supported for an approved transition period.

Deprecation Requirements

- Deprecation notice
- Migration guidance
- Governance approval
- Retirement plan

Deprecation status shall be documented.

15. API Retirement Framework

API retirement shall follow controlled procedures.

Retirement Requirements

- Consumer impact review
- Governance approval
- Retirement notice
- Historical preservation

Retired APIs shall remain traceable.

16. Migration Management Requirements

Migration plans shall include:

- Version mapping
- Consumer impact analysis
- Migration timeline
- Rollback procedures
- Validation procedures

Migration plans shall be approved.

17. Version Repository Requirements

All API versions shall be maintained within a controlled repository.

Repository Controls

- Version history
- Change history
- Approval records
- Contract preservation

Repository integrity shall be protected.

18. Contract Evolution Requirements

API contract changes shall be governed.

Governed Areas

- Resources
- Endpoints
- Schemas
- Authentication requirements
- Governance requirements

Uncontrolled contract modifications are prohibited.

19. Documentation Versioning Requirements

Each version shall maintain:

- Version identifier
- Change history
- Effective date
- Approval history
- Retirement information

Documentation shall remain synchronized with implementations.

20. Release Management Requirements

API releases shall include:

- Release identifier
- Version reference
- Deployment approval
- Validation results

Release records shall be retained.

21. Governance Review Requirements

The following changes require governance review:

- Major version changes
- Breaking changes
- Security changes
- Governance changes
- Compliance-impacting changes

Governance decisions shall be documented.

22. Security Review Requirements

Security reviews shall evaluate:

- Authentication changes
- Authorization changes
- Encryption changes
- Access control changes

Security findings shall be resolved before release.

23. Replay Preservation Requirements

Historical API versions shall remain replayable.

Preserved Artifacts

- Contracts
- Schemas

- Version metadata
- Governance records
- Security requirements

Replay reconstruction shall remain possible.

24. Audit Requirements

Audit records shall be generated for:

- Version creation
- Version approval
- Version release
- Deprecation actions
- Retirement actions
- Migration activities

Audit records shall be immutable.

25. Compliance Requirements

Version management shall support:

- SOC 2 change management controls
- ISO 27001 change control requirements
- ISO 27001 configuration management requirements
- Enterprise governance standards

Compliance mappings shall be maintained.

26. Risk Management Requirements

Version-related risks shall be assessed.

Risk Categories

- Compatibility risk
- Security risk
- Operational risk
- Governance risk
- Compliance risk

Risk findings shall be documented.

27. Architectural Constraints

All APIs shall comply with:

LCM-CON-001

Maintain version traceability.

LCM-CON-002

Support auditability.

LCM-CON-003

Support replayability.

LCM-CON-004

Protect compatibility.

LCM-CON-005

Require governance approval.

LCM-CON-006

Preserve historical versions.

LCM-CON-007

Support controlled retirement.

LCM-CON-008

Support enterprise sustainability.

28. Metrics

The following metrics shall be monitored.

MET-LCM-001

Active API versions.

MET-LCM-002

Deprecated API count.

MET-LCM-003

Retired API count.

MET-LCM-004

Compatibility issue count.

MET-LCM-005

Migration completion rate.

MET-LCM-006

Version approval cycle time.

MET-LCM-007

Breaking change volume.

MET-LCM-008

Replay version coverage.

29. Audit Controls

Lifecycle audits shall verify:

- Version governance compliance
- Compatibility evaluations
- Migration readiness
- Deprecation compliance
- Retirement compliance
- Replay preservation

Audit findings shall be remediated.

30. Section Summary

This section defines the standards governing API versioning and lifecycle management, including version numbering, compatibility controls, change governance, deprecation procedures, retirement processes, migration requirements, repository management, replay preservation, audit obligations, and compliance controls. These standards ensure that TODP APIs evolve in a controlled, transparent, traceable, auditable, and enterprise-governed manner throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 17: API Monitoring, Observability & Telemetry Specification

Document ID: API-V1-S17

Document Title: API Monitoring, Observability & Telemetry Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API monitoring, observability, telemetry collection, distributed tracing, operational diagnostics, performance analysis, service health monitoring, audit integration, and governance oversight across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all APIs remain observable, measurable, diagnosable, traceable, and operationally manageable throughout their lifecycle.

2. Objectives

OBJ-API-OBS-001

Provide comprehensive API observability.

OBJ-API-OBS-002

Support proactive operational monitoring.

OBJ-API-OBS-003

Enable rapid incident diagnosis.

OBJ-API-OBS-004

Maintain end-to-end traceability.

OBJ-API-OBS-005

Support governance oversight.

OBJ-API-OBS-006

Support audit and compliance requirements.

OBJ-API-OBS-007

Enable replay-assisted investigations.

OBJ-API-OBS-008

Improve operational reliability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The standards apply across all environments.

4. Observability Principles

All APIs shall comply with the following principles.

OBS-PRN-001

Visibility

Operational activity shall be observable.

OBS-PRN-002

Traceability

Requests shall be traceable.

OBS-PRN-003

Diagnosability

Failures shall be diagnosable.

OBS-PRN-004

Consistency

Telemetry shall follow common standards.

OBS-PRN-005

Governance

Observability controls shall be governed.

OBS-PRN-006

Auditability

Monitoring activity shall be auditable.

OBS-PRN-007

Replayability

Telemetry shall support replay investigations.

OBS-PRN-008

Scalability

Observability controls shall scale with platform growth.

5. Observability Architecture

The observability framework shall include:

- Metrics Collection
- Log Collection
- Distributed Tracing
- Event Monitoring
- Alerting
- Dashboards
- Operational Analytics

All components shall integrate into the enterprise monitoring platform.

6. Metrics Collection Standards

All APIs shall generate metrics.

Required Metric Categories

- Availability Metrics
- Performance Metrics
- Usage Metrics
- Security Metrics
- Governance Metrics
- Error Metrics

Metrics collection shall be automated.

7. Logging Standards

All APIs shall generate structured logs.

Required Log Attributes

- Timestamp
- Request ID
- Correlation ID
- Service Name
- API Endpoint
- Response Status

Logs shall support investigations and diagnostics.

8. Distributed Tracing Standards

All APIs shall participate in distributed tracing.

Required Trace Attributes

- Trace ID
- Span ID
- Parent Span ID
- Request ID
- Correlation ID

Tracing shall support cross-service visibility.

9. Service Health Monitoring Standards

All APIs shall expose health information.

Health Categories

- Service Health
- Dependency Health
- Database Health
- Integration Health

Health status shall be continuously monitored.

10. Availability Monitoring Requirements

Availability monitoring shall track:

- Uptime
- Downtime
- Service Reachability
- Dependency Availability

Availability thresholds shall be defined.

11. Performance Monitoring Requirements

Performance monitoring shall track:

- Response Latency
- Throughput
- Request Volume
- Resource Consumption

Performance baselines shall be maintained.

12. Error Monitoring Requirements

Error monitoring shall track:

- Error Rate
- Error Volume
- Error Categories
- Recovery Success

Critical errors shall trigger alerts.

13. Security Monitoring Requirements

Security monitoring shall track:

- Authentication Failures
- Authorization Failures
- Security Events
- Access Violations

Security alerts shall integrate with incident management processes.

14. Governance Monitoring Requirements

Governance monitoring shall track:

- Governance Validation Failures
- Policy Violations
- Exception Requests
- Governance Approvals

Governance metrics shall support oversight reporting.

15. Audit Monitoring Requirements

Audit monitoring shall track:

- Audit Event Volume
- Audit Integrity Validation
- Audit Export Activity
- Compliance Evidence Collection

Audit telemetry shall remain immutable.

16. Replay Monitoring Requirements

Replay monitoring shall track:

- Replay Requests
- Replay Execution Success
- Replay Failures
- Replay Reconstruction Accuracy

Replay telemetry shall support investigations.

17. Alerting Framework

The monitoring platform shall support:

- Informational Alerts
- Warning Alerts
- Critical Alerts
- Security Alerts
- Governance Alerts

Alert ownership shall be assigned.

18. Alert Escalation Requirements

Alert escalation shall support:

- Operational Escalation
- Security Escalation
- Governance Escalation
- Executive Escalation

Escalation paths shall be documented.

19. Dashboard Requirements

Dashboards shall support:

- Executive Reporting
- Operational Monitoring
- Security Monitoring
- Governance Monitoring
- Compliance Monitoring

Dashboards shall be role-based.

20. Telemetry Retention Requirements

Telemetry retention shall include:

- Metrics Retention
- Log Retention
- Trace Retention
- Alert Retention

Retention periods shall align with governance policies.

21. Operational Analytics Requirements

Operational analytics shall support:

- Trend Analysis
- Capacity Analysis
- Reliability Analysis
- Failure Analysis

Analytics outputs shall support decision-making.

22. Capacity Monitoring Requirements

Capacity monitoring shall track:

- API Growth
- Resource Utilization
- Storage Utilization
- Processing Utilization

Capacity risks shall be identified proactively.

23. Dependency Monitoring Requirements

Dependency monitoring shall track:

- Internal Services
- External Services
- Databases
- Infrastructure Components

Dependency failures shall be visible.

24. Incident Investigation Support

Observability controls shall support:

- Root Cause Analysis
- Timeline Reconstruction
- Dependency Analysis
- Replay Investigations

Investigation support shall be documented.

25. Governance Requirements

Governance validation shall apply to:

- Monitoring configuration changes
- Alert rule modifications
- Dashboard modifications
- Telemetry retention changes

Governance approvals shall be retained.

26. Audit Requirements

Audit records shall be generated for:

- Monitoring configuration changes
- Alert modifications
- Dashboard changes
- Retention changes
- Observability governance actions

Audit records shall be immutable.

27. Compliance Requirements

Observability controls shall support:

- SOC 2 monitoring controls
- ISO 27001 logging controls
- ISO 27001 monitoring controls
- Enterprise governance standards

Compliance mappings shall be maintained.

28. Architectural Constraints

All APIs shall comply with:

OBS-CON-001

Generate telemetry.

OBS-CON-002

Support distributed tracing.

OBS-CON-003

Maintain structured logging.

OBS-CON-004

Support operational monitoring.

OBS-CON-005

Support governance oversight.

OBS-CON-006

Support auditability.

OBS-CON-007

Support replay investigations.

OBS-CON-008

Support enterprise scalability.

29. Metrics

The following observability metrics shall be monitored.

MET-OBS-001

Telemetry coverage percentage.

MET-OBS-002

Tracing coverage percentage.

MET-OBS-003

Monitoring availability.

MET-OBS-004

Alert response time.

MET-OBS-005

Mean time to detect.

MET-OBS-006

Mean time to investigate.

MET-OBS-007

Mean time to recover.

MET-OBS-008

Observability compliance rate.

30. Section Summary

This section defines the standards governing API monitoring, observability, telemetry collection, logging, distributed tracing, alerting, dashboard management, operational analytics, dependency monitoring, investigation support, governance oversight, audit requirements, and compliance controls. These standards ensure that all TODP APIs remain observable, diagnosable, traceable, auditable, secure, and operationally manageable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 18: API Integration & External Connectivity Specification

Document ID: API-V1-S18

Document Title: API Integration & External Connectivity Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing external integrations, third-party connectivity, partner interfaces, enterprise system interoperability, data exchange controls, integration governance, auditability, and replay support within the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all external connectivity remains secure, governed, traceable, auditable, and aligned with enterprise architecture principles.

2. Objectives

OBJ-API-INT-001

Establish controlled integration standards.

OBJ-API-INT-002

Support secure external connectivity.

OBJ-API-INT-003

Enable enterprise interoperability.

OBJ-API-INT-004

Maintain integration traceability.

OBJ-API-INT-005

Support governance-controlled integrations.

OBJ-API-INT-006

Support audit and compliance requirements.

OBJ-API-INT-007

Enable replay-assisted investigations.

OBJ-API-INT-008

Provide scalable integration capabilities.

3. Scope

This specification applies to:

- External systems
- Enterprise platforms
- Third-party services
- Partner integrations
- Future customer integrations
- Future ecosystem integrations

The standards apply across all TODP modules.

4. Integration Principles

All integrations shall comply with the following principles.

INT-PRN-001

Security First

All integrations shall be secured.

INT-PRN-002

Governed Connectivity

All integrations shall be approved.

INT-PRN-003

Traceability

Integration activity shall be traceable.

INT-PRN-004

Auditability

Integration activity shall be auditable.

INT-PRN-005

Replayability

Integration activity shall support replay.

INT-PRN-006

Reliability

Integrations shall be operationally reliable.

INT-PRN-007

Interoperability

Standards shall support enterprise interoperability.

INT-PRN-008

Scalability

Integrations shall support growth.

5. Integration Classification Framework

Integrations shall be classified as:

INT-CLASS-001

Internal Enterprise Integration

INT-CLASS-002

Third-Party Integration

INT-CLASS-003

Partner Integration

INT-CLASS-004

Customer Integration

INT-CLASS-005

Compliance Integration

INT-CLASS-006

Operational Integration

Each integration shall have an approved classification.

6. Integration Registration Requirements

All integrations shall be registered.

Required Attributes

- Integration ID
- Integration Name
- Integration Owner
- Integration Classification
- Approval Status

Registration shall be mandatory before production use.

7. Integration Approval Process

The following approvals shall be required:

- Architecture Review
- Security Review
- Governance Review
- Operational Review

Approval records shall be retained.

8. External Connectivity Standards

External connectivity shall support:

- Secure communication
- Identity verification
- Data protection
- Monitoring controls

Connectivity requirements shall be documented.

9. Data Exchange Standards

All integrations shall use approved data exchange mechanisms.

Supported Formats

- JSON
- CSV
- Structured Export Packages

Alternative formats require approval.

10. Integration Authentication Standards

External integrations shall support:

- Service Authentication
- Token-Based Authentication
- Certificate-Based Authentication

Authentication controls shall align with SEC V1.

11. Integration Authorization Standards

Authorization requirements shall include:

- Resource-Level Access
- Role-Based Access
- Governance Validation

Authorization decisions shall be auditable.

12. Integration Security Controls

Required Controls

- Encryption in Transit
- Identity Validation
- Access Controls
- Security Monitoring

Security controls shall be validated regularly.

13. Integration Traceability Requirements

All integration activity shall generate:

- Request ID
- Correlation ID
- Integration ID
- Resource Reference
- Timestamp

Traceability records shall support investigations.

14. Integration Audit Requirements

Audit records shall be generated for:

- Integration Creation
- Configuration Changes
- Data Exchanges
- Authentication Events
- Authorization Events

Audit records shall be immutable.

15. Integration Monitoring Requirements

Monitoring shall include:

- Availability Monitoring
- Performance Monitoring
- Security Monitoring
- Dependency Monitoring

Monitoring controls shall support operational management.

16. Integration Error Handling Requirements

Errors shall support:

- Standardized Responses
- Traceability
- Audit Logging
- Recovery Procedures

Error handling shall align with API-V1-S15.

17. Integration Governance Requirements

Governance validation shall be enforced for:

- New Integrations
- Integration Changes
- Connectivity Modifications
- Data Sharing Changes

Governance exceptions require approval.

18. Integration Change Management

All integration changes shall require:

- Impact Assessment
- Security Review
- Governance Approval
- Documentation Updates

Changes shall be traceable.

19. Integration Lifecycle States

Lifecycle States

Proposed

Review

Approved

Implemented

Testing

Production

Deprecated

Retired

Archived

Lifecycle transitions shall be governed.

20. Dependency Management Requirements

Integration dependencies shall be documented.

Required Attributes

- Dependency Owner
- Dependency Type
- Criticality Level
- Recovery Strategy

Dependency records shall be maintained.

21. Replay Support Requirements

Replay support shall preserve:

- Integration Requests
- Integration Responses
- Authentication Context
- Governance Context

Replay reconstruction shall remain supported.

22. Operational Resilience Requirements

Integrations shall support:

- Retry Mechanisms
- Timeout Controls
- Failure Detection
- Recovery Procedures

Operational resilience shall be validated.

23. Compliance Requirements

Integrations shall support:

- SOC 2 requirements
- ISO 27001 requirements
- Customer contractual obligations
- Enterprise governance requirements

Compliance mappings shall be maintained.

24. Investigation Support Requirements

Integration telemetry shall support:

- Root Cause Analysis
- Timeline Reconstruction
- Dependency Analysis
- Replay Investigations

Investigation support shall be documented.

25. Data Protection Requirements

Integration data exchanges shall support:

- Data Classification Controls
- Retention Controls

- Encryption Controls
- Governance Controls

Data protection requirements shall be enforced.

26. Architectural Constraints

All integrations shall comply with:

INT-CON-001

Require registration.

INT-CON-002

Require governance approval.

INT-CON-003

Support traceability.

INT-CON-004

Support auditability.

INT-CON-005

Support replayability.

INT-CON-006

Maintain security controls.

INT-CON-007

Support operational monitoring.

INT-CON-008

Support enterprise interoperability.

27. Metrics

The following integration metrics shall be monitored.

MET-INT-001

Integration count.

MET-INT-002

Integration availability.

MET-INT-003

Integration latency.

MET-INT-004

Integration error rate.

MET-INT-005

Integration approval cycle time.

MET-INT-006

Security incident count.

MET-INT-007

Replay coverage.

MET-INT-008

Integration compliance rate.

28. Audit Controls

Integration audits shall verify:

- Registration compliance
- Security compliance
- Governance compliance
- Monitoring coverage
- Replay readiness
- Dependency management

Audit findings shall be remediated.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

RTM V1

SEC V1

GOV V1

AUD V1

REL V1

30. Section Summary

This section defines the standards governing API integrations and external connectivity, including registration requirements, security controls, authentication standards, authorization controls, monitoring requirements, governance approvals, dependency management, replay support, audit obligations, and compliance requirements. These controls ensure that all TODP integrations remain secure, governed, traceable, auditable, reliable, and enterprise-ready throughout their lifecycle.

API SPEC V1 – Enterprise API Specification

Section 19: API Data Contracts & Schema Governance Specification

Document ID: API-V1-S19

Document Title: API Data Contracts & Schema Governance Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API data contracts, schema management, schema evolution, contract validation, interoperability controls, governance approvals, auditability requirements, and replay preservation across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all API payloads remain consistent, predictable, version-controlled, traceable, and compatible with enterprise integration requirements.

2. Objectives

OBJ-API-SCH-001

Establish standardized API contracts.

OBJ-API-SCH-002

Support controlled schema evolution.

OBJ-API-SCH-003

Maintain interoperability consistency.

OBJ-API-SCH-004

Support governance-controlled schema changes.

OBJ-API-SCH-005

Maintain auditability of contract changes.

OBJ-API-SCH-006

Enable replay reconstruction.

OBJ-API-SCH-007

Support enterprise integrations.

OBJ-API-SCH-008

Ensure long-term schema sustainability.

3. Scope

This specification applies to:

- Request schemas
- Response schemas
- Shared contracts
- Domain contracts
- Integration contracts
- Governance contracts
- Audit contracts
- Replay contracts

The standards apply across all TODP APIs.

4. Contract Governance Principles

All contracts shall comply with the following principles.

SCH-PRN-001

Consistency

Schemas shall follow common standards.

SCH-PRN-002

Version Control

Schema changes shall be versioned.

SCH-PRN-003

Traceability

Contract changes shall be traceable.

SCH-PRN-004

Auditability

Contract changes shall be auditable.

SCH-PRN-005

Governance

Contract changes shall be governed.

SCH-PRN-006

Compatibility

Consumer impact shall be evaluated.

SCH-PRN-007

Replayability

Historical contracts shall be preserved.

SCH-PRN-008

Interoperability

Schemas shall support enterprise interoperability.

5. Contract Classification Framework

Contracts shall be classified as:

CONTRACT-001

Core Domain Contract

CONTRACT-002

Shared Platform Contract

CONTRACT-003

Integration Contract

CONTRACT-004

Governance Contract

CONTRACT-005

Audit Contract

CONTRACT-006

Replay Contract

Each contract shall have an assigned classification.

6. Schema Repository Requirements

All schemas shall be maintained within a controlled repository.

Repository Controls

- Version history
- Change history
- Approval records
- Contract lineage

Repository integrity shall be protected.

7. Contract Registration Requirements

Each contract shall contain:

- Contract Identifier
- Contract Name
- Contract Owner
- Version Identifier
- Approval Status

Registration shall be mandatory.

8. Request Schema Standards

Request schemas shall define:

- Required fields
- Optional fields
- Validation rules
- Metadata requirements

Request structures shall be documented.

9. Response Schema Standards

Response schemas shall define:

- Status structure
- Data structure
- Metadata structure
- Traceability structure

Response structures shall remain consistent.

10. Shared Metadata Standards

All contracts shall support:

- Request ID
- Correlation ID
- Timestamp
- Service Identifier
- Version Identifier

Metadata consistency shall be maintained.

11. Identifier Standards

Identifiers shall be:

- Globally unique
- Immutable
- Traceable
- Version independent

Identifier governance shall be enforced.

12. Schema Validation Requirements

Validation shall include:

- Structural validation
- Data type validation

- Business rule validation
- Governance validation

Validation requirements shall be documented.

13. Schema Compatibility Requirements

Compatibility reviews shall evaluate:

- Backward compatibility
- Forward compatibility
- Consumer impact
- Integration impact

Compatibility findings shall be retained.

14. Schema Evolution Framework

Schema evolution shall support:

- Additive changes
- Controlled modifications
- Deprecation procedures
- Retirement procedures

Evolution activities shall be governed.

15. Contract Approval Requirements

Contract approvals shall require:

- Architecture review
- Security review
- Governance review
- Owner approval

Approval records shall be maintained.

16. Schema Change Management

Schema changes shall require:

- Impact assessment
- Compatibility analysis
- Approval workflow
- Documentation updates

Changes shall be traceable.

17. Governance Contract Standards

Governance contracts shall support:

- Policy references
- Governance identifiers
- Approval references
- Exception references

Governance metadata shall remain traceable.

18. Audit Contract Standards

Audit contracts shall support:

- Audit identifiers
- Event references
- Correlation identifiers
- Retention references

Audit metadata shall be immutable.

19. Replay Contract Standards

Replay contracts shall support:

- Historical references
- Version references
- Replay identifiers
- Reconstruction metadata

Replay metadata shall support historical reconstruction.

20. Security Contract Standards

Security contracts shall support:

- Authentication context
- Authorization context
- Security policy references
- Security event references

Security metadata shall remain auditable.

21. Integration Contract Standards

Integration contracts shall support:

- External identifiers
- Dependency references
- Connectivity metadata
- Exchange metadata

Integration contracts shall remain governed.

22. Data Type Governance Requirements

Approved data types shall be documented.

Governed Areas

- Numeric values
- Text values
- Date values
- Boolean values
- Enumerations

Data type consistency shall be maintained.

23. Enumerated Value Governance

Enumerations shall support:

- Version control
- Approval workflows
- Documentation controls

Enumeration changes shall be governed.

24. Replay Preservation Requirements

Historical schemas shall be preserved.

Preserved Artifacts

- Schema versions
- Contract versions
- Validation rules
- Governance approvals

Replay reconstruction shall remain possible.

25. Audit Requirements

Audit records shall be generated for:

- Contract creation
- Contract modification
- Schema approval
- Schema retirement
- Validation activities

Audit records shall be immutable.

26. Compliance Requirements

Schema governance shall support:

- SOC 2 change management controls
- ISO 27001 configuration controls
- ISO 27001 integrity controls
- Enterprise governance standards

Compliance mappings shall be maintained.

27. Architectural Constraints

All contracts shall comply with:

SCH-CON-001

Require registration.

SCH-CON-002

Require version control.

SCH-CON-003

Require governance approval.

SCH-CON-004

Support traceability.

SCH-CON-005

Support auditability.

SCH-CON-006

Support replayability.

SCH-CON-007

Support interoperability.

SCH-CON-008

Support enterprise sustainability.

28. Metrics

The following schema governance metrics shall be monitored.

MET-SCH-001

Registered contract count.

MET-SCH-002

Schema change volume.

MET-SCH-003

Compatibility issue count.

MET-SCH-004

Approval cycle time.

MET-SCH-005

Validation success rate.

MET-SCH-006

Replay schema coverage.

MET-SCH-007

Contract compliance rate.

MET-SCH-008

Schema audit coverage.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

RTM V1

SEC V1

GOV V1

AUD V1

REL V1

30. Section Summary

This section defines the standards governing API data contracts and schema governance, including schema management, contract registration, validation controls, compatibility management, contract evolution, governance approvals, replay preservation, audit obligations, and compliance requirements. These controls ensure that TODP API contracts remain consistent, governed, traceable, auditable, interoperable, and sustainable throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 20: API Security Controls & Protection Specification

Document ID: API-V1-S20

Document Title: API Security Controls & Protection Specification

Version: 1.0

Status: Approved

Owner: Platform Security Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the security controls, protection mechanisms, defensive requirements, threat mitigation controls, security monitoring standards, governance requirements, audit obligations, and compliance controls applicable to all APIs within the Transparent Opportunity Decision Platform (TODP).

This specification establishes the minimum security baseline required to protect API endpoints, data exchanges, platform services, governance workflows, audit records, and replay functions from unauthorized access, misuse, compromise, or operational abuse.

2. Objectives

OBJ-API-PROT-001

Establish enterprise API security controls.

OBJ-API-PROT-002

Protect platform resources and services.

OBJ-API-PROT-003

Mitigate API-based threats.

OBJ-API-PROT-004

Support governance-controlled security operations.

OBJ-API-PROT-005

Maintain security traceability.

OBJ-API-PROT-006

Support audit and compliance requirements.

OBJ-API-PROT-007

Enable security investigation support.

OBJ-API-PROT-008

Maintain enterprise security readiness.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply across all environments.

4. Security Principles

All APIs shall comply with the following principles.

SEC-PRN-001

Least Privilege

Access shall be limited to required permissions.

SEC-PRN-002

Defense in Depth

Multiple layers of protection shall be implemented.

SEC-PRN-003

Zero Trust Validation

Requests shall be validated continuously.

SEC-PRN-004

Governed Security

Security controls shall be governed.

SEC-PRN-005

Auditability

Security activity shall be auditable.

SEC-PRN-006

Traceability

Security actions shall be traceable.

SEC-PRN-007

Replayability

Security events shall support replay investigations.

SEC-PRN-008

Continuous Monitoring

Security posture shall be continuously monitored.

5. Authentication Controls

Authentication controls shall include:

- User authentication
- Service authentication
- Session validation
- Token validation

Authentication requirements shall align with API-V1-S13.

6. Authorization Controls

Authorization controls shall support:

- Role-based access control
- Permission validation
- Resource-level authorization
- Governance-controlled authorization

Authorization decisions shall be auditable.

7. Identity Verification Requirements

Identity verification shall support:

- Unique subject identification
- Service identity validation
- Credential validation

Identity verification failures shall be logged.

8. Session Security Requirements

Sessions shall support:

- Session expiration
- Session termination
- Session monitoring
- Session auditing

Session controls shall be enforced consistently.

9. Token Security Requirements

Tokens shall support:

- Expiration controls
- Revocation controls
- Validation controls
- Traceability controls

Token activity shall be monitored.

10. Data Protection Requirements

API data exchanges shall support:

- Data classification controls
- Encryption controls
- Access controls
- Retention controls

Protected data shall be handled according to governance policies.

11. Encryption Requirements

Encryption shall be enforced for:

- API communications
- Authentication exchanges
- Sensitive metadata
- Governance data

Encryption standards shall align with SEC V1.

12. Input Validation Controls

All API inputs shall be validated.

Validation Areas

- Data format validation
- Schema validation
- Business rule validation
- Governance validation

Invalid requests shall be rejected.

13. Output Validation Controls

API responses shall support:

- Schema validation
- Data integrity validation
- Security validation

Responses shall not expose restricted information.

14. API Abuse Protection Requirements

Protection controls shall support:

- Excessive request detection
- Automated abuse detection
- Access anomaly detection

Protection events shall be logged.

15. Security Event Management

Security events shall be classified.

Event Categories

- Authentication Events
- Authorization Events
- Governance Violations
- Access Violations
- Administrative Security Events

Events shall be auditable.

16. Security Monitoring Requirements

Monitoring shall include:

- Access monitoring
- Authentication monitoring
- Authorization monitoring
- Security event monitoring

Security telemetry shall be collected continuously.

17. Threat Detection Requirements

Threat detection shall support:

- Unauthorized access attempts
- Abnormal usage patterns
- Security policy violations
- Privilege misuse

Threat events shall trigger investigation workflows.

18. Governance Security Requirements

Governance validation shall apply to:

- Security configuration changes
- Authorization policy changes
- Administrative security actions
- Security exceptions

Governance approvals shall be retained.

19. Security Exception Management

Security exceptions shall require:

- Business justification
- Risk assessment
- Governance approval
- Expiration controls

Security exceptions shall remain traceable.

20. Security Audit Requirements

Audit records shall be generated for:

- Authentication events
- Authorization events
- Security policy changes
- Security exceptions
- Security investigations

Audit records shall be immutable.

21. Incident Investigation Support

Security telemetry shall support:

- Root cause analysis
- Timeline reconstruction
- Correlation analysis
- Replay investigations

Investigation support shall be documented.

22. Security Logging Requirements

Security logs shall contain:

- Timestamp
- Request ID
- Correlation ID
- Subject ID
- Security Event ID

Logs shall support investigations.

23. Replay Preservation Requirements

Replay support shall preserve:

- Authentication context
- Authorization context
- Security events
- Security policy versions

Replay reconstruction shall remain supported.

24. Operational Security Requirements

Operational controls shall support:

- Security monitoring
- Security reporting
- Incident response
- Recovery procedures

Operational security readiness shall be maintained.

25. Compliance Requirements

Security controls shall support:

- SOC 2 Security Criteria
- ISO 27001 Security Controls
- ISO 27001 Access Controls
- Enterprise Security Standards

Compliance mappings shall be maintained.

26. Architectural Constraints

All APIs shall comply with:

SEC-CON-001

Require authentication.

SEC-CON-002

Require authorization.

SEC-CON-003

Require traceability.

SEC-CON-004

Require auditability.

SEC-CON-005

Support governance controls.

SEC-CON-006

Support replay investigations.

SEC-CON-007

Support continuous monitoring.

SEC-CON-008

Support enterprise security requirements.

27. Metrics

The following security metrics shall be monitored.

MET-SEC-001

Authentication success rate.

MET-SEC-002

Authorization success rate.

MET-SEC-003

Security event volume.

MET-SEC-004

Security violation count.

MET-SEC-005

Security exception volume.

MET-SEC-006

Incident response time.

MET-SEC-007

Replay investigation coverage.

MET-SEC-008

Security compliance rate.

28. Audit Controls

Security audits shall verify:

- Authentication controls
- Authorization controls
- Monitoring coverage
- Security logging
- Governance compliance
- Replay readiness

Audit findings shall be remediated.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

RTM V1

SEC V1

GOV V1

AUD V1

REL V1

30. Section Summary

This section defines the enterprise security controls governing TODP APIs, including authentication requirements, authorization controls, encryption standards, input validation, output validation, security monitoring, threat detection, governance oversight, audit obligations, replay preservation, and compliance requirements. These controls ensure that all TODP APIs remain protected, governed, traceable, auditable, secure, resilient, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 21: API Performance, Scalability & Capacity Management Specification

Document ID: API-V1-S21

Document Title: API Performance, Scalability & Capacity Management Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API performance management, scalability planning, capacity management, workload optimization, growth readiness, operational resilience, governance controls, audit obligations, and compliance requirements across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that TODP APIs remain performant, scalable, reliable, and capable of supporting enterprise growth while preserving transparency, governance, traceability, auditability, and replayability.

2. Objectives

OBJ-API-PERF-001

Establish API performance standards.

OBJ-API-PERF-002

Support enterprise scalability requirements.

OBJ-API-PERF-003

Enable proactive capacity management.

OBJ-API-PERF-004

Maintain operational reliability.

OBJ-API-PERF-005

Support governance-controlled scaling decisions.

OBJ-API-PERF-006

Support audit and compliance requirements.

OBJ-API-PERF-007

Enable performance investigation support.

OBJ-API-PERF-008

Maintain long-term platform sustainability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The standards apply across all deployment environments.

4. Performance Principles

All APIs shall comply with the following principles.

PERF-PRN-001

Efficiency

Resources shall be utilized efficiently.

PERF-PRN-002

Scalability

Services shall support growth.

PERF-PRN-003

Reliability

Performance shall remain predictable.

PERF-PRN-004

Observability

Performance shall be measurable.

PERF-PRN-005

Governance

Scaling decisions shall be governed.

PERF-PRN-006

Auditability

Performance changes shall be auditable.

PERF-PRN-007

Replayability

Performance events shall support replay investigations.

PERF-PRN-008

Sustainability

Capacity planning shall support long-term growth.

5. Performance Management Framework

The framework shall govern:

- Response latency
- Throughput
- Resource utilization
- Service efficiency
- Capacity consumption

Performance governance shall be documented.

6. Response Latency Requirements

Performance monitoring shall track:

- Average latency
- Peak latency
- Percentile latency
- Service latency

Latency thresholds shall be established for each API category.

7. Throughput Requirements

Throughput monitoring shall track:

- Requests per second
- Transactions per minute
- Peak transaction volume

Throughput targets shall be documented.

8. Capacity Management Framework

Capacity management shall include:

- Growth forecasting
- Utilization monitoring
- Capacity planning
- Capacity validation

Capacity reviews shall occur regularly.

9. Scalability Requirements

Scalability shall support:

- User growth
- Opportunity growth
- Evidence growth
- Governance growth
- Audit growth

Scalability assumptions shall be documented.

10. Resource Utilization Monitoring

Monitoring shall include:

- CPU utilization
- Memory utilization
- Storage utilization
- Network utilization

Resource consumption trends shall be tracked.

11. Concurrency Management Requirements

APIs shall support:

- Concurrent request handling
- Workload balancing
- Resource isolation

Concurrency controls shall be validated.

12. Load Management Requirements

Load management shall support:

- Predictable growth
- Peak demand events
- Operational surges

Load behavior shall be monitored continuously.

13. Performance Testing Requirements

Testing shall include:

- Load testing
- Stress testing
- Scalability testing
- Capacity testing

Test results shall be retained.

14. Bottleneck Identification Requirements

Performance analysis shall identify:

- Service bottlenecks
- Dependency bottlenecks
- Infrastructure bottlenecks

Findings shall be documented.

15. Service Scaling Requirements

Scaling shall support:

- Horizontal scaling
- Controlled capacity expansion
- Operational continuity

Scaling decisions shall be governed.

16. Dependency Capacity Requirements

Dependencies shall be monitored for:

- Availability
- Capacity
- Utilization
- Performance

Dependency risks shall be tracked.

17. Operational Resilience Requirements

Resilience controls shall support:

- Service continuity
- Performance recovery
- Resource recovery
- Failure isolation

Operational resilience shall be validated.

18. Growth Forecasting Requirements

Forecasting shall consider:

- User adoption
- Opportunity volume
- Evidence volume
- Governance activity

Forecast assumptions shall be documented.

19. Performance Optimization Requirements

Optimization activities shall support:

- Response improvement
- Resource efficiency
- Capacity efficiency

Optimization changes shall be governed.

20. Governance Requirements

Governance validation shall apply to:

- Capacity expansions
- Scalability changes
- Major performance optimizations
- Infrastructure growth decisions

Governance approvals shall be retained.

21. Audit Requirements

Audit records shall be generated for:

- Capacity changes
- Scalability changes
- Performance optimization actions
- Infrastructure modifications

Audit records shall be immutable.

22. Observability Requirements

Performance observability shall support:

- Metrics collection
- Alerting
- Trend analysis
- Investigation support

Observability controls shall align with API-V1-S17.

23. Security Considerations

Performance controls shall not bypass:

- Security controls
- Governance controls
- Audit controls
- Compliance requirements

Security requirements shall remain enforceable.

24. Replay Preservation Requirements

Replay support shall preserve:

- Performance metrics
- Capacity decisions
- Scaling events
- Optimization activities

Historical performance investigations shall remain possible.

25. Compliance Requirements

Performance governance shall support:

- SOC 2 availability controls
- ISO 27001 operational controls

- ISO 27001 capacity management controls
- Enterprise governance standards

Compliance mappings shall be maintained.

26. Architectural Constraints

All APIs shall comply with:

PERF-CON-001

Support observability.

PERF-CON-002

Support scalability.

PERF-CON-003

Support auditability.

PERF-CON-004

Support governance oversight.

PERF-CON-005

Support replay investigations.

PERF-CON-006

Maintain operational resilience.

PERF-CON-007

Support sustainable growth.

PERF-CON-008

Preserve security controls.

27. Metrics

The following performance metrics shall be monitored.

MET-PERF-001

Average response latency.

MET-PERF-002

Peak response latency.

MET-PERF-003

Request throughput.

MET-PERF-004

Resource utilization.

MET-PERF-005

Capacity utilization.

MET-PERF-006

Scalability readiness score.

MET-PERF-007

Performance incident count.

MET-PERF-008

Capacity forecast accuracy.

28. Audit Controls

Performance audits shall verify:

- Capacity planning compliance
- Scalability readiness
- Testing coverage
- Optimization governance
- Replay readiness
- Monitoring coverage

Audit findings shall be remediated.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the standards governing API performance, scalability, capacity management, growth forecasting, optimization activities, resilience controls, governance oversight, audit obligations, observability requirements, replay preservation, and compliance controls. These standards ensure that TODP APIs remain performant, scalable, reliable, sustainable, governed, traceable, auditable, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 23: API Compliance & Regulatory Controls Specification

Document ID: API-V1-S23

Document Title: API Compliance & Regulatory Controls Specification

Version: 1.0

Status: Approved

Owner: Compliance and Governance Office

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the compliance, regulatory, governance, audit, and control requirements applicable to all APIs within the Transparent Opportunity Decision Platform (TODP).

This specification establishes the framework for ensuring that API operations, data exchanges, governance activities, audit processes, security controls, and replay capabilities remain compliant with enterprise governance requirements and applicable regulatory obligations.

2. Objectives

OBJ-API-COMP-001

Establish enterprise compliance controls.

OBJ-API-COMP-002

Support regulatory readiness.

OBJ-API-COMP-003

Enable governance oversight.

OBJ-API-COMP-004

Maintain auditability of API activities.

OBJ-API-COMP-005

Support evidence-based compliance verification.

OBJ-API-COMP-006

Enable compliance investigations.

OBJ-API-COMP-007

Support replay-based compliance reconstruction.

OBJ-API-COMP-008

Maintain enterprise trust and accountability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply across all deployment environments and operational states.

4. Compliance Principles

All APIs shall comply with the following principles.

COMP-PRN-001

Governance First

Compliance obligations shall be governed.

COMP-PRN-002

Traceability

Compliance evidence shall be traceable.

COMP-PRN-003

Auditability

Compliance activities shall be auditable.

COMP-PRN-004

Transparency

Compliance controls shall be visible and documented.

COMP-PRN-005

Accountability

Compliance ownership shall be assigned.

COMP-PRN-006

Evidence Preservation

Compliance evidence shall be retained.

COMP-PRN-007

Replayability

Compliance decisions shall be reconstructable.

COMP-PRN-008

Continuous Validation

Compliance status shall be monitored continuously.

5. Compliance Governance Framework

The compliance framework shall include:

- Policy Management
- Control Management
- Compliance Monitoring
- Compliance Reporting
- Audit Support
- Governance Oversight

Compliance governance shall be documented.

6. Applicable Compliance Domains

The platform shall support:

- Security Compliance
- Governance Compliance
- Operational Compliance
- Audit Compliance
- Data Compliance
- Enterprise Policy Compliance

Compliance obligations shall be mapped.

7. SOC 2 Compliance Requirements

APIs shall support:

- Security Controls
- Availability Controls
- Processing Integrity Controls
- Monitoring Controls
- Audit Evidence Collection

SOC 2 mappings shall be maintained.

8. ISO 27001 Compliance Requirements

APIs shall support:

- Access Controls
- Change Controls
- Logging Controls
- Monitoring Controls
- Risk Controls

ISO 27001 mappings shall be maintained.

9. Governance Compliance Requirements

Governance controls shall support:

- Policy Validation
- Governance Reviews
- Approval Workflows
- Exception Management

Governance activities shall remain auditable.

10. Data Compliance Requirements

Data-related compliance shall support:

- Data Classification
- Data Retention
- Data Protection
- Data Lineage

Compliance controls shall be enforced.

11. Security Compliance Requirements

Security compliance shall support:

- Authentication Controls
- Authorization Controls
- Encryption Controls
- Monitoring Controls

Security compliance evidence shall be retained.

12. Audit Compliance Requirements

Audit compliance shall support:

- Audit Logging
- Audit Retention
- Audit Integrity
- Audit Traceability

Audit controls shall remain continuously operational.

13. Evidence Management Requirements

Compliance evidence shall support:

- Evidence Collection
- Evidence Preservation
- Evidence Validation
- Evidence Retrieval

Evidence integrity shall be maintained.

14. Compliance Reporting Requirements

Compliance reporting shall support:

- Executive Reporting
- Audit Reporting
- Governance Reporting
- Operational Reporting

Reports shall remain traceable.

15. Compliance Monitoring Requirements

Monitoring shall include:

- Control Effectiveness
- Compliance Status
- Exception Status
- Remediation Status

Monitoring shall be automated where possible.

16. Compliance Validation Requirements

Validation shall support:

- Control Validation
- Policy Validation
- Evidence Validation
- Governance Validation

Validation results shall be retained.

17. Exception Management Requirements

Compliance exceptions shall require:

- Business Justification
- Risk Assessment
- Governance Approval
- Expiration Controls

Exceptions shall remain traceable.

18. Remediation Management Requirements

Compliance remediation shall support:

- Issue Identification
- Corrective Actions
- Validation Activities
- Closure Approval

Remediation activities shall be documented.

19. Compliance Ownership Requirements

Compliance ownership shall be assigned for:

- Controls
- Policies
- Reports
- Evidence
- Exceptions

Ownership records shall remain current.

20. Investigation Support Requirements

Compliance telemetry shall support:

- Audit Investigations
- Governance Investigations
- Security Investigations
- Replay Investigations

Investigation readiness shall be maintained.

21. Governance Requirements

Governance validation shall apply to:

- Compliance control changes
- Compliance exceptions
- Compliance reporting changes
- Compliance policy changes

Governance approvals shall be retained.

22. Audit Requirements

Audit records shall be generated for:

- Compliance validations
- Compliance exceptions
- Compliance reports
- Compliance remediation activities
- Governance actions

Audit records shall be immutable.

23. Security Requirements

Compliance controls shall support:

- Access Controls
- Data Protection Controls
- Monitoring Controls
- Integrity Controls

Security requirements shall align with SEC V1.

24. Replay Preservation Requirements

Replay support shall preserve:

- Compliance evidence
- Compliance decisions
- Compliance exceptions
- Compliance reports

Historical compliance reconstruction shall remain possible.

25. Compliance Lifecycle Requirements

The compliance lifecycle shall include:

- Identification
- Assessment
- Validation
- Monitoring
- Reporting
- Remediation
- Closure

Lifecycle activities shall remain traceable.

26. Architectural Constraints

All APIs shall comply with:

COMP-CON-001

Support compliance monitoring.

COMP-CON-002

Support compliance reporting.

COMP-CON-003

Support governance oversight.

COMP-CON-004

Support auditability.

COMP-CON-005

Support evidence preservation.

COMP-CON-006

Support replay investigations.

COMP-CON-007

Support continuous validation.

COMP-CON-008

Support enterprise regulatory readiness.

27. Metrics

The following compliance metrics shall be monitored.

MET-COMP-001

Compliance control coverage.

MET-COMP-002

Compliance validation success rate.

MET-COMP-003

Compliance exception volume.

MET-COMP-004

Remediation completion rate.

MET-COMP-005

Compliance evidence coverage.

MET-COMP-006

Audit readiness score.

MET-COMP-007

Replay compliance coverage.

MET-COMP-008

Governance compliance rate.

28. Audit Controls

Compliance audits shall verify:

- Control effectiveness
- Evidence availability
- Monitoring coverage
- Reporting accuracy
- Governance compliance
- Replay readiness

Audit findings shall be remediated.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the compliance and regulatory controls governing TODP APIs, including compliance governance, regulatory readiness, monitoring controls, reporting requirements, evidence preservation, validation activities, exception management, remediation processes, audit obligations, replay preservation, and compliance verification. These controls ensure that all TODP APIs remain compliant, governed, traceable, auditable, accountable, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 24: API Business Continuity & Disaster Recovery Specification

Document ID: API-V1-S24

Document Title: API Business Continuity & Disaster Recovery Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing business continuity, disaster recovery, operational resilience, service restoration, data recovery, governance oversight, audit requirements, and replay preservation for all APIs within the Transparent Opportunity Decision Platform (TODP).

This specification ensures that API services remain recoverable, resilient, traceable, auditable, and operationally sustainable during disruptions, failures, incidents, and disaster scenarios.

2. Objectives

OBJ-API-BCDR-001

Establish API continuity controls.

OBJ-API-BCDR-002

Support disaster recovery readiness.

OBJ-API-BCDR-003

Enable resilient service operations.

OBJ-API-BCDR-004

Maintain governance oversight during disruptions.

OBJ-API-BCDR-005

Support auditability of recovery activities.

OBJ-API-BCDR-006

Preserve critical operational evidence.

OBJ-API-BCDR-007

Support replay-based incident reconstruction.

OBJ-API-BCDR-008

Maintain enterprise operational readiness.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply across all environments and infrastructure components supporting API operations.

4. Business Continuity Principles

All APIs shall comply with the following principles.

BCDR-PRN-001

Resilience

Critical services shall remain operational where possible.

BCDR-PRN-002

Recoverability

Services shall be recoverable.

BCDR-PRN-003

Governance

Recovery activities shall be governed.

BCDR-PRN-004

Traceability

Recovery actions shall be traceable.

BCDR-PRN-005

Auditability

Recovery activities shall be auditable.

BCDR-PRN-006

Integrity

Recovered services shall maintain integrity.

BCDR-PRN-007

Replayability

Disruption events shall support replay investigations.

BCDR-PRN-008

Preparedness

Recovery readiness shall be validated regularly.

5. Business Continuity Framework

The framework shall include:

- Service Continuity Planning
- Operational Resilience Controls
- Recovery Planning
- Incident Response Coordination
- Governance Oversight
- Compliance Validation

Continuity activities shall be documented.

6. Disaster Recovery Framework

The recovery framework shall support:

- Service Recovery
- Data Recovery
- Configuration Recovery
- Infrastructure Recovery

Recovery procedures shall be maintained.

7. Critical Service Classification

Services shall be classified according to:

- Critical
- High Priority
- Standard Priority
- Low Priority

Classification shall drive recovery prioritization.

8. Recovery Time Objective Requirements

Recovery planning shall define:

- Service Recovery Objectives
- Operational Recovery Objectives
- Dependency Recovery Objectives

Recovery objectives shall be documented and approved.

9. Recovery Point Objective Requirements

Recovery planning shall define:

- Data Recovery Objectives
- Audit Recovery Objectives
- Governance Recovery Objectives

Recovery objectives shall support business continuity requirements.

10. Backup Requirements

Backup controls shall support:

- Configuration Backups
- Metadata Backups
- Audit Backups
- Governance Backups

Backup activities shall be monitored.

11. Recovery Validation Requirements

Recovery validation shall include:

- Recovery Testing
- Data Integrity Validation
- Service Validation
- Governance Validation

Validation results shall be retained.

12. Service Restoration Requirements

Service restoration shall support:

- Controlled Recovery
- Service Verification
- Dependency Verification
- Operational Validation

Restoration activities shall be documented.

13. Data Recovery Requirements

Data recovery shall support:

- Data Integrity Validation
- Lineage Preservation
- Governance Preservation
- Audit Preservation

Recovered data shall remain trustworthy.

14. Dependency Recovery Requirements

Dependencies shall support:

- Recovery Procedures
- Recovery Prioritization
- Dependency Validation
- Operational Verification

Dependency recovery plans shall be maintained.

15. Operational Resilience Requirements

Resilience controls shall support:

- Fault Tolerance
- Service Redundancy
- Failure Isolation
- Controlled Recovery

Resilience controls shall be reviewed periodically.

16. Incident Coordination Requirements

Recovery activities shall integrate with:

- Incident Management
- Security Response
- Governance Oversight
- Audit Activities

Coordination procedures shall be documented.

17. Governance Requirements

Governance validation shall apply to:

- Recovery plan changes
- Recovery testing activities
- Continuity policy changes
- Recovery exceptions

Governance approvals shall be retained.

18. Audit Requirements

Audit records shall be generated for:

- Recovery events
- Recovery testing
- Continuity exercises
- Recovery approvals
- Governance actions

Audit records shall be immutable.

19. Security Requirements

Recovery controls shall support:

- Access Controls
- Data Protection Controls
- Recovery Authentication
- Recovery Authorization

Security requirements shall align with SEC V1.

20. Compliance Requirements

Business continuity controls shall support:

- SOC 2 Availability Criteria
- ISO 27001 Continuity Controls
- ISO 27001 Recovery Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

21. Recovery Testing Requirements

Testing activities shall include:

- Disaster Recovery Testing
- Service Restoration Testing
- Backup Restoration Testing
- Governance Recovery Testing

Testing results shall be retained.

22. Continuity Exercise Requirements

Exercises shall support:

- Operational Readiness
- Recovery Readiness
- Governance Readiness
- Investigation Readiness

Exercises shall be documented.

23. Investigation Support Requirements

Recovery telemetry shall support:

- Incident Analysis
- Timeline Reconstruction
- Root Cause Analysis
- Replay Investigations

Investigation support shall remain available.

24. Replay Preservation Requirements

Replay support shall preserve:

- Recovery Events
- Continuity Events
- Recovery Decisions
- Governance Actions

Historical reconstruction shall remain possible.

25. Documentation Requirements

Recovery documentation shall include:

- Recovery Procedures
- Continuity Procedures
- Escalation Procedures
- Validation Procedures

Documentation shall remain current.

26. Architectural Constraints

All APIs shall comply with:

BCDR-CON-001

Support continuity planning.

BCDR-CON-002

Support disaster recovery.

BCDR-CON-003

Support governance oversight.

BCDR-CON-004

Support auditability.

BCDR-CON-005

Support recovery validation.

BCDR-CON-006

Support replay investigations.

BCDR-CON-007

Maintain operational resilience.

BCDR-CON-008

Support enterprise recovery readiness.

27. Metrics

The following continuity metrics shall be monitored.

MET-BCDR-001

Recovery success rate.

MET-BCDR-002

Recovery testing coverage.

MET-BCDR-003

Recovery validation success rate.

MET-BCDR-004

Backup success rate.

MET-BCDR-005

Service restoration success rate.

MET-BCDR-006

Recovery readiness score.

MET-BCDR-007

Replay recovery coverage.

MET-BCDR-008

Continuity compliance rate.

28. Audit Controls

Continuity audits shall verify:

- Recovery readiness
- Backup effectiveness
- Testing coverage
- Recovery governance
- Replay readiness
- Compliance alignment

Audit findings shall be remediated.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the business continuity and disaster recovery controls governing TODP APIs, including continuity planning, disaster recovery readiness, service restoration, recovery validation, backup controls, operational resilience, governance oversight, audit obligations, replay preservation, and compliance requirements. These controls ensure that TODP APIs remain resilient, recoverable, traceable, auditable, governed, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 25: API Change Management & Release Governance Specification

Document ID: API-V1-S25

Document Title: API Change Management & Release Governance Specification

Version: 1.0

Status: Approved

Owner: Platform Engineering

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API change management, release governance, deployment approvals, configuration control, impact assessment, rollback management, auditability, compliance validation, and replay preservation across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all API changes are introduced in a controlled, traceable, auditable, governed, and operationally safe manner.

2. Objectives

OBJ-API-REL-001

Establish controlled API change management.

OBJ-API-REL-002

Support governed release processes.

OBJ-API-REL-003

Enable impact-based change assessment.

OBJ-API-REL-004

Maintain release traceability.

OBJ-API-REL-005

Support auditability of changes.

OBJ-API-REL-006

Support compliance validation.

OBJ-API-REL-007

Enable replay reconstruction of changes.

OBJ-API-REL-008

Maintain enterprise operational stability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply across all environments and deployment activities.

4. Change Management Principles

All API changes shall comply with the following principles.

REL-PRN-001

Governance

Changes shall be governed.

REL-PRN-002

Traceability

Changes shall be traceable.

REL-PRN-003

Auditability

Changes shall be auditable.

REL-PRN-004

Risk Awareness

Changes shall be risk-assessed.

REL-PRN-005

Controlled Deployment

Changes shall be deployed in a controlled manner.

REL-PRN-006

Rollback Readiness

Rollback capability shall be maintained.

REL-PRN-007

Replayability

Changes shall support replay reconstruction.

REL-PRN-008

Operational Stability

Changes shall minimize disruption.

5. Change Classification Framework

Changes shall be classified as:

REL-CLASS-001

Standard Change

REL-CLASS-002

Normal Change

REL-CLASS-003

Major Change

REL-CLASS-004

Emergency Change

REL-CLASS-005

Security Change

REL-CLASS-006

Governance Change

Classification shall determine approval requirements.

6. Change Registration Requirements

All changes shall be registered.

Required Attributes

- Change ID
- Change Owner
- Change Type
- Change Description
- Impact Assessment
- Approval Status

Registration shall be mandatory.

7. Impact Assessment Requirements

Assessments shall evaluate:

- Business Impact
- Operational Impact
- Security Impact
- Governance Impact
- Compliance Impact

Assessment findings shall be documented.

8. Risk Assessment Requirements

Risk reviews shall evaluate:

- Service Risk
- Data Risk
- Security Risk
- Governance Risk
- Customer Risk

Risk ratings shall be assigned.

9. Change Approval Requirements

Approvals shall require:

- Technical Review
- Architecture Review
- Security Review
- Governance Review

Approval records shall be retained.

10. Release Planning Requirements

Release planning shall include:

- Scope Definition
- Risk Evaluation
- Rollback Planning
- Validation Planning

Release plans shall be approved.

11. Deployment Authorization Requirements

Deployment authorization shall verify:

- Approved Change Record
- Approved Release Plan

- Validation Readiness
- Rollback Readiness

Unauthorized deployments are prohibited.

12. Configuration Management Requirements

Configuration controls shall support:

- Configuration Versioning
- Configuration Traceability
- Configuration Validation
- Configuration Recovery

Configuration changes shall remain governed.

13. Release Validation Requirements

Validation activities shall include:

- Functional Validation
- Security Validation
- Governance Validation
- Operational Validation

Validation results shall be retained.

14. Release Testing Requirements

Testing shall support:

- Integration Testing
- Regression Testing
- Security Testing
- Operational Testing

Testing evidence shall be preserved.

15. Rollback Requirements

Rollback procedures shall support:

- Service Restoration
- Configuration Restoration
- Version Restoration
- Operational Recovery

Rollback readiness shall be validated.

16. Emergency Change Requirements

Emergency changes shall require:

- Expedited Review
- Post-Implementation Review
- Audit Review
- Governance Review

Emergency changes shall remain traceable.

17. Security Change Requirements

Security-related changes shall require:

- Security Assessment
- Security Approval
- Validation Testing
- Monitoring Validation

Security evidence shall be retained.

18. Governance Change Requirements

Governance-related changes shall require:

- Governance Impact Analysis
- Policy Review
- Governance Approval
- Traceability Validation

Governance changes shall remain auditable.

19. Release Communication Requirements

Release communication shall support:

- Stakeholder Notification
- Operational Notification
- Governance Notification
- Audit Notification

Communications shall be documented.

20. Post-Release Validation Requirements

Post-release activities shall include:

- Service Validation
- Monitoring Validation
- Security Validation
- Governance Validation

Validation outcomes shall be recorded.

21. Governance Requirements

Governance validation shall apply to:

- Change approvals
- Release approvals
- Rollback approvals
- Emergency changes

Governance decisions shall be retained.

22. Audit Requirements

Audit records shall be generated for:

- Change registrations
- Impact assessments
- Approvals
- Deployments
- Rollbacks
- Post-release reviews

Audit records shall be immutable.

23. Security Requirements

Change controls shall support:

- Access Control
- Approval Control
- Deployment Authorization
- Security Validation

Security requirements shall align with SEC V1.

24. Compliance Requirements

Release governance shall support:

- SOC 2 Change Management Controls
- ISO 27001 Change Control Requirements
- ISO 27001 Configuration Management Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

25. Replay Preservation Requirements

Replay support shall preserve:

- Change Records
- Approval Records
- Release Records
- Rollback Activities

Historical reconstruction shall remain possible.

26. Operational Readiness Requirements

Operational readiness shall verify:

- Monitoring Coverage
- Support Readiness
- Recovery Readiness
- Governance Readiness

Readiness evidence shall be retained.

27. Architectural Constraints

All APIs shall comply with:

REL-CON-001

Require registered changes.

REL-CON-002

Require approved releases.

REL-CON-003

Support traceability.

REL-CON-004

Support auditability.

REL-CON-005

Support governance oversight.

REL-CON-006

Support rollback capability.

REL-CON-007

Support replay investigations.

REL-CON-008

Maintain operational stability.

28. Metrics

The following release governance metrics shall be monitored.

MET-REL-001

Change volume.

MET-REL-002

Release success rate.

MET-REL-003

Rollback frequency.

MET-REL-004

Emergency change volume.

MET-REL-005

Post-release defect rate.

MET-REL-006

Approval cycle time.

MET-REL-007

Replay release coverage.

MET-REL-008

Release compliance rate.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the standards governing API change management and release governance, including change classification, impact assessments, risk evaluations, approvals, deployment controls, rollback procedures, validation activities, governance oversight, audit obligations, compliance requirements, and replay preservation. These controls ensure that all TODP API changes are introduced in a controlled, traceable, auditable, governed, secure, and operationally stable manner throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 26: API Data Retention, Archival & Records Management Specification

Document ID: API-V1-S26

Document Title: API Data Retention, Archival & Records Management Specification

Version: 1.0

Status: Approved

Owner: Data Governance Office

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API-related data retention, archival management, records preservation, evidence retention, historical reconstruction, governance controls, audit obligations, and compliance requirements across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all API-generated records remain available, protected, traceable, auditable, and compliant throughout their lifecycle.

2. Objectives

OBJ-API-RRM-001

Establish enterprise retention standards.

OBJ-API-RRM-002

Support regulatory and governance requirements.

OBJ-API-RRM-003

Enable controlled archival management.

OBJ-API-RRM-004

Preserve audit and compliance evidence.

OBJ-API-RRM-005

Maintain traceability of historical records.

OBJ-API-RRM-006

Support investigation readiness.

OBJ-API-RRM-007

Enable replay reconstruction.

OBJ-API-RRM-008

Maintain enterprise records integrity.

3. Scope

This specification applies to:

- API request records
- API response records
- Audit records
- Governance records
- Security records
- Replay records
- Administrative records
- Operational records

The controls apply across all TODP modules and environments.

4. Records Management Principles

All API-related records shall comply with the following principles.

RRM-PRN-001

Retention

Records shall be retained according to policy.

RRM-PRN-002

Integrity

Records shall remain accurate and complete.

RRM-PRN-003

Traceability

Records shall support lineage and traceability.

RRM-PRN-004

Auditability

Records shall support audits and investigations.

RRM-PRN-005

Governance

Retention activities shall be governed.

RRM-PRN-006

Security

Records shall be protected.

RRM-PRN-007

Replayability

Historical records shall support replay.

RRM-PRN-008

Controlled Disposal

Records shall be disposed of according to policy.

5. Records Classification Framework

Records shall be classified as:

RRM-CLASS-001

Operational Records

RRM-CLASS-002

Audit Records

RRM-CLASS-003

Governance Records

RRM-CLASS-004

Security Records

RRM-CLASS-005

Compliance Records

RRM-CLASS-006

Replay Records

Classification shall determine retention requirements.

6. Retention Governance Framework

Retention governance shall include:

- Retention Policies
- Classification Standards
- Retention Reviews
- Disposal Controls

Governance oversight shall be maintained.

7. API Request Record Requirements

Request records shall preserve:

- Request Metadata
- Request Context
- Request Ownership
- Request Timestamp

Request records shall remain traceable.

8. API Response Record Requirements

Response records shall preserve:

- Response Metadata
- Processing Outcome
- Service Information
- Response Timestamp

Response records shall remain reconstructable.

9. Audit Record Requirements

Audit records shall preserve:

- Audit Events
- Audit References
- Audit Lineage
- Audit Metadata

Audit records shall remain immutable.

10. Governance Record Requirements

Governance records shall preserve:

- Approvals
- Rejections
- Reviews
- Exceptions

Governance records shall remain searchable.

11. Security Record Requirements

Security records shall preserve:

- Authentication Events
- Authorization Events
- Security Violations
- Security Reviews

Security records shall support investigations.

12. Compliance Record Requirements

Compliance records shall preserve:

- Compliance Evidence
- Validation Results

- Compliance Reports
- Compliance Exceptions

Compliance records shall support audits.

13. Replay Record Requirements

Replay records shall preserve:

- Historical States
- Decision Context
- Processing Context
- Reconstruction Metadata

Replay records shall remain available.

14. Archival Requirements

Archival controls shall support:

- Historical Preservation
- Controlled Storage
- Retrieval Capabilities
- Integrity Validation

Archived records shall remain protected.

15. Archive Governance Requirements

Archive activities shall require:

- Ownership Assignment
- Retention Validation
- Governance Oversight
- Audit Review

Archive controls shall remain documented.

16. Record Retrieval Requirements

Retrieval capabilities shall support:

- Search by Identifier
- Search by Date
- Search by Resource
- Search by Event

Retrieval activities shall be logged.

17. Retention Review Requirements

Retention reviews shall evaluate:

- Policy Compliance
- Record Availability
- Integrity Validation
- Disposal Eligibility

Review findings shall be retained.

18. Disposal Requirements

Record disposal shall support:

- Approval Controls
- Traceability Controls
- Audit Logging
- Compliance Validation

Unauthorized disposal is prohibited.

19. Legal Hold Requirements

Legal hold controls shall support:

- Disposal Suspension
- Evidence Preservation
- Governance Oversight
- Audit Tracking

Legal holds shall be enforced when required.

20. Investigation Support Requirements

Records shall support:

- Root Cause Analysis
- Compliance Investigations
- Security Investigations
- Replay Investigations

Investigation readiness shall be maintained.

21. Governance Requirements

Governance validation shall apply to:

- Retention policy changes
- Archive policy changes
- Disposal approvals
- Legal hold actions

Governance approvals shall be retained.

22. Audit Requirements

Audit records shall be generated for:

- Retention reviews
- Archive activities
- Record retrievals
- Disposal activities
- Legal hold actions

Audit records shall be immutable.

23. Security Requirements

Records management controls shall support:

- Access Control
- Encryption
- Integrity Validation
- Monitoring

Security requirements shall align with SEC V1.

24. Compliance Requirements

Records management shall support:

- SOC 2 Evidence Retention Controls
- ISO 27001 Records Management Controls
- ISO 27001 Audit Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

25. Replay Preservation Requirements

Replay support shall preserve:

- Historical Records
- Historical Decisions
- Historical Governance Context
- Historical Audit Context

Historical reconstruction shall remain possible.

26. Operational Requirements

Operational controls shall support:

- Archive Monitoring
- Retrieval Monitoring
- Retention Monitoring
- Disposal Monitoring

Operational metrics shall be collected.

27. Architectural Constraints

All APIs shall comply with:

RRM-CON-001

Support retention policies.

RRM-CON-002

Support archival capabilities.

RRM-CON-003

Support traceability.

RRM-CON-004

Support auditability.

RRM-CON-005

Support governance oversight.

RRM-CON-006

Support replay investigations.

RRM-CON-007

Protect record integrity.

RRM-CON-008

Support enterprise compliance requirements.

28. Metrics

The following records management metrics shall be monitored.

MET-RRM-001

Record retention compliance rate.

MET-RRM-002

Archive success rate.

MET-RRM-003

Record retrieval success rate.

MET-RRM-004

Retention review completion rate.

MET-RRM-005

Legal hold compliance rate.

MET-RRM-006

Record integrity validation rate.

MET-RRM-007

Replay record coverage.

MET-RRM-008

Records management audit score.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

RTM V1

DATA GOVERNANCE V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the standards governing API data retention, archival management, records preservation, disposal controls, legal hold requirements, governance oversight, audit obligations, compliance controls, investigation support, and replay preservation. These controls ensure that all TODP API records remain protected, traceable, auditable, compliant, recoverable, and enterprise-ready throughout the information lifecycle.

API SPEC V1 – Enterprise API Specification

Section 27: API Service Level Management & Operational Commitments Specification

Document ID: API-V1-S27

Document Title: API Service Level Management & Operational Commitments Specification

Version: 1.0

Status: Approved

Owner: Platform Operations Team

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API service levels, operational commitments, availability objectives, performance commitments, service monitoring, governance oversight, audit requirements, and compliance obligations across the Transparent Opportunity Decision Platform (TODP).

This specification establishes measurable service expectations for API consumers while ensuring operational reliability, transparency, traceability, and enterprise readiness.

2. Objectives

OBJ-API-SLM-001

Establish enterprise API service levels.

OBJ-API-SLM-002

Define operational commitments.

OBJ-API-SLM-003

Support service reliability.

OBJ-API-SLM-004

Enable service performance governance.

OBJ-API-SLM-005

Maintain operational accountability.

OBJ-API-SLM-006

Support audit and compliance requirements.

OBJ-API-SLM-007

Enable service investigation readiness.

OBJ-API-SLM-008

Support enterprise customer expectations.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply across all production service environments.

4. Service Level Principles

All API services shall comply with the following principles.

SLM-PRN-001

Availability

Services shall maintain defined availability targets.

SLM-PRN-002

Reliability

Services shall operate consistently.

SLM-PRN-003

Transparency

Service commitments shall be documented.

SLM-PRN-004

Accountability

Service ownership shall be assigned.

SLM-PRN-005

Governance

Service commitments shall be governed.

SLM-PRN-006

Auditability

Service performance shall be auditable.

SLM-PRN-007

Traceability

Operational activities shall be traceable.

SLM-PRN-008

Continuous Improvement

Service performance shall be continuously evaluated.

5. Service Level Management Framework

The framework shall govern:

- Availability Management
- Performance Management
- Capacity Management
- Reliability Management
- Service Reporting

Service levels shall be formally documented.

6. Service Classification Framework

Services shall be classified as:

SLM-CLASS-001

Critical Services

SLM-CLASS-002

High Priority Services

SLM-CLASS-003

Standard Services

SLM-CLASS-004

Support Services

Classification shall determine service targets.

7. Availability Requirements

Availability management shall support:

- Service Uptime Monitoring
- Service Availability Reporting
- Availability Validation

Availability objectives shall be approved.

8. Performance Requirements

Performance management shall support:

- Response Time Monitoring
- Throughput Monitoring
- Capacity Monitoring

Performance objectives shall be documented.

9. Reliability Requirements

Reliability controls shall support:

- Failure Detection
- Recovery Validation
- Stability Monitoring

Reliability metrics shall be collected.

10. Service Ownership Requirements

Each API service shall have:

- Service Owner
- Technical Owner
- Operational Owner
- Governance Owner

Ownership assignments shall be maintained.

11. Operational Commitment Requirements

Operational commitments shall support:

- Service Availability
- Incident Response
- Service Recovery
- Customer Communication

Commitments shall be measurable.

12. Service Monitoring Requirements

Monitoring shall include:

- Availability Monitoring
- Performance Monitoring
- Capacity Monitoring
- Dependency Monitoring

Monitoring controls shall remain operational.

13. Incident Response Requirements

Service incidents shall support:

- Detection
- Escalation
- Investigation
- Resolution

Incident handling shall be documented.

14. Escalation Requirements

Escalation procedures shall support:

- Operational Escalation
- Security Escalation

- Governance Escalation
- Executive Escalation

Escalation paths shall be maintained.

15. Service Reporting Requirements

Reporting shall support:

- Operational Reporting
- Executive Reporting
- Governance Reporting
- Compliance Reporting

Reports shall be retained.

16. Dependency Management Requirements

Service dependencies shall be monitored for:

- Availability
- Reliability
- Capacity
- Operational Health

Dependency risks shall be tracked.

17. Capacity Management Requirements

Capacity controls shall support:

- Capacity Forecasting
- Utilization Monitoring
- Growth Planning

Capacity decisions shall be documented.

18. Governance Requirements

Governance validation shall apply to:

- Service level changes
- Availability target changes
- Operational commitment changes
- Escalation policy changes

Governance approvals shall be retained.

19. Audit Requirements

Audit records shall be generated for:

- Service level changes
- Operational reviews
- Incident escalations
- Performance reviews
- Governance actions

Audit records shall be immutable.

20. Security Requirements

Service management controls shall support:

- Access Control
- Monitoring Controls
- Incident Controls
- Reporting Controls

Security requirements shall align with SEC V1.

21. Compliance Requirements

Service level controls shall support:

- SOC 2 Availability Criteria
- ISO 27001 Operational Controls
- ISO 27001 Monitoring Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

22. Service Review Requirements

Service reviews shall evaluate:

- Availability Performance
- Reliability Performance
- Capacity Performance
- Operational Risks

Review outcomes shall be documented.

23. Operational Risk Requirements

Operational risks shall be assessed for:

- Service Availability
- Capacity Constraints

- Dependency Failures
- Recovery Limitations

Risk assessments shall be retained.

24. Investigation Support Requirements

Operational telemetry shall support:

- Root Cause Analysis
- Service Failure Analysis
- Timeline Reconstruction
- Replay Investigations

Investigation readiness shall be maintained.

25. Replay Preservation Requirements

Replay support shall preserve:

- Service Performance History
- Availability History
- Incident History
- Governance Actions

Historical reconstruction shall remain possible.

26. Continuous Improvement Requirements

Improvement activities shall support:

- Service Optimization
- Reliability Improvements
- Performance Improvements
- Capacity Improvements

Improvement activities shall be governed.

27. Architectural Constraints

All APIs shall comply with:

SLM-CON-001

Support service monitoring.

SLM-CON-002

Support availability management.

SLM-CON-003

Support performance management.

SLM-CON-004

Support governance oversight.

SLM-CON-005

Support auditability.

SLM-CON-006

Support traceability.

SLM-CON-007

Support replay investigations.

SLM-CON-008

Support enterprise operational commitments.

28. Metrics

The following service level metrics shall be monitored.

MET-SLM-001

Service availability percentage.

MET-SLM-002

Average response time.

MET-SLM-003

Incident volume.

MET-SLM-004

Mean time to detect.

MET-SLM-005

Mean time to resolve.

MET-SLM-006

Service reliability score.

MET-SLM-007

Replay investigation coverage.

MET-SLM-008

Service level compliance rate.

29. Related Traceability References

This specification maintains traceability with:

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

OPS RUNBOOK V1

RTM V1

SEC V1

GOV V1

AUD V1

30. Section Summary

This section defines the standards governing API service level management and operational commitments, including availability management, performance management, reliability controls, incident response, escalation procedures, governance oversight, audit obligations, compliance requirements, service reporting, and replay preservation. These controls ensure that TODP APIs deliver reliable, measurable, governed, auditable, and enterprise-grade operational services throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 28: API Documentation, Knowledge Management & Consumer Enablement Specification

Document ID: API-V1-S28

Document Title: API Documentation, Knowledge Management & Consumer Enablement Specification

Version: 1.0

Status: Approved

Owner: Product Engineering and Platform Operations

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the standards governing API documentation, knowledge management, developer enablement, operational guidance, governance awareness, audit readiness, and consumer onboarding across the Transparent Opportunity Decision Platform (TODP).

This specification ensures that all APIs are supported by accurate, discoverable, traceable, governed, and maintainable documentation throughout their lifecycle.

2. Objectives

OBJ-API-DOC-001

Establish enterprise documentation standards.

OBJ-API-DOC-002

Support API consumer adoption.

OBJ-API-DOC-003

Enable operational knowledge management.

OBJ-API-DOC-004

Maintain documentation traceability.

OBJ-API-DOC-005

Support governance and compliance requirements.

OBJ-API-DOC-006

Enable audit readiness.

OBJ-API-DOC-007

Support replay and investigation activities.

OBJ-API-DOC-008

Promote long-term platform maintainability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs
- Administrative APIs
- Security APIs

The controls apply to all internal and future external API consumers.

4. Documentation Principles

All API documentation shall comply with the following principles.

DOC-PRN-001

Accuracy

Documentation shall reflect implemented behavior.

DOC-PRN-002

Consistency

Documentation shall follow approved standards.

DOC-PRN-003

Traceability

Documentation shall support traceability.

DOC-PRN-004

Governance

Documentation changes shall be governed.

DOC-PRN-005

Auditability

Documentation history shall be auditable.

DOC-PRN-006

Accessibility

Documentation shall be discoverable.

DOC-PRN-007

Replayability

Documentation history shall support replay reconstruction.

DOC-PRN-008

Maintainability

Documentation shall remain current.

5. Documentation Framework

The documentation framework shall include:

- API Specifications
- Architecture References
- Governance References
- Security References
- Operational Guidance
- Consumer Guides

Documentation ownership shall be assigned.

6. Documentation Categories

Documentation shall be classified as:

DOC-CLASS-001

Technical Documentation

DOC-CLASS-002

Operational Documentation

DOC-CLASS-003

Governance Documentation

DOC-CLASS-004

Security Documentation

DOC-CLASS-005

Compliance Documentation

DOC-CLASS-006

Consumer Documentation

Classification shall determine review requirements.

7. API Reference Documentation Requirements

API reference documentation shall include:

- Endpoint Definitions
- Request Structures
- Response Structures
- Error Definitions

Documentation shall remain synchronized with API versions.

8. Consumer Onboarding Requirements

Consumer onboarding shall provide:

- API Overview
- Access Procedures
- Authentication Guidance
- Usage Standards

Onboarding materials shall be maintained.

9. Authentication Documentation Requirements

Documentation shall describe:

- Authentication Methods
- Token Usage
- Session Management
- Security Expectations

Authentication guidance shall remain current.

10. Authorization Documentation Requirements

Documentation shall define:

- Access Models
- Permission Models
- Governance Requirements
- Authorization Constraints

Authorization controls shall be documented.

11. Error Handling Documentation Requirements

Documentation shall include:

- Error Categories
- Error Codes
- Recovery Guidance
- Troubleshooting Procedures

Error documentation shall remain consistent.

12. Integration Documentation Requirements

Integration guidance shall include:

- Integration Patterns
- Connectivity Requirements
- Security Controls
- Operational Expectations

Integration standards shall be documented.

13. Governance Documentation Requirements

Documentation shall include:

- Governance Policies
- Approval Requirements
- Exception Procedures
- Review Processes

Governance references shall remain traceable.

14. Security Documentation Requirements

Documentation shall include:

- Security Controls
- Security Responsibilities
- Incident Reporting Guidance
- Security Monitoring Expectations

Security references shall remain current.

15. Compliance Documentation Requirements

Documentation shall support:

- Compliance Controls
- Audit Evidence Collection
- Compliance Reporting
- Regulatory Readiness

Compliance mappings shall be maintained.

16. Operational Documentation Requirements

Operational documentation shall include:

- Monitoring Procedures
- Escalation Procedures
- Recovery Procedures
- Support Procedures

Operational guidance shall be reviewed periodically.

17. Knowledge Management Requirements

Knowledge management shall support:

- Knowledge Preservation
- Knowledge Sharing
- Knowledge Discovery
- Knowledge Maintenance

Knowledge repositories shall be maintained.

18. Documentation Review Requirements

Documentation reviews shall evaluate:

- Accuracy
- Completeness
- Consistency
- Governance Alignment

Review outcomes shall be documented.

19. Documentation Version Control Requirements

Version control shall support:

- Version History
- Change Tracking
- Approval Tracking
- Historical Preservation

Version records shall be retained.

20. Consumer Communication Requirements

Consumer communications shall support:

- Release Notifications
- Deprecation Notices
- Security Notices
- Operational Notices

Communications shall be documented.

21. Governance Requirements

Governance validation shall apply to:

- Documentation standards
- Documentation changes
- Consumer guidance changes
- Governance documentation changes

Governance approvals shall be retained.

22. Audit Requirements

Audit records shall be generated for:

- Documentation changes
- Documentation approvals
- Knowledge repository changes
- Consumer communications
- Governance actions

Audit records shall be immutable.

23. Security Requirements

Documentation controls shall support:

- Access Control
- Classification Controls
- Repository Security
- Monitoring Controls

Security requirements shall align with SEC V1.

24. Compliance Requirements

Documentation governance shall support:

- SOC 2 Documentation Controls
- ISO 27001 Documentation Controls
- ISO 27001 Knowledge Management Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

25. Investigation Support Requirements

Documentation repositories shall support:

- Root Cause Analysis
- Historical Research
- Governance Investigations
- Replay Investigations

Investigation readiness shall be maintained.

26. Replay Preservation Requirements

Replay support shall preserve:

- Documentation Versions
- Historical Specifications
- Governance References
- Operational Guidance

Historical reconstruction shall remain possible.

27. Consumer Enablement Requirements

Consumer enablement shall support:

- Self-Service Learning
- Technical Guidance
- Operational Guidance
- Governance Awareness

Enablement materials shall remain accessible.

28. Architectural Constraints

All APIs shall comply with:

DOC-CON-001

Maintain documentation accuracy.

DOC-CON-002

Support version control.

DOC-CON-003

Support traceability.

DOC-CON-004

Support auditability.

DOC-CON-005

Support governance oversight.

DOC-CON-006

Support replay investigations.

DOC-CON-007

Support consumer enablement.

DOC-CON-008

Support enterprise maintainability.

29. Metrics

The following documentation metrics shall be monitored.

MET-DOC-001

Documentation coverage percentage.

MET-DOC-002

Documentation review completion rate.

MET-DOC-003

Documentation accuracy score.

MET-DOC-004

Consumer onboarding completion rate.

MET-DOC-005

Knowledge repository utilization.

MET-DOC-006

Documentation update frequency.

MET-DOC-007

Replay documentation coverage.

MET-DOC-008

Documentation compliance rate.

30. Section Summary

This section defines the standards governing API documentation, knowledge management, consumer onboarding, governance awareness, operational guidance, compliance support, audit readiness, replay preservation, and long-term maintainability. These controls ensure that all TODP APIs remain well-documented, discoverable, governed, traceable, auditable, consumer-friendly, and enterprise-ready throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 29: API Governance, Ownership & Accountability Specification

Document ID: API-V1-S29

Document Title: API Governance, Ownership & Accountability Specification

Version: 1.0

Status: Approved

Owner: Product Governance Council

Review Authority: Architecture Review Board (ARB)

Approval Authority: Executive Governance Committee

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the governance framework, ownership model, accountability structure, decision rights, oversight responsibilities, and control mechanisms governing all APIs within the Transparent Opportunity Decision Platform (TODP).

This specification ensures that API services remain governed, accountable, traceable, auditable, compliant, and aligned with enterprise objectives throughout their lifecycle.

2. Objectives

OBJ-API-GOV-001

Establish API governance standards.

OBJ-API-GOV-002

Define ownership responsibilities.

OBJ-API-GOV-003

Establish accountability structures.

OBJ-API-GOV-004

Support governance oversight.

OBJ-API-GOV-005

Maintain auditability of governance decisions.

OBJ-API-GOV-006

Support compliance requirements.

OBJ-API-GOV-007

Enable replay-based governance reconstruction.

OBJ-API-GOV-008

Maintain enterprise decision accountability.

3. Scope

This specification applies to:

- Opportunity APIs
- Authority APIs
- Evidence APIs
- Retrieval APIs
- Ranking APIs
- Recommendation APIs
- Governance APIs
- Audit APIs
- Replay APIs

- Administrative APIs
- Security APIs

The governance model applies across all environments and operational states.

4. Governance Principles

All APIs shall comply with the following principles.

GOV-PRN-001

Accountability

Ownership shall be assigned.

GOV-PRN-002

Transparency

Governance decisions shall be documented.

GOV-PRN-003

Traceability

Governance actions shall be traceable.

GOV-PRN-004

Auditability

Governance activities shall be auditable.

GOV-PRN-005

Consistency

Governance controls shall be applied consistently.

GOV-PRN-006

Compliance

Governance decisions shall support compliance obligations.

GOV-PRN-007

Replayability

Governance history shall support replay reconstruction.

GOV-PRN-008

Enterprise Alignment

Governance decisions shall support business objectives.

5. Governance Framework

The governance framework shall include:

- Policy Governance
- Architecture Governance
- Security Governance
- Operational Governance
- Compliance Governance
- Change Governance

Governance responsibilities shall be documented.

6. Governance Structure

The governance structure shall include:

- Executive Governance Committee
- Product Governance Council
- Architecture Review Board
- Security Governance Board
- Operational Governance Team

Governance responsibilities shall be formally assigned.

7. API Ownership Model

Every API shall have:

- Business Owner
- Product Owner
- Technical Owner
- Operational Owner

Ownership assignments shall be maintained.

8. Business Owner Responsibilities

Business Owners shall be accountable for:

- Business Alignment
- Strategic Direction
- Consumer Value
- Governance Compliance

Business ownership records shall be maintained.

9. Product Owner Responsibilities

Product Owners shall be accountable for:

- API Roadmaps
- Functional Requirements
- Consumer Experience
- Product Governance

Product ownership shall be documented.

10. Technical Owner Responsibilities

Technical Owners shall be accountable for:

- API Design
- Technical Standards
- Architecture Compliance
- Operational Readiness

Technical ownership shall remain current.

11. Operational Owner Responsibilities

Operational Owners shall be accountable for:

- Service Availability
- Monitoring
- Incident Management
- Operational Reporting

Operational responsibilities shall be documented.

12. Governance Decision Rights

Governance authorities shall manage:

- Policy Decisions
- Architecture Decisions
- Security Decisions
- Compliance Decisions

Decision rights shall be documented.

13. Governance Review Requirements

Governance reviews shall evaluate:

- Architecture Alignment
- Security Compliance
- Operational Readiness
- Regulatory Compliance

Review outcomes shall be retained.

14. Policy Management Requirements

Policy governance shall support:

- Policy Creation
- Policy Approval
- Policy Review
- Policy Retirement

Policy history shall remain traceable.

15. Exception Management Requirements

Governance exceptions shall require:

- Business Justification
- Risk Assessment
- Approval Workflow
- Expiration Date

Exceptions shall remain auditable.

16. Risk Governance Requirements

Risk governance shall support:

- Risk Identification
- Risk Evaluation
- Risk Approval
- Risk Monitoring

Risk records shall be maintained.

17. Architecture Governance Requirements

Architecture governance shall support:

- Design Reviews
- Standards Compliance
- Change Reviews
- Architecture Decisions

Architecture decisions shall be preserved.

18. Security Governance Requirements

Security governance shall support:

- Security Reviews
- Security Exceptions
- Security Approvals
- Security Monitoring

Security decisions shall remain auditable.

19. Compliance Governance Requirements

Compliance governance shall support:

- Compliance Reviews
- Compliance Validation
- Compliance Reporting
- Compliance Remediation

Compliance governance records shall be maintained.

20. Change Governance Requirements

Change governance shall support:

- Change Reviews
- Approval Workflows
- Release Governance
- Rollback Governance

Change decisions shall remain traceable.

21. Governance Reporting Requirements

Governance reporting shall support:

- Executive Reporting
- Operational Reporting
- Compliance Reporting
- Risk Reporting

Reports shall be retained.

22. Audit Requirements

Audit records shall be generated for:

- Governance Decisions
- Governance Reviews

- Exception Approvals
- Risk Decisions
- Policy Changes

Audit records shall be immutable.

23. Security Requirements

Governance controls shall support:

- Access Control
- Approval Control
- Traceability Control
- Monitoring Control

Security requirements shall align with SEC V1.

24. Compliance Requirements

Governance controls shall support:

- SOC 2 Governance Controls
- ISO 27001 Governance Controls
- ISO 27001 Risk Management Controls
- Enterprise Governance Standards

Compliance mappings shall be maintained.

25. Investigation Support Requirements

Governance telemetry shall support:

- Decision Analysis
- Risk Analysis
- Compliance Investigations
- Replay Investigations

Investigation readiness shall be maintained.

26. Replay Preservation Requirements

Replay support shall preserve:

- Governance Decisions
- Governance Reviews
- Exception Records
- Risk Decisions

Historical governance reconstruction shall remain possible.

27. Accountability Requirements

Accountability controls shall support:

- Decision Ownership
- Approval Ownership
- Operational Ownership
- Compliance Ownership

Ownership lineage shall remain traceable.

28. Architectural Constraints

All APIs shall comply with:

GOV-CON-001

Require assigned ownership.

GOV-CON-002

Require governance oversight.

GOV-CON-003

Support traceability.

GOV-CON-004

Support auditability.

GOV-CON-005

Support compliance validation.

GOV-CON-006

Support replay investigations.

GOV-CON-007

Support accountability management.

GOV-CON-008

Support enterprise governance standards.

29. Metrics

The following governance metrics shall be monitored.

MET-GOV-001

Governance review completion rate.

MET-GOV-002

Policy compliance rate.

MET-GOV-003

Exception volume.

MET-GOV-004

Governance decision cycle time.

MET-GOV-005

Risk review completion rate.

MET-GOV-006

Ownership assignment coverage.

MET-GOV-007

Replay governance coverage.

MET-GOV-008

Governance audit score.

30. Section Summary

This section defines the governance, ownership, and accountability framework governing TODP APIs, including governance structures, ownership responsibilities, decision rights, policy management, risk governance, security governance, compliance governance, change governance, audit obligations, replay preservation, and accountability controls. These standards ensure that all TODP APIs remain governed, accountable, traceable, auditable, compliant, and aligned with enterprise objectives throughout the platform lifecycle.

API SPEC V1 – Enterprise API Specification

Section 30: API Specification Governance, Traceability & Document Closure

Document ID: API-V1-S30

Document Title: API Specification Governance, Traceability & Document Closure

Version: 1.0

Status: Approved

Owner: Product Governance Council

Review Authority: Architecture Review Board (ARB)
Approval Authority: Executive Governance Committee
Classification: Internal Confidential

1. Purpose

The purpose of this section is to establish the governance framework, traceability requirements, maintenance obligations, review procedures, approval controls, audit requirements, and lifecycle management processes governing the API Specification V1 document for the Transparent Opportunity Decision Platform (TODP).

This section formally closes API SPEC V1 and establishes its position as an authoritative enterprise reference document.

2. Objectives

OBJ-API-CLS-001

Establish document governance.

OBJ-API-CLS-002

Define traceability obligations.

OBJ-API-CLS-003

Maintain specification integrity.

OBJ-API-CLS-004

Support enterprise auditability.

OBJ-API-CLS-005

Enable controlled document evolution.

OBJ-API-CLS-006

Support compliance verification.

OBJ-API-CLS-007

Enable replay-based historical reconstruction.

OBJ-API-CLS-008

Provide long-term documentation sustainability.

3. Scope

This section governs:

- API SPEC V1
- Future API Specification versions
- API governance controls
- API traceability controls
- API review procedures
- API approval procedures
- API archival procedures

The controls apply throughout the API specification lifecycle.

4. API Specification Authority

API SPEC V1 shall serve as the authoritative enterprise specification for:

- API architecture
- API governance
- API security
- API operations
- API compliance
- API lifecycle management

Conflicting specifications shall require formal governance review.

5. Specification Ownership

Ownership shall be assigned to:

Primary Owner

Product Governance Council

Technical Custodian

Architecture Review Board

Operational Custodian

Platform Engineering

Security Custodian

Platform Security Team

Ownership assignments shall remain current.

6. Specification Governance Framework

Governance activities shall include:

- Periodic Reviews
- Change Management
- Version Governance
- Compliance Validation
- Audit Support

Governance activities shall be documented.

7. Review Frequency Requirements

The specification shall be reviewed:

- Annually
- Following major platform changes
- Following major governance changes
- Following major security changes

Review outcomes shall be retained.

8. Change Control Requirements

Changes to API SPEC V1 shall require:

- Impact Assessment
- Architecture Review
- Governance Approval
- Documentation Updates

Changes shall remain traceable.

9. Traceability Framework

API SPEC V1 shall maintain traceability to:

- Business Requirements
- Functional Requirements
- Technical Requirements
- Governance Requirements
- Security Requirements

Traceability relationships shall be preserved.

10. RTM Alignment Requirements

API SPEC V1 shall maintain alignment with:

RTM V1

All API requirements shall remain traceable to approved requirements and controls.

11. BRD Traceability Requirements

API specifications shall maintain traceability to:

BRD V1

Business objectives and business capabilities shall remain linked to API services.

12. SRS Traceability Requirements

API specifications shall maintain traceability to:

SRS V1

Functional requirements shall remain linked to API behaviors and interfaces.

13. SAD Traceability Requirements

API specifications shall maintain traceability to:

SAD V1

Architectural decisions and service boundaries shall remain aligned.

14. TDD Traceability Requirements

API specifications shall maintain traceability to:

TDD V1

Implementation-level design controls shall remain synchronized.

15. TRM Traceability Requirements

API specifications shall maintain traceability to:

TRM V1

Technology decisions shall remain aligned with API implementation constraints.

16. ADR Traceability Requirements

API specifications shall maintain traceability to:

ADR V1

Architectural decisions shall remain linked to API standards and constraints.

17. Security Traceability Requirements

API specifications shall maintain traceability to:

SEC V1

Security controls shall remain linked to API protection mechanisms.

18. Governance Traceability Requirements

API specifications shall maintain traceability to:

GOV V1

Governance controls shall remain linked to API governance processes.

19. Audit Traceability Requirements

API specifications shall maintain traceability to:

AUD V1

Audit controls shall remain linked to API audit requirements.

20. Compliance Traceability Requirements

API specifications shall maintain traceability to:

COMP V1

Compliance controls shall remain linked to API compliance obligations.

21. Replay Traceability Requirements

API specifications shall maintain traceability to:

REL V1

Replay and reconstruction requirements shall remain linked to API operations.

22. Audit Requirements

Audit activities shall verify:

- Specification Accuracy
- Traceability Integrity
- Governance Compliance
- Security Alignment
- Compliance Alignment

Audit findings shall be remediated.

23. Compliance Requirements

API SPEC V1 shall support:

- SOC 2 Readiness
- ISO 27001 Readiness
- Enterprise Procurement Reviews
- Customer Due Diligence Reviews

Compliance evidence shall be maintained.

24. Replay Preservation Requirements

Replay support shall preserve:

- Historical API Specifications
- Historical Version Records
- Historical Approval Records
- Historical Governance Records

Historical reconstruction shall remain possible.

25. Archival Requirements

Archived versions shall preserve:

- Version History
- Approval History
- Change History
- Traceability History

Archived artifacts shall remain accessible.

26. Version Evolution Requirements

Future versions shall support:

- Controlled Evolution
- Backward Traceability
- Governance Validation
- Compliance Validation

Version transitions shall be governed.

27. Enterprise Readiness Requirements

API SPEC V1 shall support:

- Enterprise Customers
- Regulatory Reviews

- Security Reviews
- Architecture Reviews
- Operational Reviews

Enterprise readiness shall be maintained.

28. Architectural Constraints

API SPEC V1 shall comply with:

CLS-CON-001

Maintain traceability.

CLS-CON-002

Maintain governance oversight.

CLS-CON-003

Support auditability.

CLS-CON-004

Support compliance verification.

CLS-CON-005

Support replayability.

CLS-CON-006

Support controlled evolution.

CLS-CON-007

Preserve historical integrity.

CLS-CON-008

Support enterprise readiness.

29. Metrics

The following governance metrics shall be monitored.

MET-CLS-001

Traceability coverage.

MET-CLS-002

Review completion rate.

MET-CLS-003

Audit compliance rate.

MET-CLS-004

Specification update frequency.

MET-CLS-005

Governance approval cycle time.

MET-CLS-006

Replay preservation coverage.

MET-CLS-007

Compliance readiness score.

MET-CLS-008

Enterprise documentation maturity score.

30. Section Summary

This section establishes the governance, traceability, maintenance, auditability, compliance, replay preservation, archival, and lifecycle management requirements for API SPEC V1. It formally closes the API Specification document and designates it as the authoritative enterprise reference governing all API services within TODP. The controls defined herein ensure that API SPEC V1 remains traceable, governed, auditable, compliant, replayable, maintainable, and aligned with the long-term enterprise objectives of the Transparent Opportunity Decision Platform.